

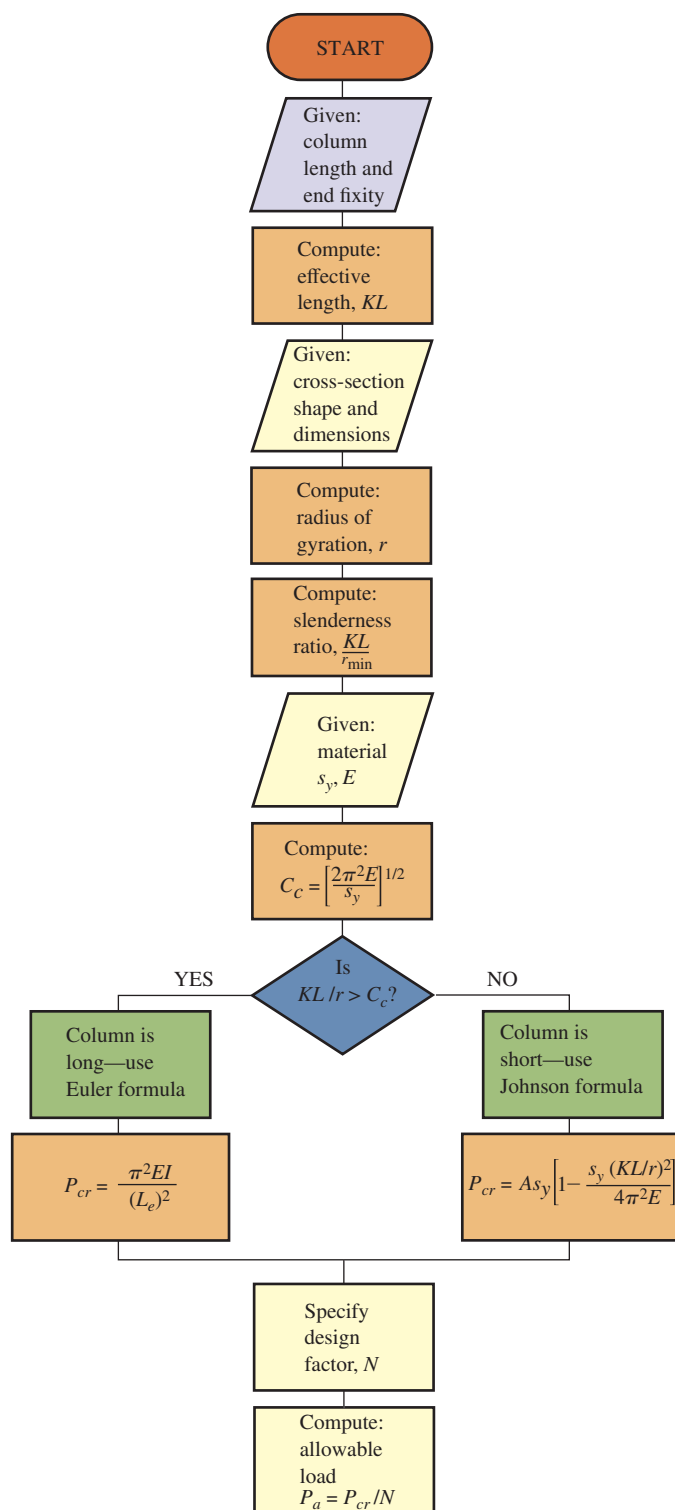


FIGURE 6-7 Analysis of a straight, centrally loaded column

Example Problem 6-1

A column has a solid circular cross section, 1.25 inches in diameter; it has a length of 4.50 ft and is pinned at both ends. If it is made from SAE 1020 cold-drawn steel, what would be a safe column loading?

Solution

Objective Specify a safe loading for the column.

Given

Solid circular cross section: diameter = $D = 1.25$ in; length = $L = 4.50$ ft.
Both ends of the column are pinned.
Material: SAE 1020 cold-drawn steel.

Analysis Use the procedure in Figure 6–7.

Results **Step 1.** For the pinned-end column, the end-fixity factor is $K = 1.0$. The effective length equals the actual length; $KL = 4.50 \text{ ft} = 54.0 \text{ in} = L_e$.

Step 2. From Appendix 1, for a solid round section,

$$r = D/4 = 1.25/4 = 0.3125 \text{ in}$$

Step 3. Compute the slenderness ratio:

$$\frac{KL}{r} = \frac{1.0(54)}{0.3125} = 173$$

Step 4. Compute the column constant from Equation (6–8). For SAE 1020 cold-drawn steel, the yield strength is 51 000 psi, and the modulus of elasticity is 30×10^6 psi. Then

$$C_c = \sqrt{\frac{2\pi^2 E}{s_y}} = \sqrt{\frac{2\pi^2(30 \times 10^6)}{51\,000}} = 108$$

Step 5. Because KL/r is greater than C_c , the column is long, and Euler's formula should be used. The area is

$$I = \frac{\pi D^4}{64} = \frac{\pi(1.25)^4}{64} = 0.1198 \text{ in}^4$$

Then the critical load is

$$P_{cr} = \frac{\pi^2 EI}{(L_e)^2} = \frac{\pi^2(30 \times 10^6)(0.1198)}{(54)^2} = 12\,169 \text{ lb}$$

At this load, the column should just begin to buckle. A safe load would be a reduced value, found by applying the design factor to the critical load. Let's use $N = 3$ to compute the *allowable load*, $P_a = P_{cr}/N$:

$$P_a = (12\,169 \text{ lb})/3 = 4056 \text{ lb}$$

Comment The safe load on the column is 4056 lb.

Example Problem 6–2

Determine the critical load on a steel column having a rectangular cross section, 12 mm by 18 mm, and a length of 280 mm. It is proposed to use SAE 1040 hot-rolled steel. The lower end of the column is inserted into a close-fitting socket and is welded securely. The upper end is pinned (see Figure 6–8).

Solution

Objective Compute the critical load for the column.

Given Solid rectangular cross section: $B = 12 \text{ mm}$; $H = 18 \text{ mm}$; $L = 280 \text{ mm}$.
 Cross sectional area = $A = BH = (12)(18) = 216 \text{ mm}^2$
 The bottom of column is fixed; the top is pinned (see Figure 6–8).
 Material: SAE 1040 hot-rolled steel.

Analysis Use the procedure in Figure 6–7.

Results **Step 1.** Compute the slenderness ratio. The radius of gyration must be computed about the axis that gives the least value. This is the Y-Y axis, for which

$$r = \frac{B}{\sqrt{12}} = \frac{12 \text{ mm}}{\sqrt{12}} = 3.46 \text{ mm}$$

The column has a fixed-pinned end fixity for which $K = 0.8$. Then

$$KL/r = [(0.8)(280)]/3.46 = 64.7$$

Step 2. Compute the transition slenderness ratio. For the SAE 1040 hot-rolled steel, $E = 207 \text{ GPa}$ and $s_y = 290 \text{ MPa}$. Then, from Equation (6–8),

$$C_c = \sqrt{\frac{2\pi^2(207 \times 10^9 \text{ Pa})}{290 \times 10^6 \text{ Pa}}} = 119$$

Step 3. Then $KL/r < C_c$; thus the column is short. Use the J. B. Johnson formula (Eqn. 6–9) to compute the critical load.

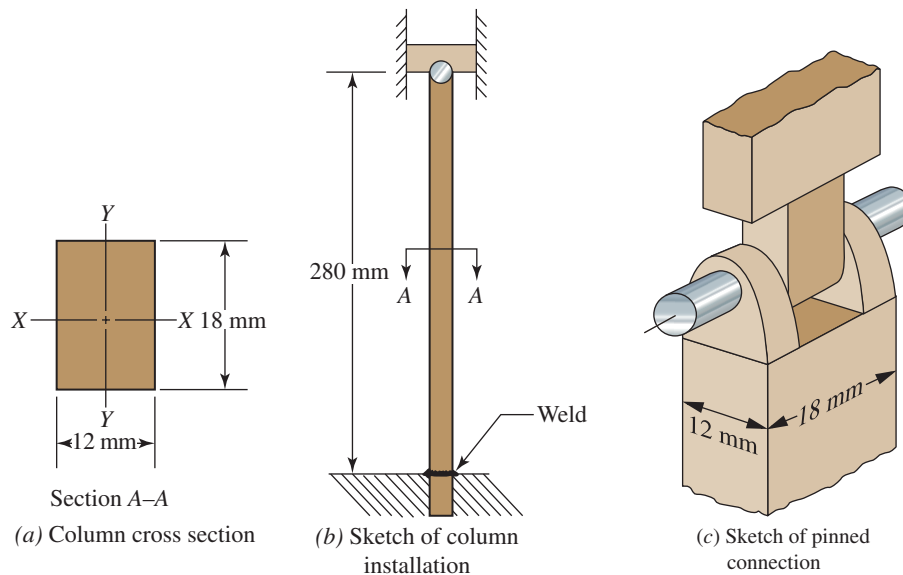


FIGURE 6-8 Steel column

$$P_{cr} = A_s \left[1 - \frac{s_y (KL/r)^2}{4\pi^2 E} \right]$$

$$P_{cr} = (216 \text{ mm}^2)(290 \text{ N/mm}^2) \left[1 - \frac{(290 \times 10^6 \text{ Pa})(64.7)^2}{4\pi^2 (207 \times 10^9 \text{ Pa})} \right]$$

$$P_{cr} = 53.3 \times 10^3 \text{ N} = 53.3 \text{ kN}$$

Comments This is the critical buckling load. We would have to apply a design factor to determine the allowable load. Specifying $N = 3$ results in $P_a = 17.8 \text{ kN}$.

6-8 COLUMN ANALYSIS SPREADSHEET

Completing the process described in Figure 6-4 using a calculator, pencil, and paper is tedious. A spreadsheet automates the calculations after you have entered the pertinent data for the particular column to be analyzed. Figure 6-9 shows the output of a spreadsheet used to solve Example Problem 6-1. The layout of the spreadsheet could be done in many ways, and you are encouraged to develop your own style. The following comments describe the features of the given spreadsheet:

1. At the top of the sheet, instructions to the user are given for entering data and for units. This sheet is for U.S. Customary units only. A different sheet would be used if SI metric data were to be used. (See Figure 6-10, which gives the solution for Example Problem 6-2.)
2. On the left side of the sheet are listed the various data that must be provided by the user to run the calculations. On the right are listed the output values. Formulas for computing L_e , C_c , KL/r , and allowable load are written directly into the cell where the computed values show. The output data for the message "Column is: *long*" and the critical buckling load are

produced by *functions* set up within *macros* written in Visual Basic and placed on a separate sheet of the spreadsheet. Figure 6-11 shows the two macros used. The first (*LorS*) carries out the decision process to test whether the column is long or short as indicated by comparison of its slenderness ratio with the column constant. The second (*Pcr*) computes the critical buckling load using either the Euler formula or the J. B. Johnson formula, depending on the result of the *LorS* macro. These functions are called by statements in the cells where "long" and the computed value of the critical buckling load (12 197 lb) are located.

3. Having such a spreadsheet can enable you to analyze several design options quickly. For example, the given problem statement indicated that the ends were pinned, resulting in an end-fixity value of $K = 1$. What would happen if both ends were fixed? Simply changing the value of that one cell to $K = 0.65$ would cause the entire sheet to be recalculated, and the revised value of critical buckling load would be available almost instantly. The result is that $P_{cr} = 28\,868 \text{ lb}$, an increase of 2.37 times the original value. With that kind of improvement, you, the designer, might be inclined to change the design to produce fixed ends.

COLUMN ANALYSIS PROGRAM		Data from:	Example Problem 6-1
Refer to Figure 6-7 for analysis logic.			
Enter data for variables in <i>italics in shaded boxes</i> .			Use consistent U.S. Customary units.
Data to Be Entered:			Computed Values:
Length and End Fixity:			
Column length, $L =$	54 in	→	Eff. length, $L_e = KL = 54.0$ in
End fixity, $K =$	1		
Material Properties:			
Yield strength, $s_y =$	51 000 psi	→	Column constant, $C_c = 107.8$
Modulus of elasticity, $E =$	3.00E + 07 psi		
Cross Section Properties:			
[Note: Enter r or compute $r = \sqrt{I/A}$.] [Always enter Area.] [Enter zero for I or r if not used.]			
Area, $A =$	1.23 in ²		
Moment of inertia, $I =$	0 in ⁴		
Or			
Radius of gyration, $r =$	0.3125 in	→	Slender ratio, $KL/r = 172.8$
			Column is: long
			Critical buckling load = 12 197 lb
Design Factor:			
Design factor on load, $N =$	3	→	Allowable load = 4 066 lb

FIGURE 6-9 Spreadsheet for column analysis with data from Example Problem 6-1

6-9 EFFICIENT SHAPES FOR COLUMN CROSS SECTIONS

An *efficient shape* is one that provides good performance with a small amount of material. In the case of columns, the shape of the cross section and its dimensions determine the value of the radius of gyration, r . From the definition of the slenderness ratio, KL/r , we can see that as r gets larger, the slenderness ratio gets smaller. In the critical load equations, a smaller slenderness ratio results in a larger critical load, the most desirable situation. Therefore, it is desirable to maximize the radius of gyration to design an efficient column cross section.

Unless end fixity varies with respect to the axes of the cross section, the column will tend to buckle with respect to the axis with the *least* radius of gyration. So a column with equal values for the radius of gyration in any direction is desirable.

Review again the definition of the radius of gyration:

$$r = \sqrt{I/A}$$

This equation indicates that for a given area of material, we should try to maximize the moment of inertia

to maximize the radius of gyration. A shape with a high moment of inertia has its area distributed far away from its centroidal axis.

Shapes that have the desirable characteristics described include circular hollow pipes and tubes, square hollow tubing, and fabricated column sections made from structural shapes placed at the outer boundaries of the section. Solid circular sections and solid square sections are also good, although not as efficient as the hollow sections. Figure 6-12(a-d) illustrates some of these shapes. The built-up section in (e) gives a rigid, boxlike section approximating the hollow square tube in larger sizes. In the case of the section in Figure 6-12(f), the angle sections at the corners provide the greatest contribution to the moment of inertia. The lacing bars merely hold the angles in position. The H-column in (g) has an equal depth and width and relatively heavy flanges and web. The moment of inertia with respect to the Y-Y axis is still smaller than for the X-X axis, but they are more nearly equal than for most other I-shapes designed to be used as beams with bending in only one direction. Thus, this shape would be more desirable for columns.

COLUMN ANALYSIS PROGRAM		Data from:	Example Problem 6-2
Refer to Figure 6-7 for analysis logic.			
Enter data for variables in <i>italics in shaded boxes</i> .			Use consistent SI metric units.
Data to Be Entered:			Computed Values:
Length and End Fixity:			
Column length, $L = 280 \text{ mm}$ End fixity, $K = 0.8$		→	Eff. length, $L_e = KL = 224.0 \text{ mm}$
Material Properties:			
Yield strength, $s_y = 290 \text{ MPa}$ Modulus of elasticity, $E = 207 \text{ GPa}$		→	Column constant, $C_c = 118.7$
Cross Section Properties:			
[Note: Enter r or compute $r = \sqrt{I/A}$.] [Always enter Area.] [Enter zero for I or r if not used.]			
Area, $A = 216 \text{ mm}^2$ Moment of inertia, $I = 0 \text{ mm}^4$ Or Radius of gyration, $r = 3.5 \text{ mm}$		→	Slender ratio, $KL/r = 64.7$
Design factor:			Column is: <i>short</i>
Design factor on load, $N = 3$		→	Critical buckling load = 53.32 kN Allowable load = 17.77 kN

FIGURE 6-10 Spreadsheet for column analysis with data from Example Problem 6-2

```

' Lors Macro
' Determines if column is long or short.
Function LorS(SR, CC)
    If SR > CC Then
        LorS = "long"
    Else
        LorS = "short"
    End if
End Function
' Critical Load Macro
' Uses Euler formula for long columns
' Uses Johnson formula for short columns
Function Pcr(LorS, SR, E, A, Sy)
Const Pi = 3.1415926
    If LorS = "long" Then
        Pcr = Pi ^ 2 * E * A / SR ^ 2
        ' Euler Equation; Eq. (6-4)
    Else
        Pcr = A * Sy(1 - (Sy * SR ^ 2 / (4 * Pi ^ 2 * E)))
        ' Johnson Equation; Eq. (6-7)
    End if
End Function

```

FIGURE 6-11 Macros used in the column analysis spreadsheet

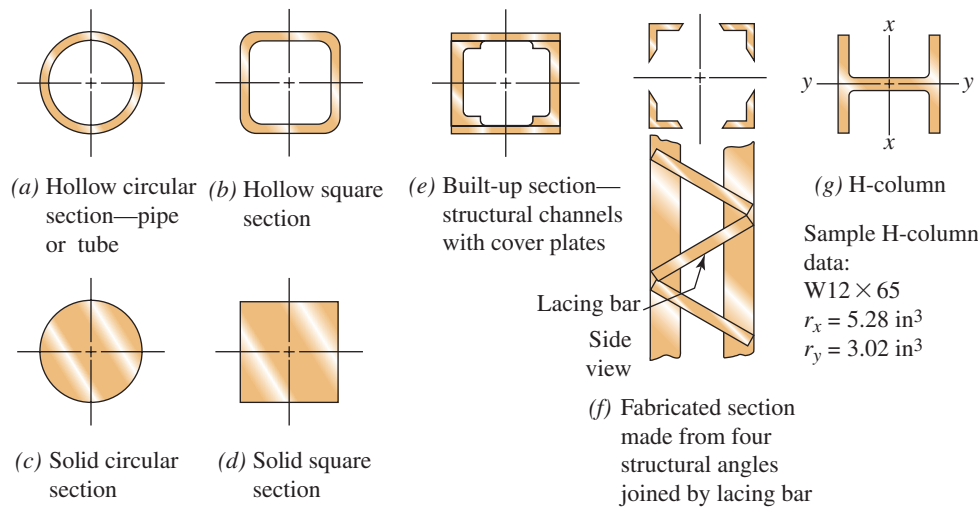


FIGURE 6-12 Column cross sections

6-10 THE DESIGN OF COLUMNS

In a design situation, the expected load on the column would be known, along with the length required by the application. The designer would then specify the following:

1. The manner of attaching the ends to the structure that affects the end fixity.
2. The general shape of the column cross section (e.g., round, square, rectangular, and hollow tube).
3. The material for the column.
4. The design factor, considering the application.
5. The final dimensions for the column.

It may be desirable to propose and analyze several different designs to approach an optimum for the application. A computer program or spreadsheet facilitates the process.

It is assumed that items 1 through 4 are specified by the designer for any given trial. For some simple shapes, such as the solid round or square section, the final dimensions are computed from the appropriate formula: the Euler formula, Equation (6-4) or (6-6), or the J. B. Johnson formula, Equation (6-9). If an algebraic solution is not possible, iteration can be done.

In a design situation, the unknown cross-sectional dimensions make computing the radius of gyration and therefore the slenderness ratio, KL/r , impossible. Without the slenderness ratio, we cannot determine whether the column is long (Euler) or short (Johnson). Thus, the proper formula to use is not known.

We overcome this difficulty by making an assumption that the column is either long or short and proceeding with the corresponding formula. Then, after the dimensions are determined for the cross section, the actual value of KL/r will be computed and compared with C_c . This will show whether the correct

formula has been used. If so, the computed answer is correct. If not, the alternate formula must be used and the computation is repeated to determine new dimensions. Figure 6-13 shows a flowchart for the design logic described here.

Design: Assuming a Long Column

Euler's formula is used if the assumption is that the column is long. Equation (6-4) would be the most convenient form because it can be solved for the moment of inertia, I :

⇨ Euler's Formula Solved for Required Value of I

$$I = \frac{P_{cr}(KL)^2}{\pi^2 E} = \frac{NP_a(KL)^2}{\pi^2 E} \quad (6-10)$$

where P_a = allowable load, usually set equal to the actual maximum expected load.

Having the required value for I , we can determine the dimensions for the shape by additional computations or by scanning tables of data of the properties of commercially available sections.

The solid circular section is one for which it is possible to derive a final equation for the characteristic dimension, the diameter. The moment of inertia is

$$I = \frac{\pi D^4}{64}$$

Substituting this into Equation (6-10) gives

$$I = \frac{\pi D^4}{64} = \frac{NP_a(KL)^2}{\pi^2 E}$$

Solving for D yields

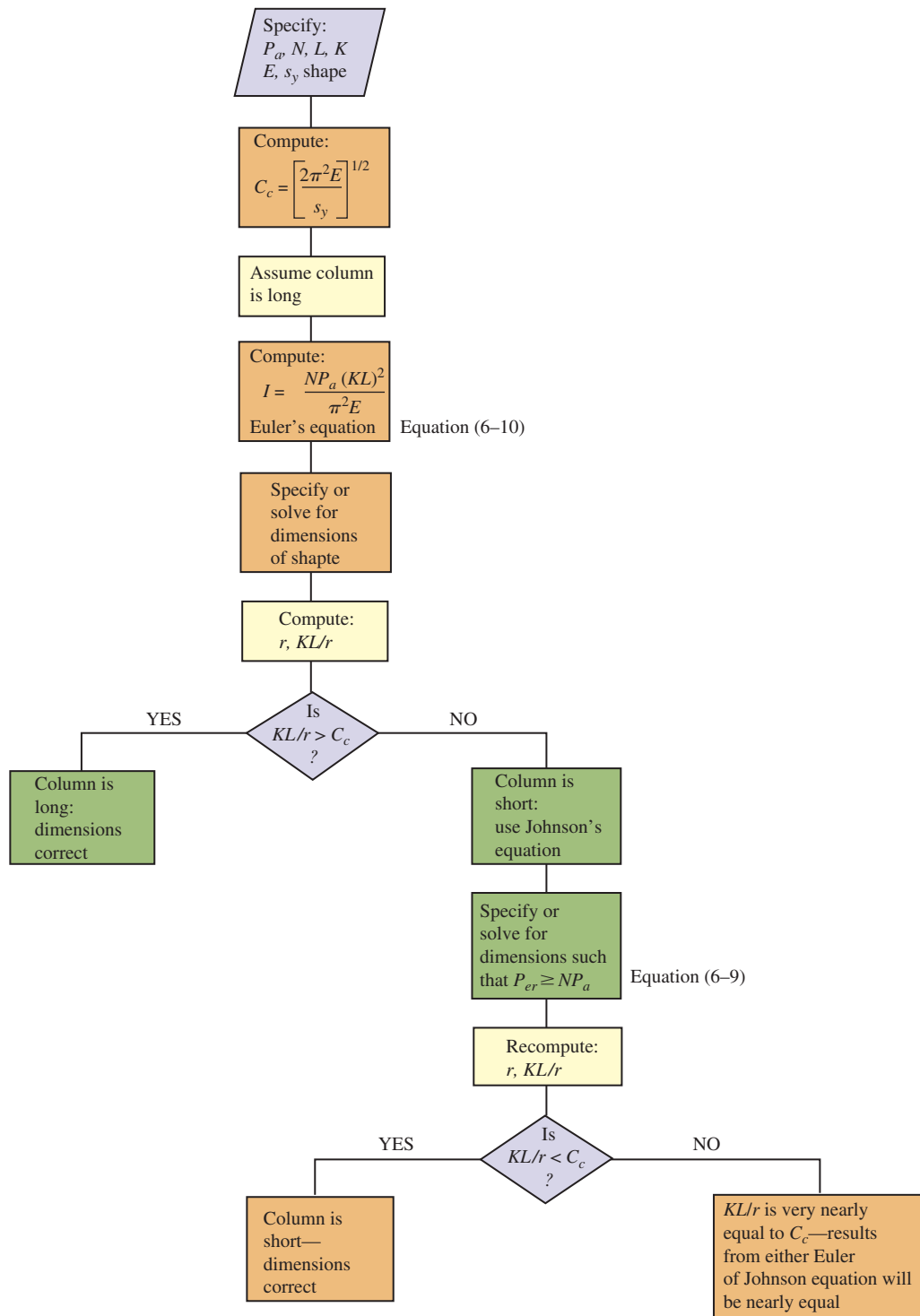


FIGURE 6-13 Design of a straight, centrally loaded column

➤ Required Diameter for a Long, Solid Circular Column

$$D = \left[\frac{64NP_a(KL)^2}{\pi^3 E} \right]^{1/4} \quad (6-11)$$

Design: Assuming a Short Column

The J. B. Johnson formula is used to analyze a short column. It is difficult to derive a convenient form for

use in design. In the general case, then, trial and error is used.

For some special cases, including the solid circular section, it is possible to solve the Johnson formula (Eqn. 6-9) for the characteristic dimension, the diameter:

$$P_{cr} = A s_y \left[1 - \frac{s_y (KL/r)^2}{4\pi^2 E} \right]$$

Since

$$A = \pi D^2/4$$

$$r = D/4 \text{ (from Appendix 1)}$$

$$P_{cr} = NP_a$$

$$\frac{4NP_a}{\pi s_y} = D^2 \left[1 - \frac{s_y(KL)^2(16)}{4\pi^2 ED^2} \right]$$

Solving for D gives

Then

$$NP_a = \frac{\pi D^2}{4} s_y \left[1 - \frac{s_y(KL)^2}{4\pi^2 E(D/4)^2} \right]$$

➤ Required Diameter for a Short, Solid Circular Column

$$D = \left[\frac{4NP_a}{\pi s_y} + \frac{4s_y(KL)^2}{\pi^2 E} \right]^{1/2} \quad (6-12)$$

Example Problem 6-3

Specify a suitable diameter of a solid, round cross section for a machine link if it is to carry 9800 lb of axial compressive load. The length will be 25 in, and the ends will be pinned. Use a design factor of 3. Use SAE 1020 hot-rolled steel.

Solution

Objective Specify a suitable diameter for the column.

Given Solid circular cross section: $L = 25$ in; use $N = 3$.
Both ends are pinned.
Material: SAE 1020 hot-rolled steel.

Analysis Use the procedure in Figure 6-13. Assume first that the column is long.

Results From Equation (6-11),

$$D = \left[\frac{64NP_a(KL)^2}{\pi^3 E} \right]^{1/4} = \left[\frac{64(3)(9800)(25)^2}{\pi^3(30 \times 10^6)} \right]^{1/4}$$

$$D = 1.06 \text{ in}$$

The radius of gyration can now be found:

$$r = D/4 = 1.06/4 = 0.265 \text{ in}$$

The slenderness ratio is

$$KL/r = [(1.0)(25)]/0.265 = 94.3$$

For the SAE 1020 hot-rolled steel, $s_y = 30\,000$ psi. The graph in Figure 6-5 shows C_c to be approximately 138. Thus, the actual KL/r is less than the transition value, and the column must be redesigned as a short column, using Equation (6-12) derived from the Johnson formula:

$$D = \left[\frac{4NP_a}{\pi s_y} + \frac{4s_y(KL)^2}{\pi^2 E} \right]^{1/2}$$

$$D = \left[\frac{4(3)(9800)}{(\pi)(30\,000)} + \frac{4(30\,000)(25)^2}{\pi^2(30 \times 10^6)} \right]^{1/2} = 1.23 \text{ in}$$

Checking the slenderness ratio again, we have

$$KL/r = [(1.0)(25)]/(1.23/4) = 81.3$$

Comments This is still less than the transition value, so our analysis is acceptable. A preferred size of $D = 1.25$ in could be specified.

An alternate method of using spreadsheets to design columns is to use an analysis approach similar to that shown in Figure 6-9 but to use it as a convenient “trial and error” tool. You could compute data by hand, or you could look them up in a table for A , I , and r for any desired cross-sectional shape and dimensions and insert the values into the spreadsheet. Then you could compare the computed allowable load with the required value and choose smaller or larger sections to bring the computed value close to the required value. Many iterations could be completed in a short amount of time. For shapes that allow computing r and A fairly simply, you could add a new section to the spreadsheet to calculate these values. An example is shown in Figure 6-14, where the differently shaded box shows the calculations for the properties of a round cross section. The data are from Example Problem 6-3, and the result shown was arrived at with only four iterations.

CIRCULAR COLUMN ANALYSIS		Data from:	Example Problem 6-3
Refer to Figure 6-7 for analysis logic.			
Enter data for variables in <i>italics in shaded boxes</i> .			Use consistent U.S. Customary units.
Data to Be Entered:			Computed Values:
Length and End Fixity:			
Column length, $L = 25\text{ in}$		→	Eff. Length, $L_e = KL = 25.0\text{ in}$
End fixity, $K = 1$			
Material Properties:			
Yield strength, $s_y = 30\,000\text{ psi}$		→	Column constant, $C_c = 140.5$
Modulus of elasticity, $E = 3.00E + 07\text{ psi}$			
Cross Section Properties:			
[Note: A and r computed from] [dimensions for circular cross section] [in following section of this spreadsheet.]			
Area, $A = 1.188\text{ in}^2$			
Radius of gyration, $r = 0.3075\text{ in}$		→	Slender ratio, $KL/r = 81.3$
Properties for Round Column:			
Diameter for round column = 1.23 in			Column is: <i>short</i>
Area, $A = 1.188\text{ in}^2$			
Radius of gyration, $r = 0.3075\text{ in}$			Critical buckling load = 29 679 lb
Design Factor:			
Design factor on load, $N = 3$		→	Allowable load = 9893 lb

FIGURE 6-14 Spreadsheet for column analysis used as a tool to design a column with a round cross section

6-11 CROOKED COLUMNS

The Euler and Johnson formulas assume that the column is straight and that the load acts in line with the centroid of the cross section of the column. If the column is somewhat crooked, bending occurs in addition to the column action (see Figure 6-15).

The crooked column formula allows an initial crookedness, a , to be considered (see References 6, 7, and 8):

⇒ Crooked Column Formula

$$P_a^2 - \frac{1}{N} \left[s_y A + \left(1 + \frac{ac}{r^2} \right) P_{cr} \right] P_a + \frac{s_y A P_{cr}}{N^2} = 0 \quad (6-13)$$

where c = distance from the neutral axis of the cross section about which bending occurs to its outer edge

P_{cr} is defined to be the critical load found from the *Euler formula*.

Although this formula may become increasingly inaccurate for shorter columns, it is not appropriate to switch to the Johnson formula as it is for straight columns.

The crooked column formula is a quadratic with respect to the allowable load P_a . Evaluating all constant

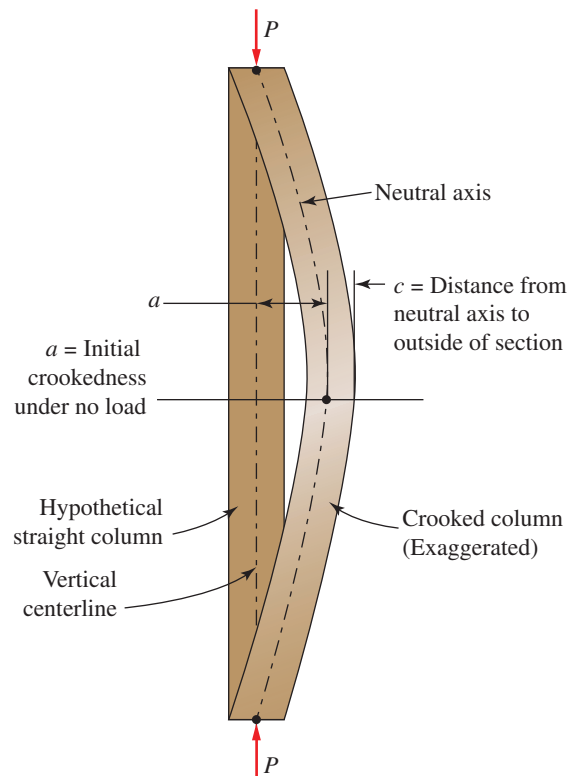


FIGURE 6-15 Illustration of crooked column

terms in Equation (6-13) produces an equation of the form

$$P_a^2 + C_1 P_a + C_2 = 0$$

Then, from the solution for a quadratic equation,

$$P_a = 0.5 [-C_1 \pm \sqrt{C_1^2 - 4C_2}]$$

The smaller of the two possible solutions is selected.

Figure 6-16 shows the solution of Example Problem 6-4 using a spreadsheet. Whereas its appearance is similar to that of the earlier column analysis spreadsheets, the details follow the calculations needed to solve Equation (6-13). On the lower left, two special data values are needed: (1) the crookedness a and (2) the distance c

from the neutral axis for buckling to the outer surface of the cross section. In the middle of the right part are listed some intermediate values used in Equation (6-13): C_1 and C_2 as defined in the solution to Example Problem 6-4. The result, the allowable load, P_a , is at the lower right of the spreadsheet. Above that, for comparison, the computed value of the critical buckling load is given for a straight column of the same design. Note that this solution procedure is most accurate for long columns. If the analysis indicates that the column is *short* rather than *long*, the designer should take note of how short it is by comparing the slenderness ratio, KL/r , with the column constant, C_c . If the column is quite short, the designer should not rely on the accuracy of the result from Equation (6-13).

Example Problem 6-4

A column has both ends pinned and has a length of 32 in. It has a circular cross section with a diameter of 0.75 in and an initial crookedness of 0.125 in. The material is SAE 1040 hot-rolled steel. Compute the allowable load for a design factor of 3.

Solution

Objective Specify the allowable load for the column.

Given Solid circular cross section: $D = 0.75$ in; $L = 32$ in; use $N = 3$.
Both are ends pinned. Initial crookedness = $a = 0.125$ in.
Material: SAE 1040 hot-rolled steel.

Analysis Use Equation (6-13). First evaluate C_1 and C_2 . Then solve the quadratic equation for P_a .

Results

$$\begin{aligned} s_y &= 42\,000 \text{ psi} \\ A &= \pi D^2/4 = (\pi)(0.75)^2/4 = 0.442 \text{ in}^2 \\ r &= D/4 = 0.75/4 = 0.188 \text{ in} \\ c &= D/2 = 0.75/2 = 0.375 \text{ in} \\ KL/r &= [(1.0)(32)]/0.188 = 171 \\ P_{cr} &= \frac{\pi^2 EA}{(KL/r)^2} = \frac{\pi^2 (30\,000\,000)(0.442)}{(171)^2} = 4476 \text{ lb} \\ C_1 &= \frac{-1}{N} \left[s_y A + \left(1 + \frac{ac}{r^2} \right) P_{cr} \right] = -9649 \\ C_2 &= \frac{s_y A P_{cr}}{N^2} = 9.232 \times 10^6 \end{aligned}$$

The quadratic is therefore

$$P_a^2 - 9649 P_a + 9.232 \times 10^6 = 0$$

Comment From this, $P_a = 1076$ lb is the allowable load.

6-12 ECCENTRICALLY LOADED COLUMNS

An *eccentric load* is one that is applied away from the centroidal axis of the cross section of the column, as shown in Figure 6-17. The eccentricity, e , is the perpendicular distance from the line of action between the applied end-loads to the centroidal axis of the initially straight column. Such a load exerts bending in addition to the column action that results in the deflected shape shown in the figure. The maximum stress in the deflected column occurs in the outermost fibers of the cross

section at the midlength of the column where the maximum deflection, $y_{\max}$, occurs. Let's denote the stress at this point as $\sigma_{L/2}$. Then, for any applied load, P ,

Secant Formula for Eccentrically Loaded Columns

$$\sigma_{L/2} = \frac{P}{A} \left[1 + \frac{ec}{r^2} \sec \left(\frac{KL}{2r} \sqrt{\frac{P}{AE}} \right) \right] \quad (6-14)$$

(see References 3, 5, and 9.) Note that this stress is *not* directly proportional to the load. When evaluating the secant in this formula, note that its argument in the

CROOKED COLUMN ANALYSIS		Data from:	Example Problem 6-4
Solves Equation (6-13) for allowable load.			
Enter data for variables in <i>italics in shaded boxes</i> .			Use consistent U.S. Customary units.
Data to Be Entered:			Computed Values:
Length and End Fixity:			
Column length, $L =$	32 in	→	Eff. length, $L_e = KL =$ 32.0 in
End fixity, $K =$	1		
Material Properties:			
Yield strength, $s_y =$	42 000 psi	→	Column constant, $C_c =$ 118.7
Modulus of elasticity, $E =$	3.00E + 07 psi		Euler buckling load = 4,491 lb
Cross Section Properties:			
[Note: Enter r or compute $r = \sqrt{I/A}$.] [Always enter Area.] [Enter zero for I or r if not used.]			C_1 in Eq. (6-11) = -9,678 C_2 in Eq. (6-11) = 9.259E+06
Area, $A =$	0.442 in ²	→	Slender ratio, $KL/r =$ 170.7
Moment of inertia, $I =$	0 in ²		Column is: <i>long</i>
Radius of gyration, $r =$	0.188 in		Straight Column
Values for Eq. (6-11):			Critical buckling load = 4491 lb
Initial crookedness = $a =$	0.125 in		Crooked Column
Neutral axis to outside = $c =$	0.375 in		Allowable load = 1076 lb
Design Factor:			
Design factor on load, $N =$	3	→	

FIGURE 6-16 Spreadsheet for analysis of crooked columns

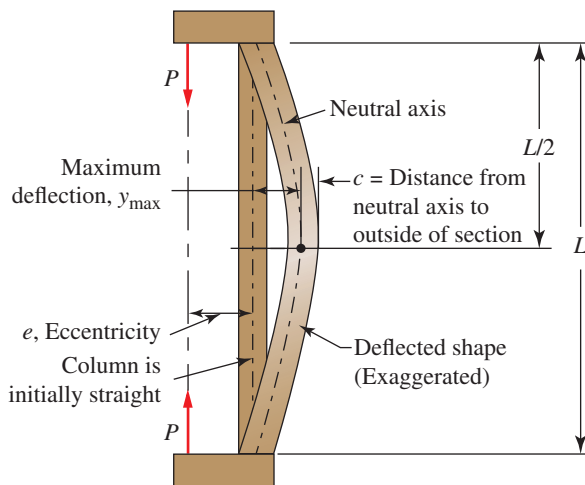


FIGURE 6-17 Illustration of eccentrically loaded columns

parentheses is in *radians*. Also, because most calculators do not have the secant function, recall that the secant is equal to $1/\cosine$.

For design purposes, we would like to specify a design factor, N , that can be applied to the *failure load* similar to that defined for straight, centrally loaded

columns. However, in this case, failure is predicted when the maximum stress in the column exceeds the yield strength of the material. Let's now define a new term, P_y , to be the load applied to the eccentrically loaded column when the maximum stress is equal to the yield strength. Equation (6-14) then becomes

$$s_y = \frac{P_y}{A} \left[1 + \frac{ec}{r^2} \sec \left(\frac{KL}{2r} \sqrt{\frac{P_y}{AE}} \right) \right]$$

Now, if we define the *allowable load* to be

$$P_a = P_y/N$$

or

$$P_y = NP_a$$

this equation becomes

Design Equation for Eccentrically Loaded Columns

$$\text{Required } s_y = \frac{NP_a}{A} \left[1 + \frac{ec}{r^2} \sec \left(\frac{KL}{2r} \sqrt{\frac{NP_a}{AE}} \right) \right] \quad (6-15)$$

This equation cannot be solved analytically for either A or P_a . Therefore, an iterative solution is required, as will be demonstrated in Example Problem 6-6.

Example Problem 6-5

For the column of Example Problem 6-4, compute the maximum stress and deflection if a load of 1075 lb is applied with an eccentricity of 0.75 in. The column is initially straight.

Solution

Objective Compute the stress and the deflection for the eccentrically loaded column.

Given Data from Example Problem 6-4, but eccentricity = $e = 0.75$ in.
 Solid circular cross section: $D = 0.75$ in; $L = 32$ in; Initially straight
 Both ends are pinned; $KL = 32$ in; $r = 0.188$ in; $c = D/2 = 0.375$ in.
 Material: SAE 1040 hot-rolled steel; $E = 30 \times 10^6$ psi, $s_y = 42$ psi

Analysis Use Equation (6-14) to compute maximum stress. Then use Equation (6-16) to compute maximum deflection.

Results All terms have been evaluated before. Then the maximum stress is found from Equation (6-14):

$$\sigma_{L/2} = \frac{1075}{0.422} \left[1 + \frac{(0.75)(0.375)}{(0.188)^2} \sec \left(\frac{32}{2(0.188)} \sqrt{\frac{1075}{(0.442)(30 \times 10^6)}} \right) \right]$$

$$\sigma_{L/2} = 29\,300 \text{ psi}$$

The maximum deflection is found from Equation (6-16):

$$y_{\max} = 0.75 \left[\sec \left(\frac{32}{2(0.188)} \sqrt{\frac{1075}{(0.442)(30 \times 10^6)}} \right) - 1 \right] = 0.293 \text{ in}$$

Comments The maximum stress is 29 300 psi at the midlength of the column. The deflection there is 0.293 in from the original straight central axis of the column.

Example Problem 6-6

The stress in the column found in Example Problem 6-5 seems high for the SAE 1040 hot-rolled steel. Redesign the column to achieve a design factor of at least 3.

Solution

Objective Redesign the eccentrically loaded column of Example Problem 6-5 to reduce the stress and achieve a design factor of at least 3.

Given Data from Example Problems 6-4 and 6-5.

Analysis Use a larger diameter. Use Equation (6-15) to compute the required strength. Then compare that with the strength of SAE 1040 hot-rolled steel. Iterate until the stress is satisfactory.

Results Appendix 3 gives the value for the yield strength of SAE 1040 HR to be 42 000 psi. If we choose to retain the same material, the cross-sectional dimensions of the column must be increased to decrease the stress. Equation (6-15) can be used to evaluate a design alternative.

The objective is to find suitable values for A , c , and r for the cross section such that $P_a = 1075$ lb; $N = 3$; $L_e = 32$ in; $e = 0.75$ in; and the value of the entire right side of the equation is less than 42 000 psi. The original design had a circular cross section with a diameter of 0.75 in. Let's try increasing the diameter to $D = 1.00$ in. Then

$$A = \pi D^2/4 = \pi(1.00 \text{ in})^2/4 = 0.785 \text{ in}^2$$

$$r = D/4 = (1.00 \text{ in})/4 = 0.250 \text{ in}$$

$$r^2 = (0.250 \text{ in})^2 = 0.0625 \text{ in}^2$$

$$c = D/2 = (1.00 \text{ in})/2 = 0.50 \text{ in}$$

Now let's call the right side of Equation (6-15) s'_y . Then

$$s'_y = \frac{3(1075)}{0.785} \left[1 + \frac{(0.75)(0.50)}{(0.0625)} \sec \left(\frac{32}{2(0.250)} \sqrt{\frac{(3)(1075)}{(0.785)(30 \times 10^6)}} \right) \right]$$

$$s'_y = 37\,740 \text{ psi} = \text{required value of } s_y$$

This is a satisfactory result because it is just slightly less than the value of s_y of 42 000 psi for the steel.

Now we can evaluate the expected maximum deflection with the new design using Equation (6–16):

$$y_{\max} = 0.75 \left[\sec \left(\frac{32}{2(0.250)} \sqrt{\frac{1075}{(0.785)(30 \times 10^6)}} \right) - 1 \right]$$

$$y_{\max} = 0.076 \text{ in}$$

Comments The diameter of 1.00 in is satisfactory. The maximum deflection for the column is 0.076 in.

ECCENTRIC COLUMN ANALYSIS		Data from:	Example Problem 6–6
Solves Equation (6–15) for design stress and Equation (6–16) for maximum deflection.			
Enter data for variables in <i>italics in shaded boxes</i> .			Use consistent U.S. Customary units.
Data to Be Entered:			Computed Values:
Length and End Fixity:			
Column length, $L =$	32 in	→	Eff. length, $L_e = KL = 32.0$ in
End fixity, $K =$	1		
Material Properties:			
Yield strength, $s_y =$	42 000 psi	→	Column constant, $C_c = 118.7$
Modulus of elasticity, $E =$	3.00E + 07 psi		
Cross Section Properties:			
[Note: Enter r or compute $r = \sqrt{I/A}$.] [Always enter Area.] [Enter zero for I or r if not used.]			Argument of sec = 0.749 for strength Value of secant = 1.3654
Area, $A =$	0.785 in ²		Argument of sec = 0.432 for deflection Value of secant = 1.1014
Moment of inertia, $I =$	0 in ⁴		
OR		→	Slender ratio, $KL/r = 128.0$
Radius of gyration, $r =$	0.250 in		
Values for Eqs. (6–15) and (6–16)			Column is: <i>long</i>
Eccentricity, $e =$	0.75 in		
Neutral axis to outside, $c =$	0.5 in		
Allowable load, $P_a =$	1,075 lb		FINAL RESULTS
Design Factor:			Req'd yield strength = 37 764 psi Must be less than actual yield strength: $s_y = 42\,000$ psi
Design factor on load, $N =$	3		Max deflection, $y_{\max} = 0.076$ in

FIGURE 6–18 Spreadsheet for analysis of eccentric columns

Another critical factor may be the amount of deflection of the axis of the column due to the eccentric load:

➤ Maximum Deflection in an Eccentrically Loaded Column

$$y_{\max} = e \left[\sec \left(\frac{KL}{2r} \sqrt{\frac{P}{AE}} \right) - 1 \right] \quad (6-16)$$

Note that the argument of the secant is the same as that used in Equation (6–14).

Figure 6–18 shows the solution of the eccentric column problem of Example Problem 6–6 using a spreadsheet to evaluate Equations (6–15) and (6–16). It is a design aid that facilitates the iteration required to

determine an acceptable geometry for a column to carry a specified load with a desired design factor. Note that the data are in U.S. Customary units. At the lower left of the spreadsheet, data required for Equations (6–15) and (6–16) are entered by the designer, along with the other data discussed for earlier column analysis spreadsheets. The “FINAL RESULTS” at the lower right show the computed value of the required yield strength of the material for the column and compare it with the given value entered by the designer near the upper left. The designer must ensure that the actual value is greater than the computed value. The last part of the right side of the spreadsheet gives the computed maximum deflection of the column that occurs at its midlength.

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8. A steel pipe has an outside diameter of 1.60 in, a wall thickness of 0.109 in, and a length of 6.25 ft. Compute the critical load for each of the end conditions shown in Figure 6–2. Use SAE 1020 HR steel.
9. Compute the required diameter of a circular bar to be used as a column carrying a load of 8500 lb with pinned ends. The length is 50 in. Use SAE 4140 OQT 1000 steel and a design factor of 3.0.
10. Repeat Problem 9 using SAE 1020 HR steel.
11. Repeat Problem 9 with aluminum 2014-T4.
12. In Section 6–10, equations were derived for the design of a solid circular column, either long or short. Perform the derivation for a solid square cross section.
13. Repeat the derivations called for in Problem 12 for a hollow circular tube for any ratio of inside to outside diameter. That is, let the ratio $R = ID/OD$, and solve for the required OD for a given load, material, design factor, and end fixity.
14. Determine the required dimensions of a column with a square cross section to carry an axial compressive load of 6500 lb if its length is 64 in and its ends are fixed. Use a design factor of 3.0. Use aluminum 6061-T6.
15. Repeat Problem 14 for a hollow aluminum tube (6061-T6) with the ratio of $ID/OD = 0.80$. Compare the weight of this column with that of Problem 14.
16. A toggle device is being used to compact scrap steel shavings, as illustrated in Figure P6–16. Design the two links of the toggle to be steel, SAE 5160 OQT 1000, with a circular cross section and pinned ends. The force P required to crush the shavings is 5000 lb. Use $N = 3.50$.

PROBLEMS

1. A column has both ends pinned and has a length of 32 in. It is made of SAE 1040 HR steel and has a circular shape with a diameter of 0.75 in. Determine the critical load.
2. Repeat Problem 1 using a length of 15 in.
3. Repeat Problem 1 with the bar made of aluminum 6061-T4.
4. Repeat Problem 1 assuming both ends fixed.
5. Repeat Problem 1 using a square cross section, 0.65 in on a side, instead of the circular cross section.
6. Repeat Problem 1 with the bar made from high-impact acrylic plastic.
7. A rectangular steel bar has a cross section 0.50 by 1.00 in and is 8.5 in long. The bar has pinned ends and is made of SAE 4150 OQT 1000 steel. Compute the critical load.
17. Repeat Problem 16, but propose a design that will be lighter than the solid circular cross section.
18. A sling, sketched in Figure P6–18, is to carry 18 000 lb. Design the spreader.
19. For the sling in Problem 18, design the spreader if the angle shown is changed from 30° to 15° .
20. A rod for a certain hydraulic cylinder behaves as a fixed-free column when used to actuate a compactor of industrial waste. Its maximum extended length will be 10.75 ft. If it is to be made of SAE 1144 OQT 1300 steel, determine the required diameter of the rod for a design factor of 2.5 for an axial load of 25 000 lb.
21. Design a column to carry 40 000 lb. One end is pinned, and the other is fixed. The length is 12.75 ft.

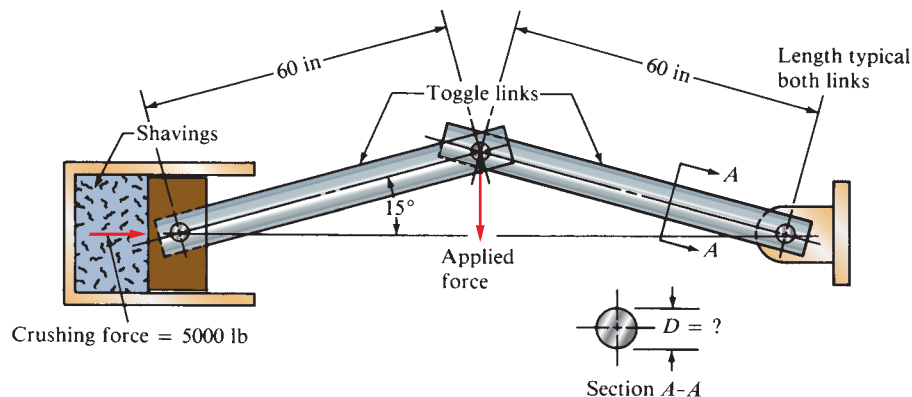


FIGURE P6–16 (Problems 16 and 17)

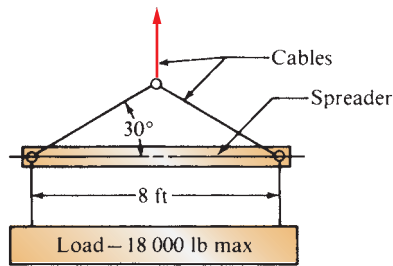


FIGURE P6-18 (Problems 18 and 19)

22. Repeat Problem 21 using a length of 4.25 ft.
23. Repeat Problem 1 if the column has an initial crookedness of 0.08 in. Determine the allowable load for a design factor of 3.
24. Repeat Problem 7 if the column has an initial crookedness of 0.04 in. Determine the allowable load for a design factor of 3.
25. Repeat Problem 8 if the column has an initial crookedness of 0.15 in. Determine the allowable load for a design factor of 3 and pinned ends only.
26. An aluminum (6063-T4) column is 42 in long and has a square cross section, 1.25 in on a side. If it carries a compressive load of 1250 lb, applied with an eccentricity of 0.60 in, compute the maximum stress in the column and the maximum deflection.
27. A steel (SAE 1020 hot-rolled) column is 3.2 m long and is made from a standard 3-in Schedule 40 steel pipe (see Table A15-17). If a compressive load of 30.5 kN is applied with an eccentricity of 150 mm, compute the maximum stress in the column and the maximum deflection.
28. A link in a mechanism is 14.75 in long and has a square cross section, 0.250 in on a side. It is made from annealed SAE 410 stainless steel. Use $E = 28 \times 10^6$ psi. If it carries a compressive load of 45 lb with an eccentricity of 0.30 in, compute the maximum stress and the maximum deflection.
29. A hollow square steel tube, 40 in long, is proposed for use as a prop to hold up the ram of a punch press during installation of new dies. The ram weighs 75 000 lb. The prop is made from 4×4×1/4 structural tubing. It is made from steel similar to structural steel, ASTM A500 Grade C. If the load applied by the ram could have an eccentricity of 0.50 in, would the prop be safe?
30. Determine the allowable load on a column 16.0 ft long made from a wide-flange beam shape, W5×19. The load will be centrally applied. The end conditions are somewhat between fixed and hinged, say, $K = 0.8$. Use a design factor of 3. Use ASTM A36 structural steel.
31. Determine the allowable load on a fixed-end column having a length of 66 in if it is made from a steel American Standard Beam, S4×7.7. The material is ASTM A36 structural steel. Use a design factor of 3.
32. Compute the maximum stress and deflection that can be expected in the steel machine member carrying an eccentric load as shown in Figure P6-32. The load P is 1000 lb. If a design factor of 3 is desired, specify a suitable steel.
33. Specify a suitable steel tube from Table A15-14 to support one side of a platform as shown in Figure P6-33. The material has a yield strength of 36 ksi. The total load on the platform is 55 000 lb, uniformly distributed.
34. Compute the allowable axial load on a steel channel, C5×9, made from ASTM A36 structural steel. The channel is 112 in long and can be considered to be pinned at its ends. Use a design factor of 3.
35. Repeat Problem 34 with the ends fixed rather than pinned.
36. Repeat Problem 34, except consider the load to be applied along the outside of the web of the channel instead of being axial.
37. Figure P6-37 shows a 4×4×1/2 steel column made from ASTM A500 Grade B structural steel. To accommodate a special mounting restriction, the load is applied eccentrically as shown. Determine the amount of load that the column can safely support. The column is supported laterally by the structure.
38. The device shown in Figure P6-38 is subjected to opposing forces F . Determine the maximum allowable load to achieve a design factor of 3. The device is made from aluminum 6061-T6.
39. A hydraulic cylinder is capable of exerting a force of 5200 N to move a heavy casting along a conveyor. The design of the pusher causes the load to be applied eccentrically to the piston rod as shown in Figure P6-39. Is the piston rod safe under this loading if it is made from SAE 416 stainless steel in the Q&T 1000 condition? Use $E = 200$ GPa.

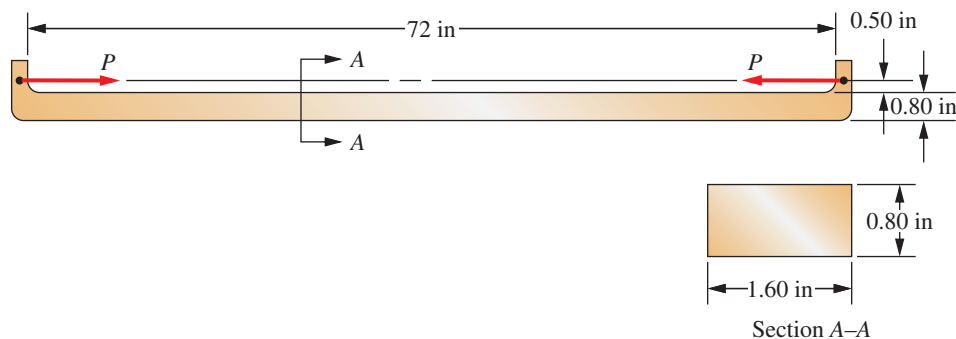


FIGURE P6-32 (Problem 32)

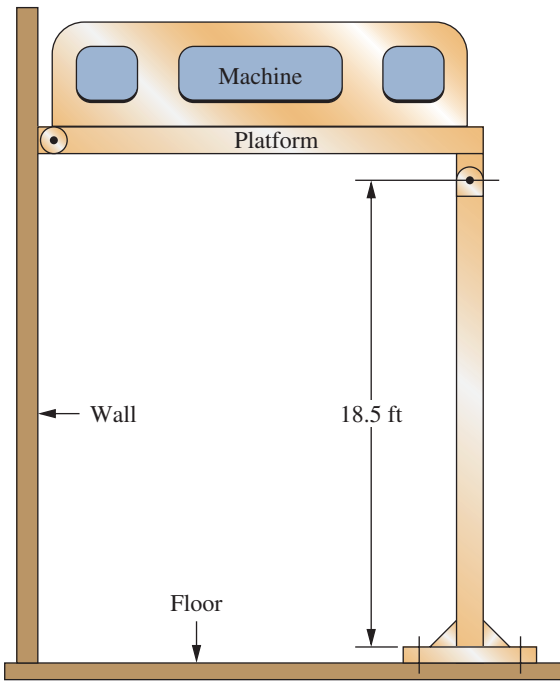


FIGURE P6-33 (Problem 33)

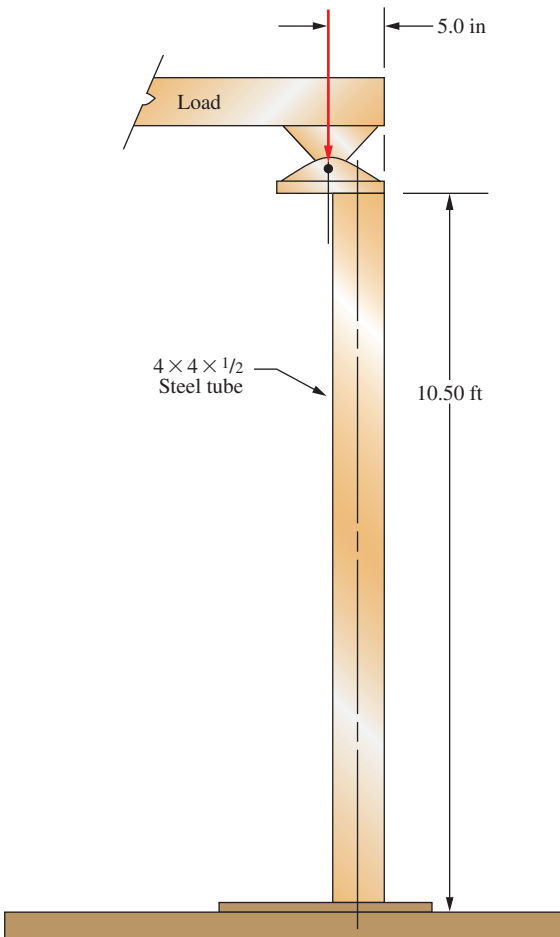


FIGURE P6-37 (Problem 37)

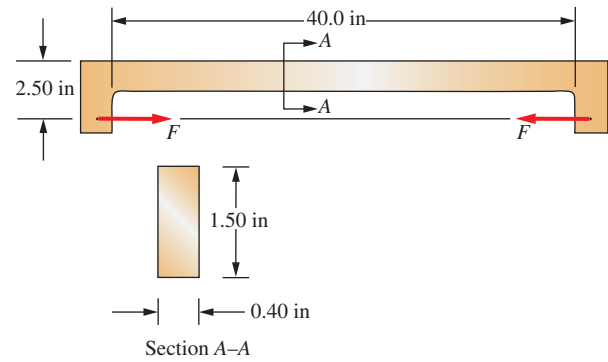


FIGURE P6-38 (Problem 38)

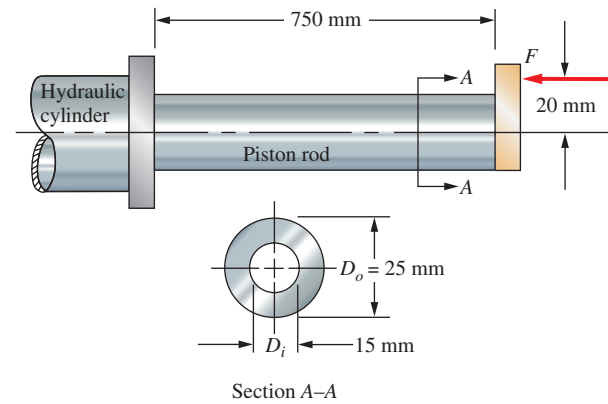


FIGURE P6-39 (Problem 39)

40. A standard 2-in schedule 40 steel pipe is proposed to be used to support the roof of a porch during renovation. Its length is 13.0 ft. The pipe is made from ASTM A501 structural steel.
- Determine the safe load on the pipe to achieve a design factor of 3 if the pipe is straight.
 - Determine the safe load if the pipe has an initial crookedness of 1.25 in.

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DESIGN OF A MECHANICAL DRIVE

OBJECTIVES AND CONTENT OF PART II

Part II of this book contains nine chapters (Chapters 7–15) that help you gain experience in approaching the design of an important complete device—a *mechanical drive*. The drive, sometimes called a *power transmission*, serves the following functions:

- It receives power from some kind of rotating source such as an electric motor, an internal combustion engine, a gas turbine engine, a hydraulic or pneumatic motor, a steam or water turbine, or even hand rotation provided by the operator.
- The drive typically causes some change in the speed of rotation of the shafts that make up the drive so that the output shaft operates more slowly or faster than the input shaft. Speed reducers are more prevalent than speed increasers.
- The active elements of the drive transmit the power from the input to the output shafts.
- When there is a speed reduction, there is a corresponding increase in the torque transmitted. Conversely, a speed increase causes a reduction in torque at the output compared with the input of the drive.

The chapters of Part II provide the detailed descriptions of the various machine elements that are typically used in power transmissions: *belt drives*, *chain drives*, *gears*, *shafts*, *bearings*, *keys*, *couplings*, *seals*, and *housings to hold all the elements together*. You will learn the important features of these elements and the methods of analyzing and designing them.

Of equal importance is the information provided on how the various elements interact with each other. You must be sensitive, for example, to how gears are mounted on shafts, how the shafts are supported by bearings, and how the bearings must be mounted securely and precisely in a housing that holds the system together. The final completed design must function as an integrated unit.

THE PROCESS OF DESIGNING A MECHANICAL DRIVE

In the design of a power transmission, you would typically know the following:

- **The nature of the driven machine:** It might be a machine tool in a factory that cuts metal parts for engines; an electric drill used by professional carpenters or home craft workers; the axle of a farm tractor; the propeller shaft of a turbojet for an airplane; the propeller shaft for a large ship; the wheels of a toy train; a mechanical timing mechanism; or any other of the numerous products that need a controlled-speed drive.
- **The level of power to be transmitted:** From the examples just listed, the power demanded may range from thousands of horsepower for a ship, hundreds of horsepower for a large farm tractor or airplane, or a few watts for a timer or a toy.

- **The rotational speed of the drive motor or other prime mover:** Typically the prime mover operates at a rather high speed of rotation. The shafts of standard electric motors rotate at about 1200, 1800, or 3600 revolutions per minute (rpm). Actual full-load operating speeds are somewhat less than these speeds, say 1150, 1750, or 3550 rpm. Automotive engines operate from about 1000 to 6000 rpm. Universal motors in some hand tools (drills, saws, and routers) and household appliances (mixers, blenders, and vacuum cleaners) operate from 3500 to 20 000 rpm. Gas turbine engines for aircraft rotate many thousands of rpm.
- **The desired output speed of the transmission:** This is highly dependent on the application. Some gear motors for instruments rotate less than 1.0 rpm. Production machines in factories may run a few hundred rpm. Drives for assembly conveyors may run fewer than 100 rpm. Aircraft propellers may operate at several thousand rpm.

You, the designer, must then do the following:

- Choose the type of power transmission elements to be used: gears, belt drives, chain drives, or other types. In fact, some power transmission systems use two or more types in series to optimize the performance of each.
- Specify how the rotating elements are arranged and how the power transmission elements are mounted on shafts.
- Design the shafts to be safe under the expected torques and bending loads and properly locate the power transmission elements and the bearings. It is likely that the shafts will have several diameters and special features to accommodate keys, couplings, retaining rings, and other details. The dimensions of all features must be specified, along with the tolerances on the dimensions and surface finishes.
- Specify suitable bearings to support the shafts and determine how they will be mounted on the shafts and how they will be held in a housing.
- Specify keys to connect the shaft to the power transmission elements; couplings to connect the shaft from the driver to the input shaft of the transmission or to connect the output shaft to the driven machine; seals to effectively exclude contaminants from entering the transmission; and other accessories.
- Place all of the elements in a suitable housing that provides for the mounting of all elements and for their protection from the environment and their lubrication.

CHAPTERS THAT MAKE UP PART II

To guide you through this process of designing a mechanical drive, Part II includes the following chapters:

Chapter 7: Belt Drives, Chain Drives and Wire Rope emphasizes recognizing the variety of commercially available belt and chain drives, the critical design parameters, and the methods used to specify reasonably optimum components of the drive systems.

Chapter 8: Kinematics of Gears describes and defines the important geometric features of gears. Methods of manufacturing gears are discussed, along with the importance of precision in the operation of the gears. The details of how a pair of gears operates are described, and the design and the operation of two or more gear pairs in a gear train are analyzed.

Chapter 9: Spur Gear Design illustrates how to compute forces exerted by one tooth of a gear on its mating teeth. Methods of computing the stresses in the gear teeth are presented, and design procedures for specifying gear-tooth geometry and material are given to produce a safe, long-life gear transmission system.

Chapter 10: Helical Gears, Bevel Gears, and Wormgearing contains approaches similar to those described for spur gears, with special attention to the unique geometry of these types of gears.

Chapter 11: Keys, Couplings, and Seals discusses how to design keys to be safe under the prevailing loads caused by the torque transmitted by them from the shaft to the gears or other elements. Couplings must be specified that accommodate the possible

misalignment of connected shafts while transmitting the required torque at operating speeds. Seals must be specified for shafts that project through the sides of the housing and for bearings that must be kept free of contaminants. The maintenance of a reliable supply of clean lubricant for the active elements is essential.

Chapter 12: Shaft Design discusses the fact that in addition to being designed to safely transmit the required torque levels at given speeds, the shafts will probably have several diameters and special features to accommodate keys, couplings, retaining rings, and other details. Shaft deflections, vibration, and the dynamic performance of the shaft assembly must also be considered. The dimensions of all features must be specified, along with the tolerances on the dimensions and surface finishes. Completion of these tasks requires some of the skills developed in following chapters. So you will have to come back to this task later.

Chapter 13: Tolerances and Fits discusses the fit of elements that are assembled together and that may operate on one another; this fit is critical to the performance and life of the elements. In some cases, such as fitting the inner race of a ball or roller bearing onto a shaft, the bearing manufacturer specifies the allowable dimensional variation on the shaft so that it is compatible with the tolerances to which the bearing is produced. There is typically an interference fit between the bearing inner race and the shaft diameter where the bearing is to be mounted. But there is a close sliding fit between the outer race and the housing that holds the bearing in place. In general, it is important for you to take charge of specifying the tolerances for all dimensions to ensure proper operation while allowing economical manufacture.

Chapter 14: Rolling Contact Bearings focuses on commercially available rolling contact bearings such as ball bearings, roller bearings, and tapered roller bearings. You must be able to compute or specify the loads that the bearings will support, their speed of operation, and their expected life. From these data, standard bearings from manufacturers' catalogs will be specified. Then you must review the design process for the shafts as described for Chapter 12 to complete the specification of dimensions and tolerances. It is likely that iteration among the design processes for the power transmission elements, the shafts, and the bearings will be needed to achieve an optimum arrangement.

Chapter 15: Completion of the Design of a Power Transmission merges all of the preceding topics together. You will resolve the details of the design of each element and ensure the compatibility of mating elements. You will review all previous design decisions and assumptions and verify that the design meets specifications. After the individual elements have been analyzed and the iteration among them is complete, they must be packaged in a suitable housing to hold them securely, to protect them from contaminants, and to protect the people who may work around them. The housing must also be designed to be compatible with the driver and the driven machine. That often requires special fastening provisions and means of locating all connected devices relative to one another. Assembly and service must be considered. Then you will present a final set of specifications for the entire power transmission system and document your design with suitable drawings and a written report.

BELT DRIVES, CHAIN DRIVES, AND WIRE ROPE

The Big Picture

You Are the Designer

- 7-1 Objectives of This Chapter
- 7-2 Kinematics of Belt and Chain Drive Systems
- 7-3 Types of Belt Drives
- 7-4 V-Belt Drives
- 7-5 Synchronous Belt Drives
- 7-6 Chain Drives
- 7-7 Wire Rope

THE BIG PICTURE

Belt Drives, Chain Drives and Wire Rope

Discussion Map

- Belts and chains are the major types of flexible power transmission elements. V-belts operate on smooth sheaves or pulleys, whereas the cogs of synchronous belts operate on sprockets. Chains operate on toothed wheels also called sprockets.
- Wire rope, sometimes called cable, is another type of flexible machine element, used primarily for lifting heavy loads.

Discover

Look around and identify at least one mechanical device having a belt drive, one having a chain drive system, and one using wire rope for lifting.

Describe each drive system, and make a sketch showing how it receives power from some source and how it transfers power to a driven machine.

Describe the differences between belt drives and chain drives.

Describe the application of the wire rope lifting system you have found.

In this chapter, you will learn how to select suitable components for belt drives and chain drives from commercially available designs. You will also learn about available types of wire rope and principles of their application.

Belt drives and chain drives represent the major types of flexible power transmission elements. Figure 7-1 shows a typical industrial application of these elements combined with a gear-type speed reducer. This application illustrates where belts, gear drives, and chains are each used to best advantage.

Rotary power is developed by the electric motor, but motors typically operate at too high a speed and deliver too low a torque to be appropriate for the final drive application. Remember, for a given power transmission, the torque is increased in proportion to the amount that rotational speed is reduced. So some speed reduction is often desirable. The high speed

of the motor makes belt drives somewhat ideal for that first stage of reduction. A smaller drive sheave is attached to the motor shaft, while a larger diameter sheave is attached to a parallel shaft that operates at a correspondingly lower speed. Sheaves for belt drives are also called *pulleys*.

However, if very large ratios of speed reduction are required in the drive, gear reducers are desirable because they can typically accomplish large reductions in a rather small package. The output shaft of the gear-type speed reducer is generally at low speed and high torque. If both speed and torque are satisfactory for the application, it could be directly coupled to the driven machine.

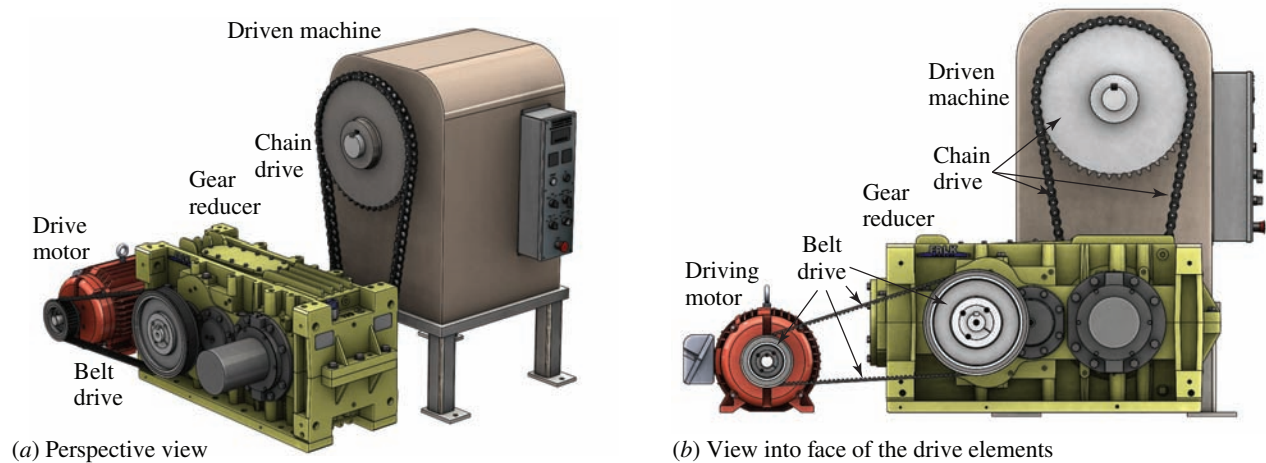


FIGURE 7-1 Drive system for an industrial application employing a belt drive, a gear reducer, and a chain drive

However, because gear reducers are available only at discrete reduction ratios, the output must often be reduced more before meeting the requirements of the driven machine. At the low-speed, high-torque condition, chain drives become desirable. The high torque causes high tensile forces to be developed in the chain. The elements of the chain are typically metal, and they are sized to withstand the high forces. The links of chains are engaged in toothed wheels called *sprockets* to provide positive mechanical drive, desirable at the low-speed, high-torque conditions.

In general, belt drives are applied where the rotational speeds are relatively high, as on the first stage of speed reduction from an electric motor or engine. The linear speed of a belt is usually 2500–6500 ft/min, which results in relatively low tensile forces in the belt. At lower speeds, the tension in the belt becomes too large for typical belt cross sections, and slipping may occur between the sides of the belt and the sheave or pulley that carries it. At higher speeds, dynamic effects such as centrifugal forces, belt whip, and vibration reduce the effectiveness of the drive and its life. A speed of 4000 ft/min is generally ideal.

The *synchronous belt drive* employs precision cogs on the inside surface of the belt that engage in matching grooves in the pulleys to enhance their ability to transmit high forces at low speeds. There is also precise timing between the driver and driven pulleys so this type of belt is sometimes called a *timing belt*. Such belt drives often compete with chain drives and gear drives in some applications.

Where have you seen belt drives? Consider mechanical devices around your home or office;

vehicles; construction equipment; heating, air-conditioning, and ventilation systems; and industrial machinery. Describe their general appearance. To what was the input sheave attached? Was it operating at a fairly high speed? What was the size of the next sheave? Did it cause the second shaft to rotate at a slower speed? How much slower? Were there more stages of reduction accomplished by belts or by some other reducer? Was the belt of the V-belt type operating in smooth v-shaped grooves or was it a synchronous belt having cogs that mate with grooves in sprockets? Make a sketch of the layout of the drive system. Make measurements if you can get access to the equipment safely.

Where have you seen chain drives? One obvious place is likely to be the chain on a bicycle where the sprocket attached to the pedal-crank assembly is fairly large and that attached to the rear wheel is smaller. The drive sprocket and/or the driven sprocket assemblies may have several sizes to allow the rider to select many different speed ratios to permit optimum operation under different conditions of speed and hill-climbing demands. Where else have you seen chain drives? Again consider vehicles, construction equipment, and industrial machinery. Describe and sketch at least one chain drive system.

This chapter will help you learn to identify the typical design features of commercially available belt and chain drives. You will be able to specify suitable types and sizes to transmit a given level of power at a certain speed and to accomplish a specified speed ratio between the input and the output of the drive. Installation considerations are also described so that you can put your designs into successful systems.

YOU ARE THE DESIGNER

A plant in Hawaii that produces sugar needs a drive system designed for a machine that chops long pieces of sugar cane into short lengths prior to processing. The machine's drive shaft is to rotate slowly at 30 rpm so that the cane is chopped smoothly and not beaten. The large machine requires a torque of 31 500 lb · in to drive the chopping blades.

Your company is asked to design the drive, and you are given the assignment. What kind of power source should be used? You might consider an electric motor, a gasoline engine, or a hydraulic motor. Most of these run at relatively high speeds, significantly higher than 30 rpm. Therefore, some type of speed reduction is needed. Perhaps you decide to use a drive similar to that shown in Figure 7-1.

Three stages of speed reduction are used. The input sheave of the belt drive rotates at the speed of the motor, while the larger driven sheave rotates at a slower speed and delivers the power to the input of the gear reducer. The larger part of the speed reduction is likely to be accomplished in the gear reducer, with the output shaft rotating slowly and providing a large torque. Remember, as the speed of rotation of a rotating shaft decreases, the torque delivered increases for a given power transmitted. But because there are only

a limited number of reducer designs available, the output speed of the reducer will probably not be ideal for the cane chopper input shaft. The chain drive then provides the last stage of reduction.

As the designer, you must decide what type and size of belt drive to use and what the speed ratio between the driving and the driven sheave should be. How is the driving sheave attached to the motor shaft? How is the driven sheave attached to the input shaft of the gear reducer? Where should the motor be mounted in relation to the gear reducer, and what will be the resulting center distance between the two shafts? What speed reduction ratio will the gear reducer provide? What type of gear reducer should be used: helical gears, a worm and worm-gear drive, or bevel gears? How much additional speed reduction must the chain drive provide to deliver the proper speed to the cane-chopper shaft? What size and type of chain should be specified? What is the center distance between the output of the gear reducer and the input to the chopper? Then what length of chain is required? Finally, what motor power is required to drive the entire system at the stated conditions? The information in this chapter will help you answer questions about the design of power transmission systems incorporating belts and chains. Gear reducers are discussed in Chapters 8–10.

7-1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Describe the basic features of a belt drive system.
2. Describe several types of belt drives.
3. Specify suitable types and sizes of belts and sheaves to transmit a given level of power at specified speeds for the input and output sheaves.
4. Specify the primary installation variables for belt drives, including center distance and belt length.
5. Describe the basic features of a chain drive system.
6. Describe several types of chain drives.
7. Specify suitable types and sizes of chains and sprockets to transmit a given level of power at specified speeds for the input and output sprockets.
8. Specify the primary installation variables for chain drives, including center distance between the sheaves, chain length, and lubrication requirements.
9. Describe the basic types of commercially available wire rope, the preferred methods of applying them, and the typical working loads.

7-2 KINEMATICS OF BELT AND CHAIN DRIVE SYSTEMS

Mechanical drives are used to transmit power and rotational motion. A drive system can include chain drives, belt drives, and gear drives. Gear drives will be covered in later chapters. The requirements of the application will determine the selection of the type of drive system. A chain drive may be used when high rotational power is transmitted at relatively slow speeds and when lifting

heavy loads, as on a fork-lift truck. Belt drives are typically used for higher speeds and on applications requiring accurate registration between the shafts of driving and driven machines.

Belts and chains are both flexible elements that transmit power from a driver, such as an electric motor, engine, or turbine to a driven machine. Details of their design are presented in later sections of this chapter. The belts and chains are placed on rotating elements attached to the output shaft of the driver and to the input shaft of the driven machine.

- Belts have continuous cross sections that are mounted on *sheaves*, sometimes called *pulleys*. This chapter discusses two types of belt drives:
 - *V-belts* that operate without slipping on smooth, V-shaped grooves in the sheaves
 - *Synchronous belts* that have molded cogs that engage grooved sprockets producing positive driving
- Chains are comprised of a set of discrete links that engage teeth on *sprockets*, producing positive driving.

The kinematics of the drive system ensures correct relative positioning, angular velocity, and acceleration of the input driver and the driven machine. In each case, the flexible belt or chain has a uniform linear velocity as it passes around its sheave or sprocket. The linear velocity is often called *belt speed* or *chain speed*. The following development relates the linear velocity to the rotational speeds of the driver and the driven machine. Other relevant geometry of the drive system is also developed. Refer to Figures 7-2 to 7-4.

Figure 7-2 shows a simple disk rotating about its center. If the disk (2) is rotating at an angular velocity (ω_A),

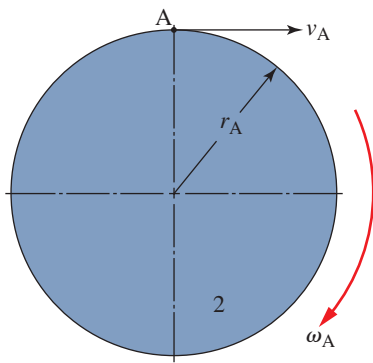


FIGURE 7-2 Linear velocity related to angular velocity of a rotating disk

the linear velocity of point A (v_A), located a distance (r_A) from its axis of rotation is given by the equation

$$v_A = r_A \omega_A \quad (7-1)$$

where

v_A = Linear velocity of point A [in/s], [m/s], [ft/min or fpm]

r_A = Radius to point A [in], [m], [ft]

ω_A = Angular velocity of the disk [rad/s]

This equation allows us to relate the angular velocity (ω) to the linear velocity (v) based on the distance (r) from its axis of rotation. This motion corresponds to the motion of a belt or chain drive system, where the linear velocity of point A is equivalent to the belt or chain linear velocity and the angular velocity of disk 2 is equivalent to the angular velocity of the rotating sprocket or sheave of the drive system. The radius, r , is equal to the radius of the *pitch diameter of the sheave or sprocket*, PD, the kinematically characteristic diameter.

Figure 7-3 shows the pitch diameters on a chain sprocket, synchronous belt sprocket, and a V-belt sheave. Note the following:

- The pitch diameter of a chain sprocket, also called the *pitch circle diameter*, goes through the center of the chain bearing pin when the chain is wrapped around the sprocket.
- The pitch diameter of the synchronous belt sprocket is along the theoretical belt pitch line defined by the belt manufacturer and is always greater than the outside diameter of the grooves on the sprocket.
- The pitch diameter of the V-belt sheave is slightly inside the top of the cross section of the belt near where the tensile cords are placed.

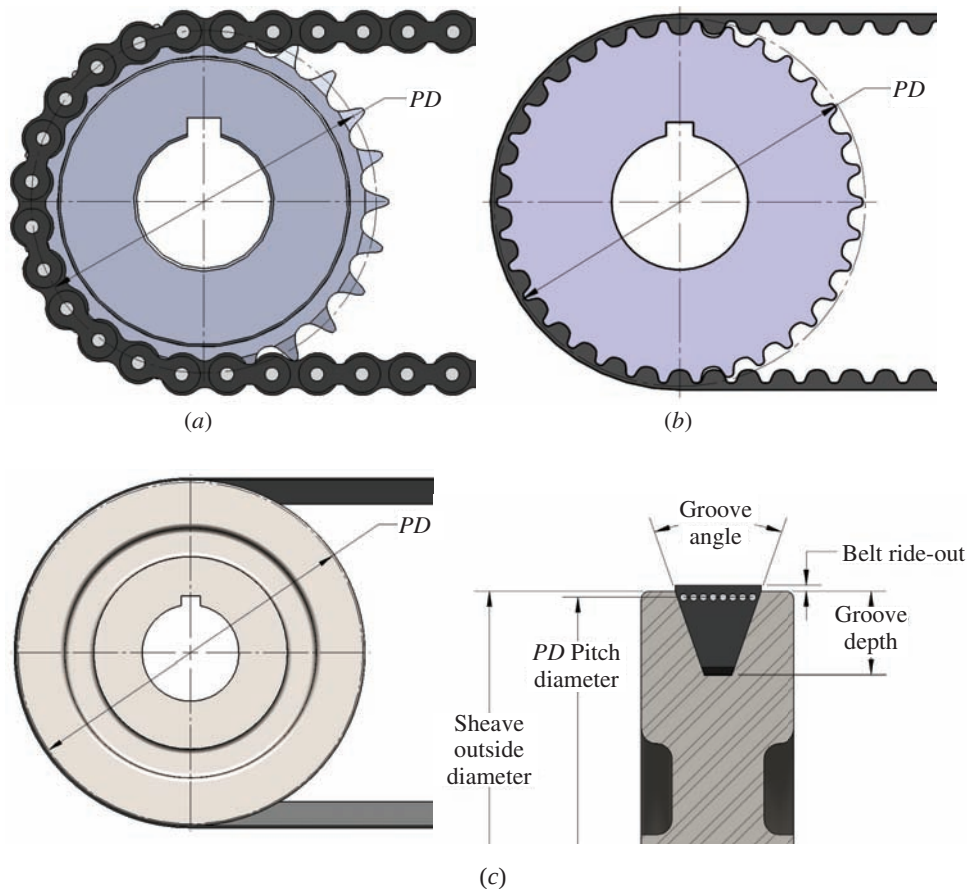


FIGURE 7-3 Pitch diameter on a (a) chain sprocket, (b) synchronous belt sprocket, and (c) V-belt sheave with section view

Typically the input sprocket/sheave is rotating at a faster speed than the output sprocket/sheave. This is called a *speed reducer* and is very common in many applications. When the belt or chain is used for a speed reduction, the smaller sprocket/sheave is mounted on the high-speed shaft, such as an electrical motor. The larger sprocket/sheave is mounted on the driven machine where the shaft is turning at a slower speed. The other scenario is when the output sprocket is rotating faster than the input sprocket and is called a *speed increaser*. In this case, the larger sprocket/sheave is mounted on the input shaft and the smaller sprocket/sheave is mounted on the output shaft.

The belt or chain linear velocity is uniform throughout its length, and it can be related to the angular velocities of the driving sprocket and driven sprocket using the following equation, based on Equation (7-1) developed earlier. Figure 7-4 shows the basic layout where ω_1 is equal to the input angular velocity (ω_{driving}) and ω_2 is equal to the output angular velocity (ω_{driven}).

$$v \frac{\text{belt}}{\text{chain}} = \frac{PD_{\text{driving}}}{2} \cdot \omega_{\text{driving}} = \frac{PD_{\text{driven}}}{2} \cdot \omega_{\text{driven}} \quad (7-2)$$

Letting the driving member to be 1 as used in Figure 7-4 and the driven member to be 2, we can restate Equation (7-2) as

$$v \frac{\text{belt}}{\text{chain}} = \frac{PD_1}{2} \cdot \omega_1 = \frac{PD_2}{2} \cdot \omega_2 \quad (7-3)$$

Since the belt or chain linear velocity is the same for both the driving and driven sprockets, we can equate the last two parts of Equation (7-2) and define the velocity ratio, VR , which is the driving angular velocity over the driven angular velocity. The angular velocity ratio can

also be expressed by the ratios of the pitch diameters or the numbers of teeth of the driving and driven sprockets.

$$VR = \frac{\omega_{\text{driving}}}{\omega_{\text{driven}}} = \frac{PD_{\text{driven}}}{PD_{\text{driving}}} = \frac{N_{\text{driven}}}{N_{\text{driving}}} \quad (7-4)$$

The angle of wrap on the driving and driven sprockets can be found from,

$$\sin(\phi) = \frac{\left(\frac{PD_2 - PD_1}{2} \right)}{CD} \quad (7-5)$$

The angle of wrap on sprocket (1)

$$\theta_1 = 180^\circ - 2 \cdot \phi \quad (7-6)$$

The angle of wrap on sprocket (2)

$$\theta_2 = 180^\circ + 2 \cdot \phi \quad (7-7)$$

The length of the belt or chain wrap on sprocket 1 and sprocket 2 is the arc length s_1 and s_2 .

$$s_1 = \frac{PD_1}{2} \cdot \theta_1 \quad (7-8)$$

$$s_2 = \frac{PD_2}{2} \cdot \theta_2 \quad (7-9)$$

where θ_1 and θ_2 are in radians.

The distance, d , represents the belt or chain length that is tangent to sprocket 1 and sprocket 2.

$$d = CD \cdot \cos \phi \quad (7-10)$$

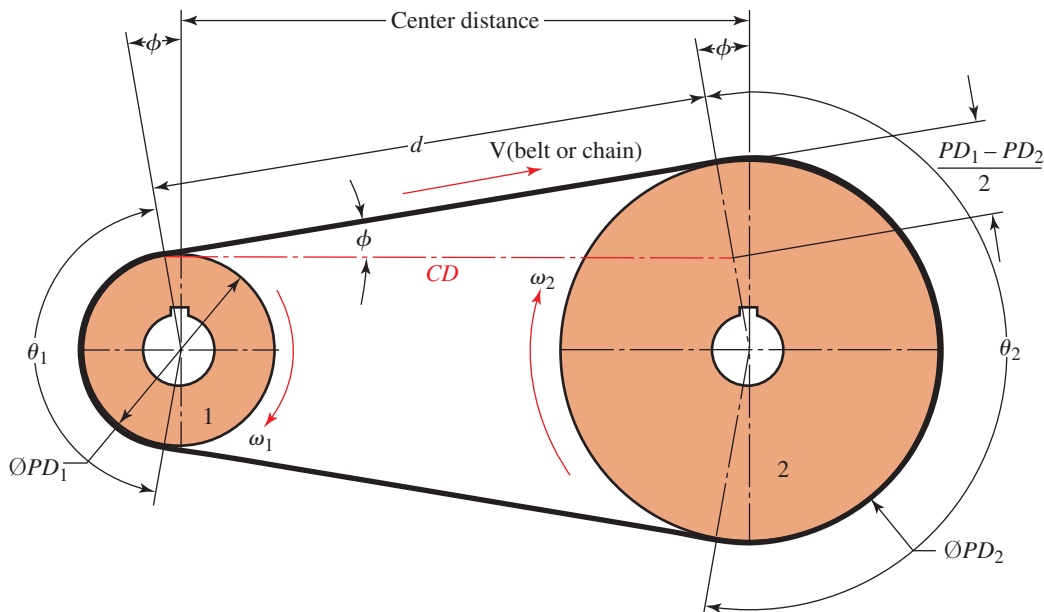


FIGURE 7-4 Belt/chain drive configuration

The belt or chain perimeter length, L_p , can now be calculated.

$$\begin{aligned} \text{Belt or chain perimeter length} &= L_p \\ L_p &= 2 \cdot d + s_1 + s_2 \quad (7-11) \end{aligned}$$

The distance, d , is often called the *span* of the belt or chain drive system, giving the unsupported length of the belt or chain. Long spans sometimes lead to vibration during operation, called *whip*, and should be avoided where possible.

Example Problem 7-1

A synchronous belt drive system shown in Figure 7-5 is used as a speed reducer. The input shaft rotates at 1000 rpm and has a 36-tooth sprocket with a pitch diameter of 3.609 in. The output shaft has a 72-tooth sprocket with a pitch diameter of 7.218 in. The center distance between the two shafts is 15.01 in.

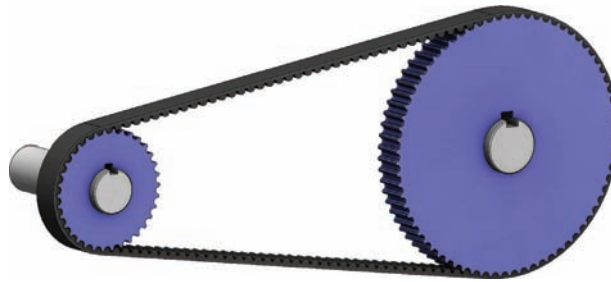


FIGURE 7-5 Speed reducer belt drive

Determine the following for the belt drive:

- The velocity ratio
- The angular velocity of the output shaft
- The linear belt speed
- The belt wrap on the input and output sprockets
- The belt perimeter length
- Optional: Draw the schematic of the belt drive system using CAD

Solution

Objective Use the equations developed in this section to solve for the kinematics of the belt drive system.

$$\begin{aligned} \text{Given} \quad N_{\text{driving}} &= 36 & PD_{\text{driving}} &= 3.609 \text{ in} & n_{\text{driving}} &= 1000 \text{ rpm} \\ N_{\text{driven}} &= 72 & PD_{\text{driven}} &= 7.218 \text{ in} \\ CD &= 15.01 \text{ in} \end{aligned}$$

Results a. Use Equation (7-4) to calculate the velocity ratio.

$$\begin{aligned} VR &= \frac{\omega_{\text{driving}}}{\omega_{\text{driven}}} = \frac{PD_{\text{driven}}}{PD_{\text{driving}}} = \frac{N_{\text{driven}}}{N_{\text{driving}}} \\ VR &= \frac{72}{36} = 2.0 \end{aligned}$$

This tells us the input sprocket turns twice as fast as the output sprocket and the belt drive is a speed reducer. The velocity ratio can be used to find the reduced speed of the output shaft.

b. Manipulating Equation (7-4), we can now solve for the driven speed of the output shaft.

$$\begin{aligned} \omega_{\text{driven}} &= \frac{\omega_{\text{driving}}}{VR} \\ \omega_{\text{driven}} &= \frac{1000 \text{ rpm}}{2} \\ \omega_{\text{driven}} &= 500 \text{ rpm} \end{aligned}$$

- c. The belt linear velocity can be calculated using Equation (7-2)

$$v_{belt} = \frac{PD_{driving}}{2} \cdot \omega_{driving} = \frac{PD_{driven}}{2} \cdot \omega_{driven}$$

Using the input sprocket,

$$v_{belt} = \frac{3.609 \text{ in}}{2} \cdot 1000 \text{ rpm} \cdot \frac{2 \cdot \pi \text{ rad}}{1 \text{ rev}} \cdot \frac{1 \text{ ft}}{12 \text{ in}} = 944.83 \text{ fpm}$$

Using the output sprocket,

$$v_{belt} = \frac{7.218 \text{ in}}{2} \cdot 500 \text{ rpm} \cdot \frac{2 \cdot \pi \text{ rad}}{1 \text{ rev}} \cdot \frac{1 \text{ ft}}{12 \text{ in}} = 944.83 \text{ fpm}$$

- d. In order to solve for the angle of the belt wrap on both sprockets, first we need to find the angle $\emptyset$ using Equation (7-5)

$$\begin{aligned} \sin(\emptyset) &= \frac{\left(\frac{PD_2 - PD_1}{2} \right)}{CD} \\ \sin(\emptyset) &= \frac{\left(\frac{7.218 \text{ in} - 3.609}{2} \right)}{15.01 \text{ in}} = 0.120 \\ \emptyset &= \sin^{-1}(0.120) \\ \emptyset &= 6.905^\circ \end{aligned}$$

The angle of wrap on sprocket (1) is given by Equation (7-6)

$$\begin{aligned} \theta_1 &= 180^\circ - 2 \cdot \emptyset \\ \theta_1 &= 180^\circ + 2 \cdot 6.905^\circ \\ \theta_1 &= 166.2^\circ \end{aligned}$$

The angle of wrap on sprocket (2) is given by Equation (7-7)

$$\begin{aligned} \theta_2 &= 180^\circ + 2 \cdot \emptyset \\ \theta_2 &= 180^\circ - 2 \cdot 6.905^\circ \\ \theta_2 &= 193.8^\circ \end{aligned}$$

- e. To find the belt perimeter length, the belt span distance and the arc lengths of belt wrap on both sprockets need to be found. Use Equation (7-10) to first calculate the belt span distance, d

$$\begin{aligned} d &= CD \cdot \cos \emptyset \\ d &= 15.01 \text{ in} \cdot \cos(6.905^\circ) \\ d &= 14.90 \text{ in} \end{aligned}$$

Next solve for the arc length s_1 using Equation (7-8),

$$\begin{aligned} s_1 &= \frac{PD_1}{2} \cdot \theta_1 \\ s_1 &= \frac{3.609 \text{ in}}{2} \cdot 166.2^\circ \cdot \frac{2 \cdot \pi \text{ rad}}{360^\circ} \\ s_1 &= 5.23 \text{ in} \end{aligned}$$

Now solve for the arc length s_2 using Equation (7-9),

$$s_2 = \frac{PD_2}{2} \cdot \theta_2$$

$$s_2 = \frac{7.218 \text{ in}}{2} \cdot 193.8^\circ \cdot \frac{2 \cdot \pi \text{ rad}}{360^\circ}$$

$$s_2 = 12.21 \text{ in}$$

Lastly, use Equation (7-11) to add the belt span distance and the two arc lengths to solve for the belt perimeter length,

$$\text{Belt perimeter length} = 2 \cdot d + s_1 + s_2$$

$$\text{Belt perimeter length} = 2 \cdot 14.90 \text{ in} + 5.23 \text{ in} + 12.21 \text{ in}$$

$$\text{Belt perimeter length} = 47.24 \text{ in}$$

The belt perimeter length would be used to purchase the correct belt length from the belt manufacturer. Since belt manufacturer make belts of specific lengths, the calculated center distance and belt perimeter length must correspond to the manufacturer's belt specifications.

- f. The layout of the belt drive system using a CAD package is shown in Figure 7-6.

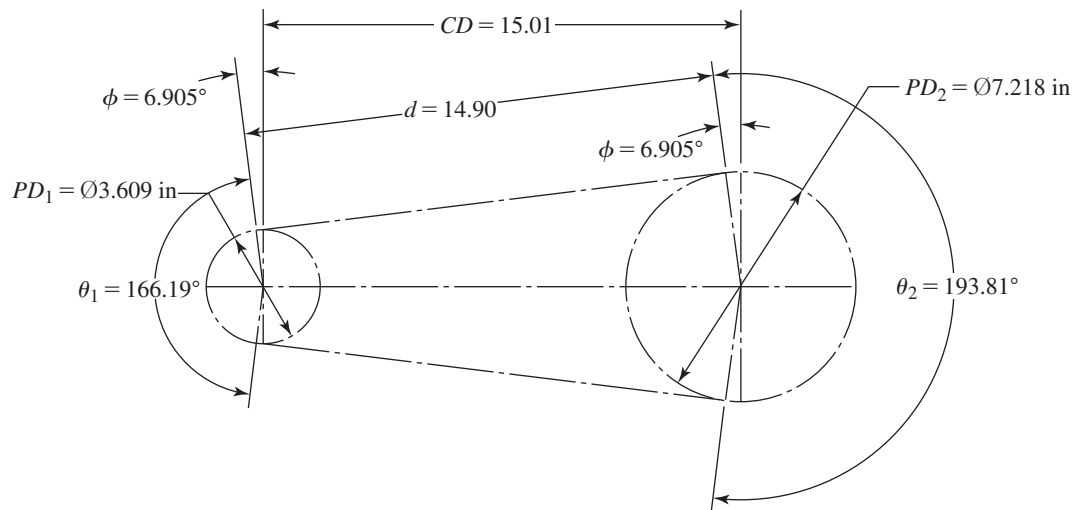


FIGURE 7-6 Schematic drawing of the speed reducer belt drive for Example Problem 7-1

7-3 TYPES OF BELT DRIVES

A belt is a flexible power transmission element that seats tightly on a set of pulleys, sprockets, or sheaves. Figure 7-4 shows the basic layout.

Many types of belts are available: flat belts, grooved or cogged belts, standard V-belts, double-angle V-belts, and others. See Figure 7-7 for examples. References 3a-g and 4-8 give more examples and technical data. See also Internet sites 2-6, 8-11, 14, and 16 for industry data.

The *flat belt* is the simplest type, often made from leather or rubber-coated fabric. The sheave surface is also flat and smooth, and the driving force

is therefore limited by the pure friction between the belt and the sheave. Some designers prefer flat belts for delicate machinery because the belt *will* slip if the torque tends to rise to a level high enough to damage the machine. The design of flat belt drives is not covered in this book.

Synchronous belts, sometimes called *timing belts* [see Figure 7-7(b)], ride on sprockets having mating grooves into which the teeth on the belt seat. This is a positive drive, limited only by the tensile strength of the belt and the shear strength of the teeth. See Section 7-6.

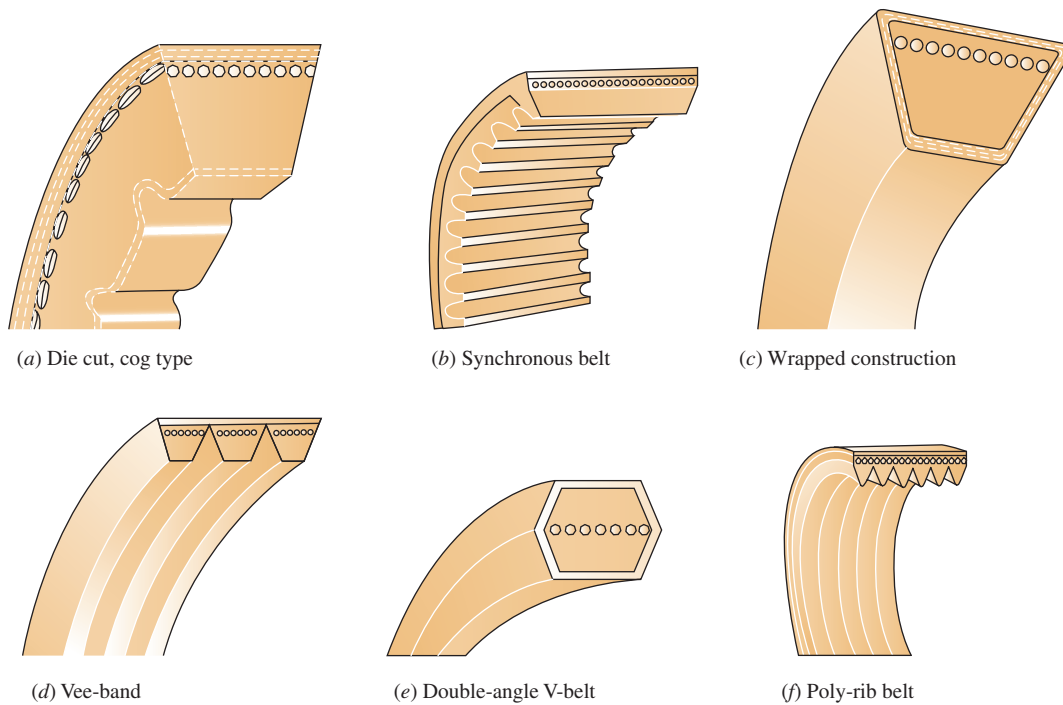


FIGURE 7-7 Examples of belt construction

The belt is installed by placing it around the two sheaves while the center distance between them is reduced. Then the sheaves are moved apart, placing the belt in a rather high initial tension. When the belt is transmitting power, friction causes the belt to grip the driving sheave, increasing the tension in one side, called the “tight side,” of the drive. The tensile force in the belt exerts a tangential force on the driven sheave, and thus a torque is applied to the driven shaft. The opposite side of the belt is still under tension, but at a smaller value. Thus, it is called the “slack side.”

Some cog belts, such as that shown in Figure 7-7(a), are applied to standard V-grooved sheaves. The cogs give the belt greater flexibility and higher efficiency compared with standard belts. They can operate on smaller sheave diameters.

A widely used type of belt, particularly in industrial drives and vehicular applications, is the *V-belt drive*, shown in Figures 7-7(a) and 7-7(c). Figure 7-8 shows the V-belt section seated in its groove in the sheave. The V-shape causes the belt to wedge tightly into the groove, increasing friction and allowing high torques to be transmitted before slipping occurs. Most belts have high-strength cords positioned at the pitch diameter of the belt cross section to increase the tensile strength of the belt. The cords, made from natural fibers, synthetic strands, or steel, are embedded in a firm rubber compound to provide the flexibility needed to allow the belt to pass around the sheave. Often an outer fabric cover is added to give the belt good durability. The belt is designed to ride around the two sheaves without slipping.

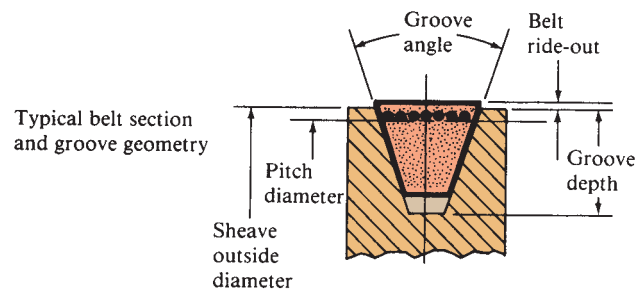


FIGURE 7-8 Cross section of V-belt and sheave groove

The groove angle ranges from 34° to 42° depending on the belt cross-section style and the pitch diameter. Refer to manufacturer's data for sheaves.

7-4 V-BELT DRIVES

The typical arrangement of the elements of a V-belt drive is shown in Figure 7-4. The geometry and kinematics of the drive were described in Section 7-2. Equations 7-1 through 7-11 can be used as part of the belt drive design process. Other relationships are added in this section. The important design-related observations to be derived from this arrangement are summarized as follows:

1. The relationships between perimeter length, L_p , center distance, CD , and the sheave diameters are given as

$$L_p = 2CD + 1.57(D_2 + D_1) + \frac{(D_2 - D_1)^2}{4CD} \quad (7-12)$$

$$CD = \frac{B + \sqrt{B^2 - 32(D_2 - D_1)^2}}{16} \quad (7-13)$$

where $B = 4L_p - 6.28(D_2 + D_1)$

2. The angle of contact of the belt on each sheave can be found directly by combining Equations (7-5) to (7-7) from Section 7-2, yielding

$$\theta_1 = 180^\circ - 2 \sin^{-1} \left[\frac{D_2 - D_1}{2CD} \right] \quad (7-14)$$

$$\theta_2 = 180^\circ + 2 \sin^{-1} \left[\frac{D_2 - D_1}{2CD} \right] \quad (7-15)$$

These angles are important because commercially available belts are rated with an assumed contact angle of 180° . This will occur only if the drive ratio is 1 (no speed change). The angle of contact on the smaller of the two sheaves of unequal diameters will always be less than 180° , requiring a lower power rating.

3. In place of Equation (7-10) for the span, d , the following equation can be used. The length of the span between the two sheaves, over which the belt is unsupported, is

$$d = \sqrt{CD^2 - \left[\frac{D_2 - D_1}{2} \right]^2} \quad (7-16)$$

This is important for two reasons: You can check the proper belt tension by measuring the amount of force required to deflect the belt at the middle of the span by a given amount. Also, the tendency for the belt to vibrate or whip is dependent on this length as reported in Section 7-2.

4. The contributors to the stress in the belt are as follows:
 - a. The tensile force in the belt, maximum on the tight side of the belt.
 - b. The bending of the belt around the sheaves, maximum as the tight side of the belt bends around the smaller sheave.
 - c. Centrifugal forces created as the belt moves around the sheaves.

The maximum total stress occurs where the belt enters the smaller sheave, and the bending stress is a major part. Thus, there are recommended minimum sheave diameters for standard belts. Using smaller sheaves drastically reduces belt life.

5. The design value of the ratio of the tight side tension to the slack side tension is 5.0 for V-belt drives. The actual value may range as high as 10.0.

Standard V-Belt Cross Sections

Commercially available V-belts are made to one of the standards shown in Figures 7-9 through 7-12. The alignment between the inch sizes and the metric sizes indicates that the paired sizes are actually the same cross section. A “soft conversion” was used to rename the familiar inch sizes with the number for the metric sizes giving the nominal top width in millimeters.

The nominal value of the included angle between the sides of the V-groove ranges from 34° to 42° . The angle on the belt may be slightly different to achieve a tight fit in the groove. Some belts are designed to “ride out” of the groove somewhat.

The designations shown for the various cross sections apply when the belt construction is like that shown in Figure 7-7(c). For the cog-type belt [Figure 7-7(a)] of the same cross section, the letter X is added to the designation. For example, a 5V belt has a smooth inner surface, while a 5VX belt is of the cog type. Power transmission ratings are typically higher for the cogged type because they bend more easily around the sheaves with less stress in the belt.

Single automotive V-belts have cross sections ranging across the nine sizes shown in Figure 7-12 and may have either the smooth or cogged (X) type inner surfaces. Many applications employ the vee-band [Figure 7-7(d)] or the poly-rib style [Figure 7-7(f)]. Refer to References 3a, 3f, 5, or 8.

V-Belt Drive Design

The factors involved in the selection of a V-belt and the driving and driven sheaves and proper installation of the drive are summarized in this section. Abbreviated examples of the data available from suppliers are given for illustration. Catalogs contain extensive data, and step-by-step instructions are given for their use. The basic data required for drive selection are the following:

- The rated power of the driving motor or other prime mover
- The service factor based on the type of driver and driven load
- The center distance
- The power rating for one belt as a function of the size and speed of the smaller sheave
- The belt length
- The size of the driving and driven sheaves
- The correction factor for belt length
- The correction factor for the angle of wrap on the smaller sheave
- The number of belts
- The initial tension on the belt

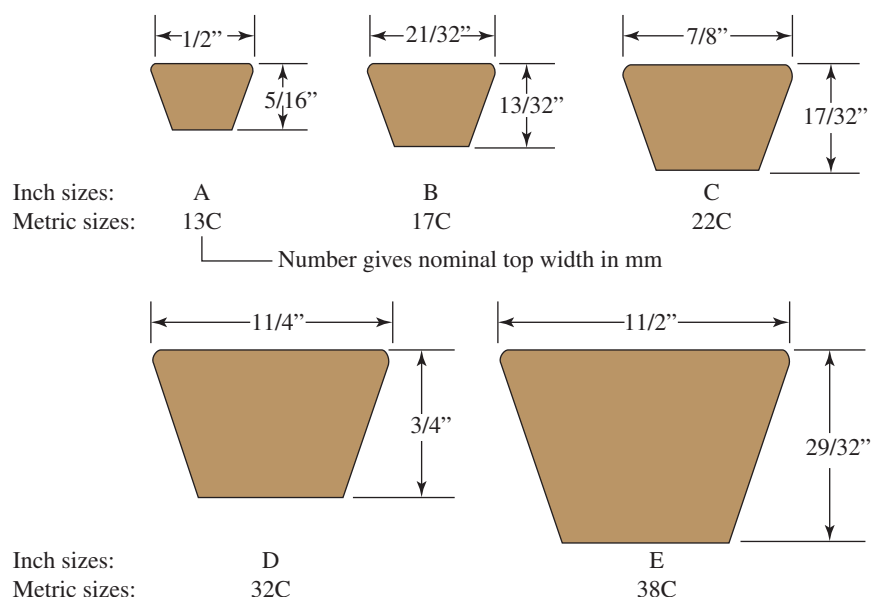


FIGURE 7-9 Heavy-duty industrial V-belts

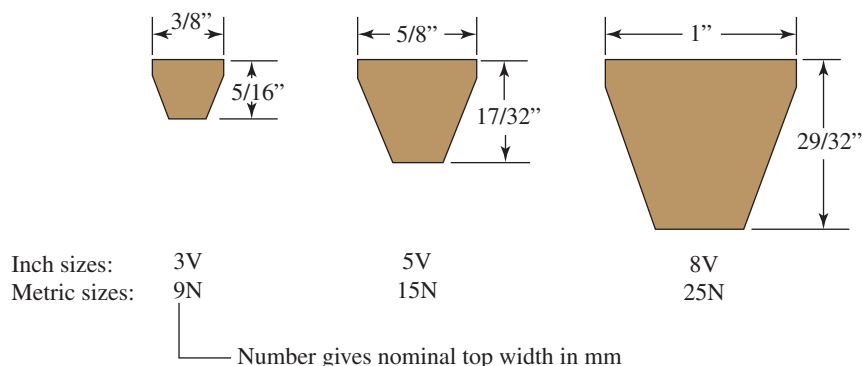


FIGURE 7-10 Industrial narrow-section V-belts

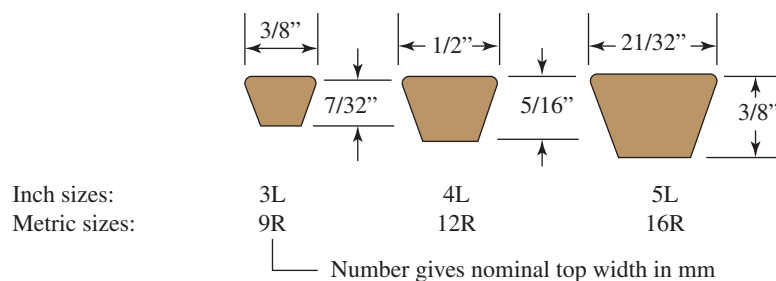


FIGURE 7-11 Light-duty, fractional horsepower (FHP) V-belts

Many design decisions depend on the application and on space limitations. A few guidelines are given as follows:

- The recommended maximum reduction ratio for a plain V-belt drive is 6:1. For cogged belts it is 7:1.

For higher desired ratios use two or more stages of reduction.

- Adjustment for the center distance must be provided in both directions from the nominal value. The center

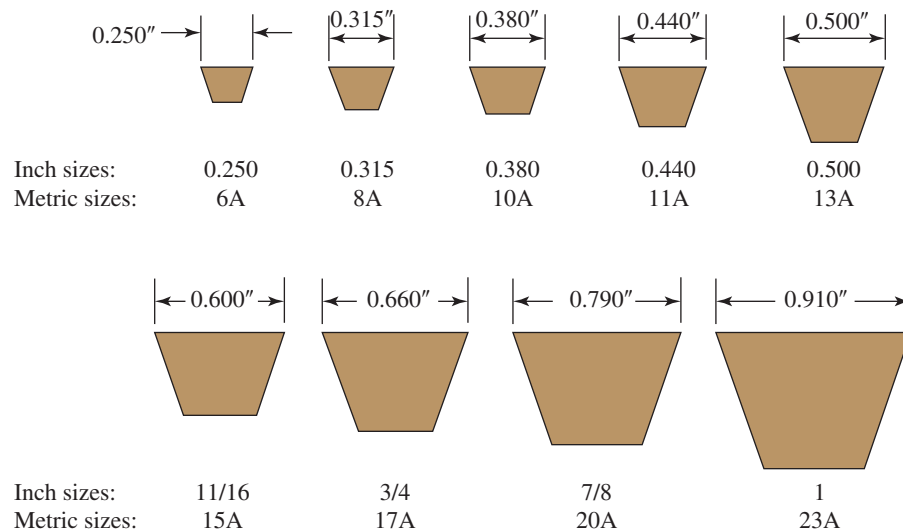


FIGURE 7-12 Automotive V-belts

distance must be shortened at the time of installation to enable the belt to be placed in the grooves of the sheaves without force. Provision for increasing the center distance must be made to permit the initial tensioning of the drive and to take up for belt stretch. Manufacturers' catalogs give the data. One convenient way to accomplish the adjustment is the use of a take-up unit, as shown in Figures 14-10(b) and (c).

- If fixed centers are required, idler pulleys should be used. It is best to use a grooved idler on the inside of the belt, close to the large sheave. Adjustable tensioners are commercially available to carry the idler. More discussion of the use of idlers is included in Section 7-6.
- The nominal range of center distances should be

$$D_2 < CD < 3(D_2 + D_1) \quad (7-17)$$

- The angle of wrap on the smaller sheave should be greater than 120° .
- Because of balancing, centrifugal stresses, belt whip, and other dynamic considerations, belt speeds should be under 5000 ft/min or the supplier of the sheaves should be consulted. A recommended maximum belt speed is 6500 ft/min.
- Consider an alternative type of drive, such as a gear type, synchronous belt drive, or chain, if the belt speed is less than 1000 ft/min.
- Avoid elevated temperatures around belts.
- Ensure that the shafts carrying mating sheaves are parallel and that the sheaves are in alignment so that the belts track smoothly into the grooves.
- In multibelt installations, matched belts are required. Match numbers are printed on industrial belts, with 50 indicating a belt length very close to nominal.

Longer belts carry match numbers above 50; shorter belts below 50.

- Belts must be installed with the initial tension recommended by the manufacturer. Tension should be checked after the first few hours of operation because seating and initial stretch occur.
- Reported power transmission ratings typically are based on belt life of approximately 5000–7000 hours of operation and about 25 000 hours for the sheaves.

Design Data

Catalogs from commercial belt drive manufacturers typically give several dozen pages of design data for the various sizes of belts and sheave combinations to ease the job of drive design. See Internet sites 3-6, 8, and 16 for examples. The data typically are given in tabular form. Graphical form is used here so that you can get a feel for the variation in performance with design choices. Any design made from the data in this book should be checked against a particular manufacturer's ratings before use.

The data given here are for the narrow-section belts: 3V, 5V, and 8V. These three sizes cover a wide range of power transmission capacities. Note that the cogged versions of these narrow section belts—3VX, 5VX, and 8VX—have higher power ratings and are reported separately in catalogs. Figure 7-13 can be used to choose the basic size for the belt cross section. Note that the power axis is *design power*, the rated power of the prime mover times the service factor from Table 7-1.

Figures 7-13 to 7-19 and Tables 7-1 and 7-2 are for use in problem solving in this book and they are similar to those found in many manufacturers' catalogs, but they

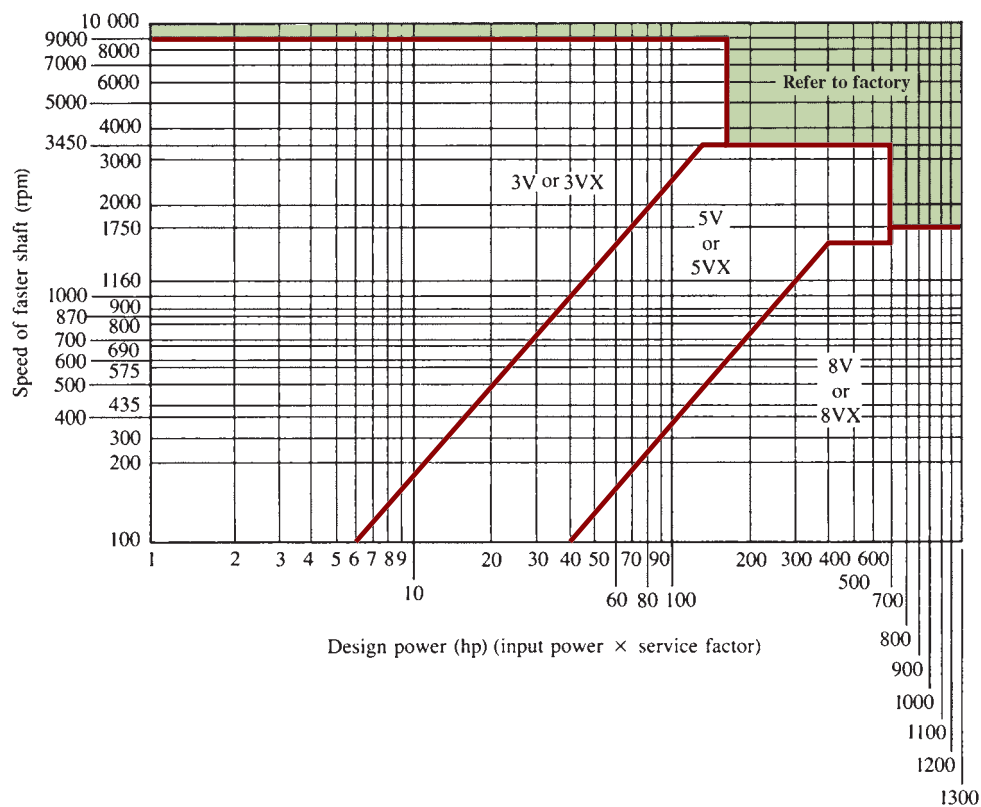


FIGURE 7-13 Sample selection chart for narrow-section industrial V-belts

TABLE 7-1 V-Belt Service Factors¹

Driven machine type	Driver type					
	AC motors: Normal torque ² DC motors: Shunt-wound Engines: Multiple-cylinder			AC motors: High torque ³ DC motors: Series-wound, or compound-wound Engines: 4-cylinder or less		
	<6 h per day	6–15 h per day	>15 h per day	<6 h per day	6–15 h per day	>15 h per day
Smooth loading Agitators, light conveyors, centrifugal pumps fans and blowers under 10 hp (7.5 kW)	1.0	1.1	1.2	1.1	1.2	1.3
Light shock loading Generators, machine tools mixers, fans and blowers over 10 hp (7.5 kW) gravel conveyors	1.1	1.2	1.3	1.2	1.3	1.4
Moderate shock loading Bucket elevators, piston pumps textile machinery, hammer mills heavy conveyors, pulverizers	1.2	1.3	1.4	1.4	1.5	1.6
Heavy shock loading Crushers, ball mills, hoists rubber mills, and extruders	1.3	1.4	1.5	1.5	1.6	1.8
Machinery that can choke	2.0	2.0	2.0	2.0	2.0	2.0

¹Factors given are for speed reducers. For speed increases, multiply listed factors by 1.2.²Synchronous, split-phase, three-phase with starting torque or breakdown torque less than 175% of full-load torque.³Single-phase, three-phase with starting torque or breakdown torque greater than 175% of full-load torque.

do not represent any particular company's data. Refer to Internet Sites 3–6, 8, 9, 14, and 16 for examples of V-belt drive products and specific design data.

Reference 3c gives the generic formula for the power rating of a V-belt as

$$P_{\text{nom}} = K(P_b + \Delta P_R + \Delta P_L)$$

Where,

K = factor based on the angle of wrap on the sheave

P_b = Basic power rating for a ratio of 1.0 and a set belt length

ΔP_R = Added power capacity based on speed ratio

ΔP_L = Added power capacity based on belt length

Additional detail is provided in the reference for the individual terms. Manufacturers provide rating data for the particular styles and quality factors for their products. This section gives sample data to demonstrate the process for specifying a particular belt design.

Figures 7–14 to 7–16 give the rated power per belt for the three cross sections as a function of the pitch diameter of the smaller sheave and its speed of rotation.

The labeled vertical lines in each figure give the standard sheave pitch diameters available.

The basic power rating for a speed ratio of 1.00 is given as the solid curve. A given belt can carry a greater power as the speed ratio increases, up to a ratio of approximately 3.38. Further increases have little effect and may also lead to trouble with the angle of wrap on the smaller sheave. Figure 7–17 is a plot of the data for power to be added to the basic rating as a function of speed ratio for the 5V belt size. The catalog data are given in a step-wise fashion. The maximum power added, for ratios of above 3.38, was used to draw the dashed curves in Figures 7–14 to 7–16. In most cases, a rough interpolation between the two curves is satisfactory.

Figure 7–18 gives the value of a correction factor, C_θ , as a function of the angle of wrap of the belt on the small sheave.

Figure 7–19 gives the value of the correction factor, C_L , for belt length. A longer belt is desirable because it reduces the frequency with which a given part of the belt encounters the stress peak as it enters the small sheave. Only certain standard belt lengths are available. For problems in this book, we use those listed in Table 7–2.

Example Problem 7–1 illustrates the use of the design data.

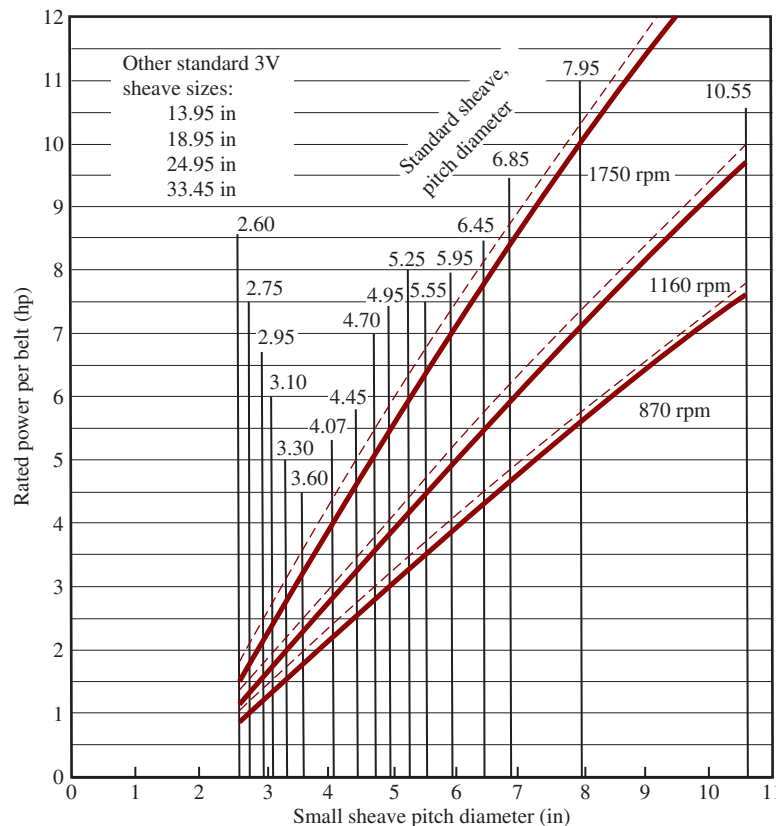


FIGURE 7–14 Power rating: 3V belts

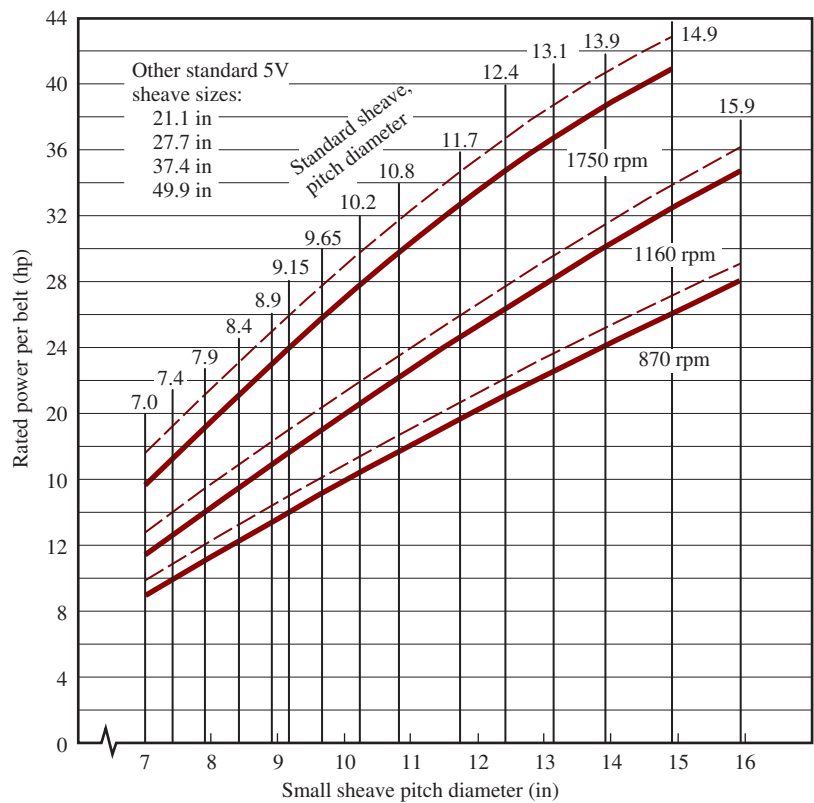


FIGURE 7-15 Power rating: 5V belts

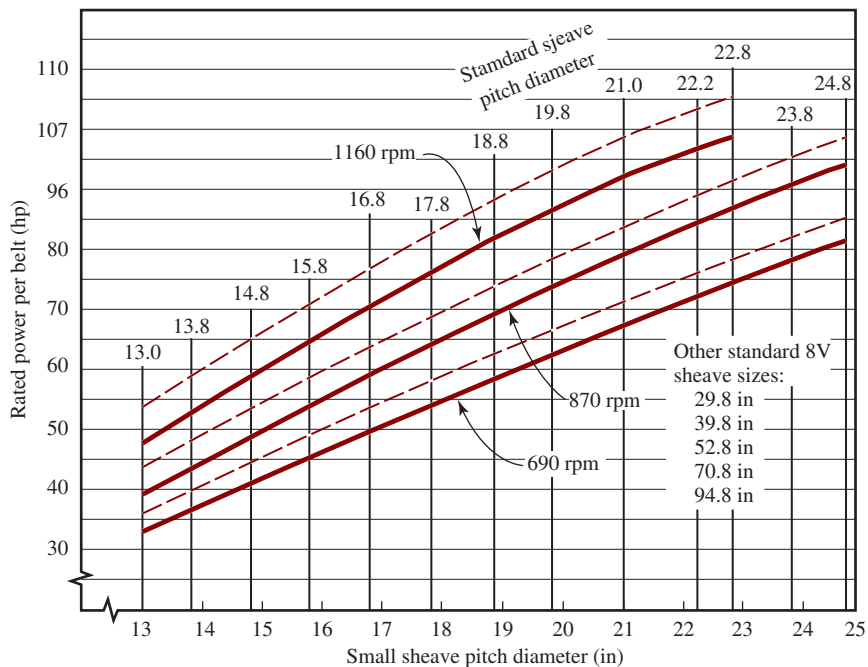


FIGURE 7-16 Power rating: 8V belts

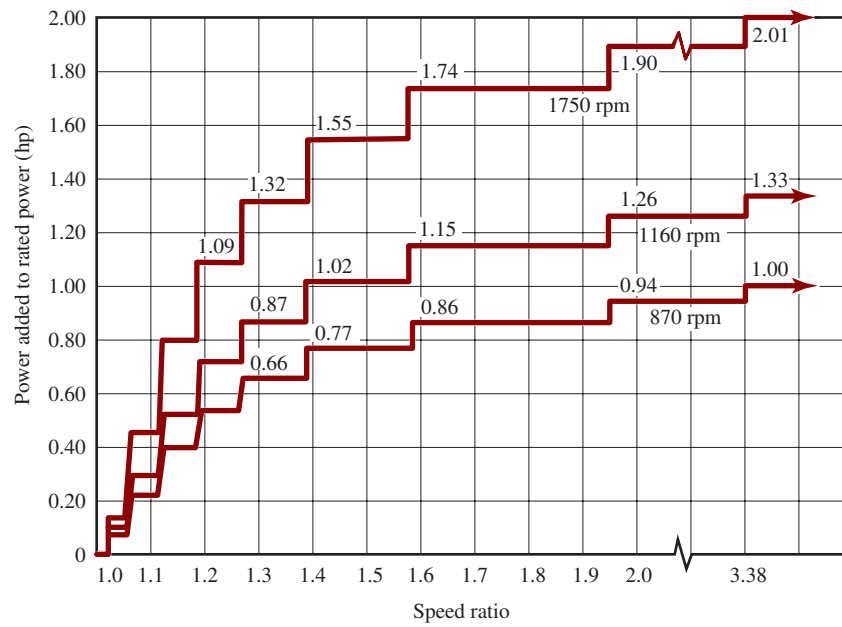


FIGURE 7-17 Power added versus speed ratio: 5V belts

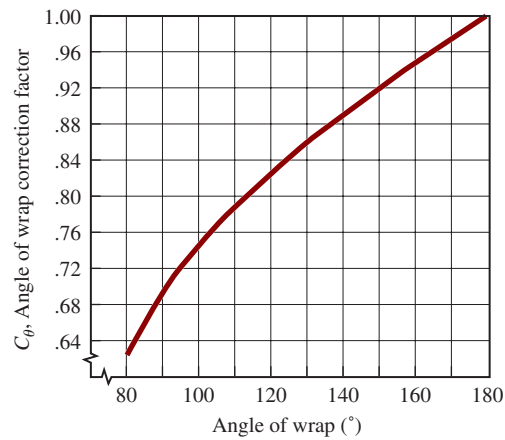
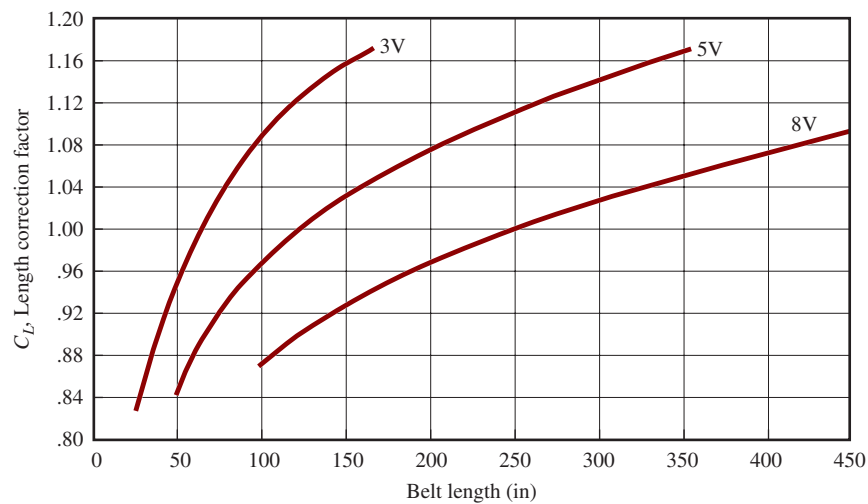

 FIGURE 7-18 Angle of wrap correction factor, C_θ

 FIGURE 7-19 Belt length correction factor, C_L

TABLE 7-2 Standard Belt Lengths for 3V, 5V, and 8V Belts (in)

3V only	3V and 5V	3V, 5V, and 8V	5V and 8V	8V only
25	50	100	150	375
26.5	53	106	160	400
28	56	112	170	425
30	60	118	180	450
31.5	63	125	190	475
33.5	67	132	200	500
35.5	71	140	212	
37.5	75		224	
40	80		236	
42.5	85		250	
45	90		265	
47.5	95		280	
			300	
165			315	
			335	
			355	

Example Problem 7-2

Design a V-belt drive that has the input sheave on the shaft of an electric motor (normal torque) rated at 50.0 hp at 1160-rpm, full-load speed. The drive is to a bucket elevator in a potash plant that is to be used 12 hours (h) daily at approximately 675 rpm.

Solution

Objective Design the V-belt drive.

Given Power transmitted = 50 hp to bucket elevator
Speed of motor = 1160 rpm; output speed = 675 rpm

Analysis Use the design data presented in this section. The solution procedure is developed within the Results section of the problem solution.

Results **Step 1.** Compute the design power. From Table 7-1, for a normal torque electric motor running 12 h daily driving a bucket elevator, the service factor is 1.30. Then the design power is $1.30(50.0 \text{ hp}) = 65.0 \text{ hp}$.

Step 2. Select the belt section. From Figure 7-13, a 5V belt is recommended for 70.0 hp at 1160-rpm input speed.

Step 3. Compute the nominal speed ratio:

$$\text{Ratio} = 1160/675 = 1.72$$

Step 4. Compute the driving sheave size that would produce a belt speed of $v_b = 4000 \text{ ft/min}$, as a guide to selecting a standard sheave, adapted from Equation (7-2):

$$\text{Belt speed} = v_b = \frac{D_1 n_1}{2}$$

Then the required diameter to give $v_b = 4000 \text{ ft/min}$ is

$$D_1 = \frac{2v_b}{n_1} = \frac{2(4000 \text{ ft/min})}{1160 \text{ rev/min}} \times (12 \text{ in/ft})(1 \text{ rev}/(2\pi \text{ rad})) = 13.17 \text{ in}$$

Step 5. Select trial sizes for the input sheave, and compute the desired size of the output sheave. Select a standard size for the output sheave, and compute the actual ratio and output speed.

TABLE 7-3 Trial Sheave Sizes for Example Problem 7-1

Standard driving sheave size, D_1	Approximate driven sheave size ($1.72D_1$)	Nearest standard sheave, D_2	Actual output speed (rpm)
13.10	22.5	21.1	720
12.4	21.3	21.1	682
11.7	20.1	21.1	643
10.8	18.6	21.1	594
10.2	17.5	15.9	744
9.65	16.6	15.9	704
9.15	15.7	15.9	668
8.9	15.3	14.9	693

For this problem, the trials are given in Table 7-3 (diameters are in inches).

The two trials in **boldface** in Table 7-3 give only about 1% variation from the desired output speed of 675 rpm, and the speed of a bucket elevator is not critical. Because no space limitations were given, let's choose the larger size; $D_1 = 12.4$ in; $D_2 = 21.1$ in.

We can now compute the actual belt speed:

$$v_b = D_1 n_1 / 2 = (12.4 \text{ in} / 2)(1160 \text{ rev/min})(2\pi \text{ rad/rev})(1 \text{ ft} / 12 \text{ in}) = 3766 \text{ ft/min}$$

Step 6. Determine the rated power from Figures 7-14 to 7-16.

For the 5V belt that we have selected, Figure 7-15 is appropriate. For a 12.4-in sheave at 1160 rpm, the basic rated power is 26.4 hp. Multiple belts will be required. The ratio is relatively high, indicating that some added power rating can be used. This value can be estimated from Figure 7-15 or taken directly from Figure 7-17 for the 5V belt. Power added is 1.15 hp. Then the actual rated power is $26.4 + 1.15 = 27.55$ hp.

Step 7. Specify a trial center distance, CD .

We can use Equation (7-17) to determine a nominal acceptable range for CD :

$$D_2 < CD < 3(D_2 + D_1)$$

$$21.1 < CD < 3(21.1 + 12.4)$$

$$21.1 < CD < 100.5 \text{ in}$$

In the interest of conserving space, let's try $CD = 24.0$ in.

Step 8. Compute the required belt length for our preliminary choices for pitch diameters of the sheaves and this center distance from Equation (7-12):

$$L = 2CD + 1.57(D_2 + D_1) + \frac{(D_2 - D_1)^2}{4C}$$

$$L = 2(24.0) + 1.57(21.1 + 12.4) + \frac{(21.1 - 12.4)^2}{4(24.0)} = 101.4 \text{ in}$$

Step 9. Select a standard belt length from Table 7-2, and compute the resulting actual center distance from Equation (7-13).

In this problem, the nearest standard length is 100.0 in. Then, from Equation (7-13),

$$B = 4L - 6.28(D_2 + D_1) = 4(100) - 6.28(21.1 + 12.4) = 189.6$$

$$CD = \frac{189.6 + \sqrt{(189.6)^2 - 32(21.1 - 12.4)^2}}{16} = 23.30 \text{ in}$$

Step 10. Compute the angle of wrap of the belt on the small sheave from Equation (7-14):

$$\theta_1 = 180^\circ - 2 \sin^{-1} \left[\frac{D_2 - D_1}{2CD} \right] = 180^\circ - 2 \sin^{-1} \left[\frac{21.1 - 12.4}{2(23.30)} \right] = 158^\circ$$

Step 11. Determine the correction factors from Figures 7–18 and 7–19. For $\theta = 158^\circ$, $C_\theta = 0.94$. For $L = 100$ in, $C_L = 0.96$.

Step 12. Compute the corrected rated power per belt and the number of belts required to carry the design power:

$$\text{Corrected power rating per belt} = C_\theta C_L P = (0.94)(0.96)(27.55 \text{ hp}) = 24.86 \text{ hp}$$

$$\text{Minimum number of belts} = \text{Design power/Corrected power rating}$$

$$\text{Minimum number of belts} = 65.0 \text{ hp}/24.86 \text{ hp} = 2.61 \text{ belts (Use 3 belts.)}$$

Comments

Summary of Design

Input: Electric motor, 50.0 hp at 1160 rpm

Service factor: 1.3

Design power: 65.0 hp

Belt: 5V cross section, 100-in length, 3 belts

Sheaves: Driver, 12.4-in pitch diameter, 3 grooves, 5V. Driven, 21.1-in pitch diameter, 3 grooves, 5V

Actual output speed: 682 rpm

Center distance: 23.30 in

Belt Tension

The initial tension given to a belt is critical because it ensures that the belt will not slip under the design load. At rest, the two sides of the belt have the same tension. As power is being transmitted, the tension in the tight side increases while the tension in the slack side decreases. Without the initial tension, the slack side would go totally loose, and the belt would not seat in the groove; thus, it would slip. Manufacturers' catalogs give data for the proper belt-tensioning procedures.

7-5 SYNCHRONOUS BELT DRIVES

Synchronous belts are constructed with ribs or teeth across the underside of the belt, as shown in Figure 7–7(b). The teeth mate with corresponding grooves in the driving and driven pulleys, called sprockets, providing a positive drive without slippage. Therefore, there is a fixed relationship between the speed of the driver and the speed of the driven sprocket. For this reason synchronous belts are often called timing belts and when properly designed will have an efficiency as high as 98%. In contrast, V-belts can creep or slip with respect to their mating sheaves, especially under heavy loads and varying power demand and typically are 95%–98% efficient. The efficiency of properly maintained chain drives range between 92% and 98%. Synchronous action is critical to the successful operation of such systems as printing, material handling, packaging, and assembly. Synchronous belt drives are increasingly being considered for applications in which gear drives or chain drives had been used previously. Synchronous

belts provide a positive and trouble-free transmission of power and offer these advantages:

- High capacity
- Highly accurate registration
- Low vibration
- Low noise
- No lubrication required
- No stretching due to wear
- Corrosion resistance
- Abrasion resistance
- Clean operation

Figure 7–20 shows a synchronous belt mating with the toothed driving sprocket and driven sprocket.

Figure 7–21 shows commonly available commercial shapes for synchronous belts. Two series are in use,

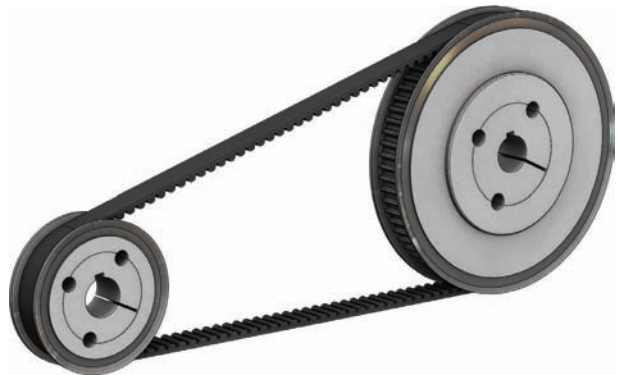


FIGURE 7–20 Synchronous belt on driving and driven sprockets

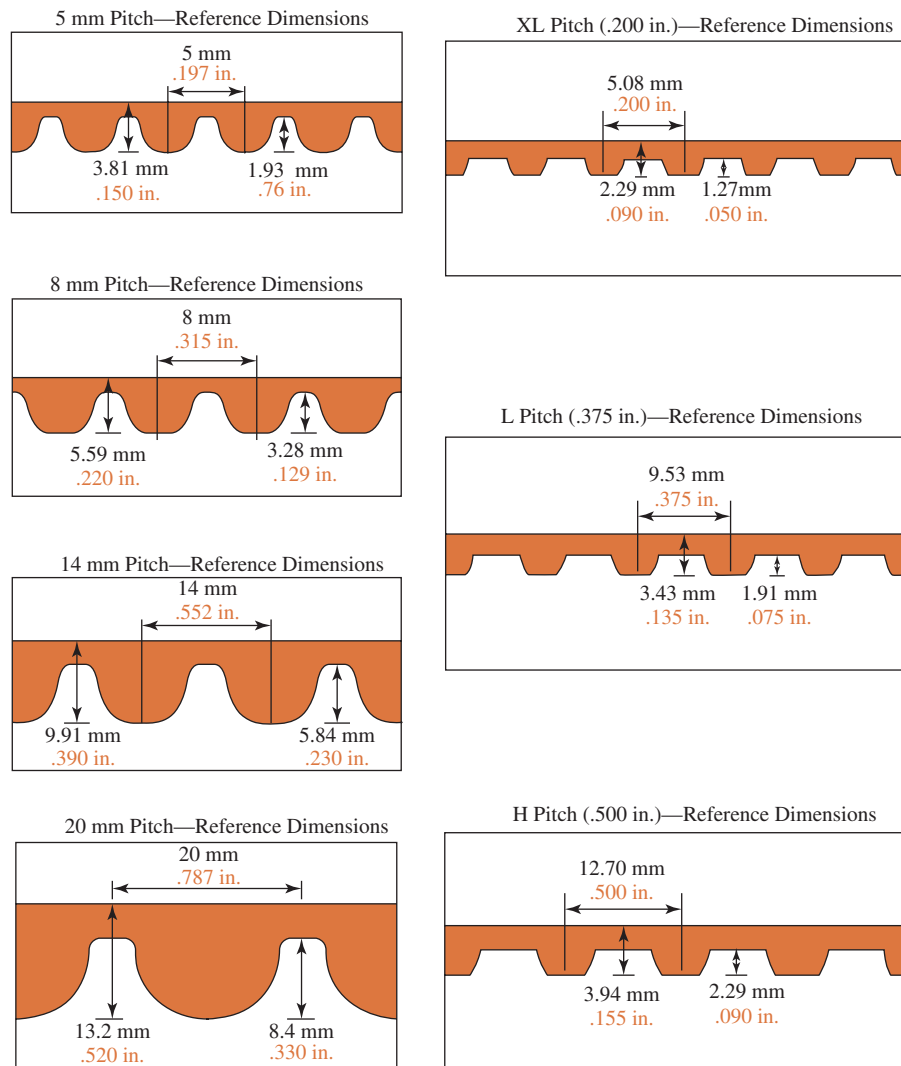


FIGURE 7-21 Sizes and shapes for synchronous belt-cogs.

Metric sizes (mm) and U.S. customary sizes (in.). The shapes of the cross sections are drawn full size, showing the pitch lengths from the center of one tooth to the next adjacent tooth. Check individual manufacturers' catalogs for available stock sizes.

The trapezoidal shape of timing belts in Figure 7-22(c) offer better timing and indexing for systems that require

good registration control. The HTD standard curvilinear tooth profile belt in Figure 7-22(b) has a high load-carrying capacity and is used in high-torque applications. HTD belt drives require increased clearance between the belt tooth and the sprocket groove to operate properly. This reduces the registration accuracy of HTD belts. The GT style belt in Figure 7-22(a) has a modified

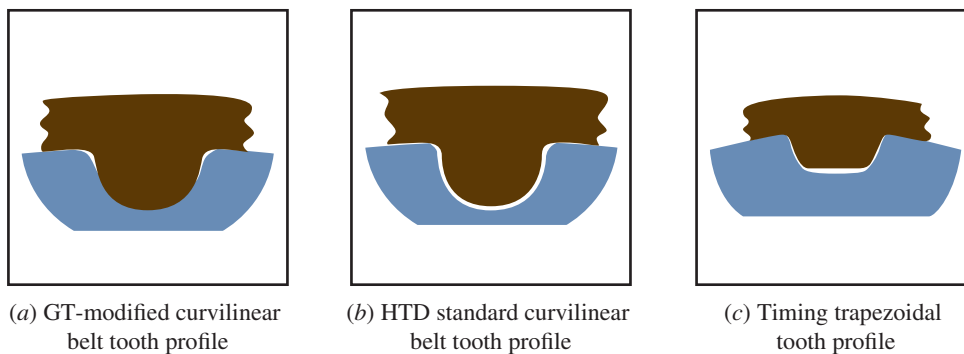


FIGURE 7-22 Belt tooth and pulley groove contact

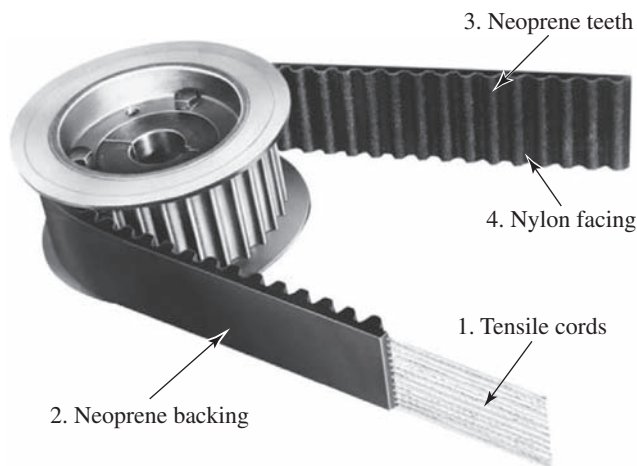


FIGURE 7-23 Synchronous belt construction

curvilinear tooth profile that offers improved indexing accuracy over the HTD style belt, higher load-carrying capacity, and longer life than timing belts.

Figure 7-23 shows detail of the construction of the cross section of a synchronous belt. The tensile strength is provided predominantly by high-strength cords made from fiberglass or similar materials. The cords are encased in a rubber backing material, and the teeth are formed integrally with the backing. Often the fabric covering is used on those parts of the belt that contact the sprockets to provide additional wear resistance and high net shear strength for the teeth.

Various widths of belts are available for each given pitch to provide a wide range of power transmission capacity. Belts are also available in various perimeter lengths. This allows a two sprocket belt drive system to have a wide range of center distances. An example of a belt designation is

Belt Designation: 1760-8MGT-30

The first set of numbers (in mm) represents the belt perimeter, the second set represents the belt pitch with the style of the tooth profile, and the last number represents the belt width. Therefore, this belt has a length of 1760 mm, a belt pitch of 8 mm with a GT tooth profile, and a belt width of 30 mm. The belt drive center distance is determined by the belt perimeter and the pitch diameters of the driven and driving sprocket combination used.

Typical driving and driven sprockets with taper-lock bushings are shown in Figure 7-24. At least one of the two sprockets will have side flanges to ensure that the belt does not move axially. Commercially available sprockets typically employ a split-taper bushing in their hubs with a precise bore. The taper-lock bushing is available in different bore sizes to fit a range of shaft diameters for a given bushing. The keyway in the bushing is a standard size for the shaft diameter and prevents rotational movement of the sprocket and bushing on the shaft. The bore of the sprocket and the outside diameter of the bushing are both tapered. As the sprocket, split



FIGURE 7-24 Driven and driving sprockets with taper-lock bushings for synchronous belt drive

taper-lock bushing, and key are positioned on the shaft and tightened together, the taper-lock bushing squeezes the shaft and creates a clamping force which prevents axial movement on the shaft.

Table 7-4 shows the available sprockets with a pitch of 8 mm. The pitch diameter and flange diameter are given for the corresponding sprocket number of teeth and are common for all belt widths. The 8-mm pitch sprocket is available for 20 mm, 30 mm, 50 mm, and 85-mm belt widths. The corresponding taper-lock bushing number is given for each sprocket and is based on the sprocket width. An example of a sprocket designation is

Sprocket Designation: P72-8MGT-50

The first set of numbers (in mm) represents the number of teeth, the second set represents the belt pitch with the style of tooth profile, and the last number is the belt width. Therefore this sprocket has 72 teeth, a 8-mm belt pitch with GT tooth profile, and a belt width of 50 mm. The pitch diameter of the 72-tooth sprocket is 7.218 in and the flange diameter is 7.598 in. Table 7-5 shows the available taper-lock bushings with their minimum and maximum bore diameters. The taper-lock bushing minimum and maximum bore diameters will limit the size of the shaft for a specific sprocket. Figure 7-25 shows a typical taper-lock bushing. The three half holes are aligned with the three half holes in the sprocket and are used to tighten the taper-lock bushing to the sprocket and remove the taper-lock bushing from the sprocket. The P72-8MGT-50 sprocket requires taper-lock bushing number 2517. The 2517 taper-lock bushing has a minimum bore diameter of 0.50 in and a maximum bore diameter of 2.25 in and would not fit any shaft diameter larger than 2.25 in.

Table 7-6 gives some overall data for the range of belt widths, number of teeth in pulleys, and belt lengths available for both metric and English belt pitches.

TABLE 7-4 Sprockets with 8 mm Belt Pitch

Dim's all widths			20-mm Wide belt		30-mm Wide belt		50-mm Wide belt		85-mm Wide belt	
No. of teeth	Pitch dia.	Flange dia.	Sprocket number	Bushing size	Sprocket number	Bushing size	Sprocket number	Bushing size	Sprocket number	Bushing size
22	2.206	2.559	P22-8MGT-20	1108	P22-8MGT-30	1108	N/A	N/A	N/A	N/A
24	2.406	2.756	P24-8MGT-20	1108	P24-8MGT-30	1108	N/A	N/A	N/A	N/A
26	2.607	2.953	P26-8MGT-20	1108	P26-8MGT-30	1108	N/A	N/A	N/A	N/A
28	2.807	3.15	P28-8MGT-20	1108	P28-8MGT-30	1108	P28-8MGT-50	MPB	N/A	N/A
30	3.008	3.346	P30-8MGT-20	1210	P30-8MGT-30	1210	P30-8MGT-50	1210	N/A	N/A
32	3.208	3.543	P32-8MGT-20	1210	P32-8MGT-30	1210	P32-8MGT-50	1210	N/A	N/A
34	3.409	3.819	P34-8MGT-20	1610	P34-8MGT-30	1610	P34-8MGT-50	1610	P34-8MGT-85	1615
36	3.609	3.937	P36-8MGT-20	1610	P36-8MGT-30	1610	P36-8MGT-50	1610	P36-8MGT-85	1615
38	3.810	4.134	P38-8MGT-20	1610	P38-8MGT-30	1610	P38-8MGT-50	1610	P38-8MGT-85	1610
40	4.010	4.331	P40-8MGT-20	1610	P40-8MGT-30	2012	P40-8MGT-50	2012	P40-8MGT-85	2012
44	4.411	4.764	P44-8MGT-20	2012	P44-8MGT-30	2012	P44-8MGT-50	2012	P44-8MGT-85	2012
48	4.812	5.157	P48-8MGT-20	2012	P48-8MGT-30	2012	P48-8MGT-50	2012	P48-8MGT-85	2012
56	5.614	5.945	P56-8MGT-20	2012	P56-8MGT-30	2012	P56-8MGT-50	2517	P56-8MGT-85	2517
64	6.416	6.772	P64-8MGT-20	2012	P64-8MGT-30	2517	P64-8MGT-50	2517	P64-8MGT-85	2517
72	7.218	7.598	P72-8MGT-20	2012	P72-8MGT-30	2517	P72-8MGT-50	2517	P72-8MGT-85	3020
80	8.020	8.386	P80-8MGT-20	2517	P80-8MGT-30	2517	P80-8MGT-50	2517	P80-8MGT-85	3020
90	9.023	N/A	P90-8MGT-20	2517	P90-8MGT-30	2517	P90-8MGT-50	3020	P90-8MGT-85	3020
112	11.229	N/A	N/A	N/A	P112-8MGT-30	2517	P112-8MGT-50	3020	P112-8MGT-85	3020
144	14.437	N/A	N/A	N/A	P144-8MGT-30	2517	P144-8MGT-50	3020	P144-8MGT-85	3535
192	19.249	N/A	N/A	N/A	N/A	N/A	P192-8MGT-50	3020	P192-8MGT-85	3535

TABLE 7-5 Taper-Lock Bushing

Bushing size	Min bore	Max bore
1008	0.500	0.875
1108	0.500	1.000
1210	0.500	1.250
1610	0.500	1.500
1615	0.500	1.500
2012	0.500	1.875
2517	0.500	2.250
3020	0.875	2.750
3525	1.188	3.250
3535	1.188	3.250
4030	1.438	3.625
4040	1.438	3.625
4535	1.938	4.250
4545	1.938	4.250
5040	2.438	4.500
6050	4.438	6.000
7060	4.938	7.000

Installation of sprockets and the belt requires a nominal amount of center distance adjustment to enable the belt teeth to slide into the sprocket grooves without force, as shown in Figure 7-26. Subsequently, the center distance will be adjusted inward to install the belt over the flanges of the sprockets and then be adjusted

**FIGURE 7-25** Taper-lock bushing

TABLE 7-6 Types and Sizes of Synchronous Belt Drives

Metric Sizes (Dimensions in mm)

Pitch	Belt widths (typical)	Range of number of teeth on sprockets (typical)	Range of belt pitch lengths	
			(mm)	No. of teeth
5	15, 25	32–112	350–2000	70–400
8	20, 30, 50, 85	22–192	480–4400	60–550
14	40, 55, 85, 115, 170	28–216	966–6860	69–490
20	115, 170, 230, 290, 340	34–216	2000–6600	100–330

U.S. Customary Sizes (Dimensions in inches)

Pitch	Belt widths (typical)	Range of number of teeth in sprockets	Range of belt pitch lengths	
			(in)	No. of teeth
XL-0.200	0.25, 0.375	10–72	5.00–77.00	25–385
L-0.375	0.50, 0.75, 1.00	10–120	12.38–90.00	33–240
H-0.500	0.75, 1.00, 1.50, 2.00, 3.00	14–156	21.00–170.0	42–340
XH-0.875	2.00, 3.00, 4.00, 5.00, 6.00, 8.00	18–120	50.75–175.0	58–200
XXH-1.250	2.00, 3.00, 4.00, 5.00, 6.00, 8.00	18–90		

Note: Check manufacturers' catalogs for available stock sizes.

outward to provide a suitable amount of initial tension as defined by the manufacturer. The initial tension is typically less than required for a V-belt drive since the synchronous belt is a positive drive.

In operation, the final tension in the tight side of the belt is much less than that developed by a V-Belt and the slack side tension is virtually zero. The results are lower net forces on the belt, lower side loads on the shafts carrying the sprockets, and reduced bearing loads.

The value for the belt perimeter length, L_p , can be calculated using the Equation (7-11) in Section (7-2) or by applying Equations (7-12) and (7-13) in Section (7-4). Similarly, for drives with two different diameters for the sprockets, the angle of belt wrap can be calculated using Equations (7-6) and (7-7) in Section (7-2) or Equations (7-14) and (7-15) in Section (7-4).

Various center distances, calculated based on belt lengths and the velocity ratios of the sprocket combinations, have been compiled in Table 7-7. This table

only shows a sample of the combinations available. See Internet sites 3, 5, 8, 14, or 16 to see the entire table for a given manufacturer.

General Selection Procedure for Synchronous Belt Drives

1. Specify the speed of the driving sprocket (typically a motor or engine) and the desired speed of the driven sprocket.
2. Specify the rated power for the driving motor or engine. The rated power of the motor or engine is based on the calculated power of the driven machine. For the belt drive problems we will be solving, the rated power will be a given value.
3. Determine a service factor (SF) using Table 7-8. The service factor is based on the type of driving motor, the nature of the driven machine, and the required hours of operation of the application.

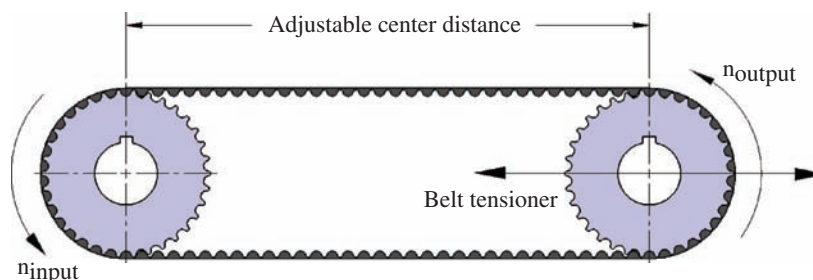


FIGURE 7-26 Belt drive with an adjustable center distance

TABLE 7-7 8-mm Pitch GT Drive Selection Table

Sprocket combinations		Center distance (inches)																
Driver	Driven	Velocity ratio	920-8MGT P.L. 36.220 115 teeth	960-8MGT P.L. 37.795 120 teeth	1040-8MGT P.L. 40.945 130 teeth	1064-8MGT P.L. 41.890 133 teeth	1120-8MGT P.L. 44.094 140 teeth	1160-8MGT P.L. 45.669 145 teeth	1200-8MGT P.L. 47.244 150 teeth	1224-8MGT P.L. 48.189 153 teeth	1280-8MGT P.L. 50.394 160 teeth	1440-8MGT P.L. 56.693 180 teeth	1512-8MGT P.L. 59.528 189 teeth	1584-8MGT P.L. 62.362 198 teeth	1600-8MGT P.L. 62.992 200 teeth	1760-8MGT P.L. 69.291 220 teeth	1800-8MGT P.L. 70.866 225 teeth	2000-8MGT P.L. 78.740 250 teeth
22	22	1.000	14.65	15.43	17.01	17.48	18.58	19.37	20.16	20.63	21.73	24.88	26.30	27.72	28.03	31.18	31.97	35.90
24	24	1.000	14.33	15.12	16.69	17.17	18.27	19.06	19.84	20.32	21.42	24.57	25.98	27.40	27.72	30.87	31.65	35.59
26	26	1.000	14.02	14.80	16.38	16.85	17.95	18.74	19.53	20.00	21.10	24.25	25.67	27.09	27.40	30.55	31.34	35.28
28	28	1.000	13.70	14.49	16.06	16.54	17.64	18.43	19.21	19.69	20.79	23.94	25.35	26.77	27.09	30.24	31.02	34.96
30	30	1.000	13.39	14.17	15.75	16.22	17.32	18.11	18.90	19.37	20.47	23.62	25.04	26.46	26.77	29.92	30.71	34.65
32	32	1.000	13.07	13.86	15.43	15.91	17.01	17.80	18.58	19.06	20.16	23.31	24.72	26.14	26.46	29.61	30.39	34.33
34	34	1.000	12.76	13.54	15.12	15.59	16.69	17.48	18.27	18.74	19.84	22.99	24.41	25.83	26.14	29.29	30.08	34.02
36	36	1.000	12.44	13.23	14.80	15.28	16.38	17.17	17.95	18.43	19.53	22.68	24.09	25.51	25.83	28.98	29.76	33.70
38	38	1.000	12.13	12.91	14.49	14.96	16.06	16.85	17.64	18.11	19.21	22.36	23.78	25.20	25.51	28.66	29.45	33.39
40	40	1.000	11.67	12.46	14.03	14.50	15.61	16.39	17.18	17.65	18.76	21.91	23.32	24.74	25.06	28.21	28.99	32.93
44	44	1.000	11.18	11.97	13.54	14.02	15.12	15.91	16.69	17.17	18.27	21.42	22.83	24.25	24.57	27.72	28.50	32.44
48	48	1.000	10.55	11.34	12.91	13.39	14.49	15.28	16.06	16.54	17.64	20.79	22.21	23.62	23.94	27.09	27.87	31.81
56	56	1.000	9.29	10.08	11.65	12.13	13.23	14.02	14.80	15.28	16.38	19.53	20.95	22.36	22.68	25.83	26.61	30.55
64	64	1.000	8.03	8.82	10.39	10.87	11.97	12.76	13.54	14.02	15.12	18.27	19.69	21.10	21.42	24.57	25.35	29.29
72	72	1.000	-	-	9.13	9.61	10.71	11.50	12.28	12.76	13.86	17.01	18.43	19.84	20.16	23.31	24.10	28.03
80	80	1.000	-	-	-	-	9.45	10.24	11.02	11.50	12.60	15.75	17.17	18.58	18.90	22.05	22.84	26.77
24	30	1.250	13.85	14.64	16.22	16.69	17.79	18.58	19.37	19.84	20.94	24.09	25.51	26.93	27.24	30.39	31.18	35.12
32	40	1.250	12.43	13.22	14.80	15.27	16.37	17.16	17.95	18.42	19.52	22.67	24.09	25.51	25.82	28.97	29.76	33.70
64	80	1.250	-	-	9.10	9.57	10.68	11.47	12.26	12.73	13.84	16.99	18.41	19.83	20.14	23.29	24.08	28.02
72	90	1.250	-	-	-	-	9.25	10.04	10.83	11.30	12.41	15.56	16.98	18.40	18.72	21.87	22.66	26.60
24	32	1.333	13.70	14.48	16.06	16.53	17.63	18.42	19.21	19.68	20.78	23.93	25.35	26.77	27.08	30.23	31.02	34.96
30	40	1.333	12.59	13.38	14.95	15.42	16.53	17.32	18.10	18.58	19.68	22.83	24.25	25.66	25.98	29.13	29.92	33.85

(continued)

TABLE 7-7 (continued)

Sprocket combinations			Center distance (inches)																		
			920-8MGT P.L. 36.220 115 teeth	960-8MGT P.L. 37.795 120 teeth	1040-8MGT P.L. 40.945 130 teeth	1064-8MGT P.L. 41.890 133 teeth	1120-8MGT P.L. 44.094 140 teeth	1160-8MGT P.L. 45.669 145 teeth	1200-8MGT P.L. 47.244 150 teeth	1224-8MGT P.L. 48.189 153 teeth	1280-8MGT P.L. 50.394 160 teeth	1440-8MGT P.L. 56.693 180 teeth	1512-8MGT P.L. 59.528 189 teeth	1584-8MGT P.L. 62.362 198 teeth	1600-8MGT P.L. 62.992 200 teeth	1760-8MGT P.L. 69.291 220 teeth	1800-8MGT P.L. 70.866 225 teeth	2000-8MGT P.L. 78.740 250 teeth			
Driver	Driven	Velocity ratio	36	48	1.333	11.48	12.27	13.85	14.32	15.42	16.21	17.00	17.47	18.57	21.72	23.14	24.56	24.87	28.03	28.81	32.75
48	64	1.333	9.26	10.05	11.63	12.10	13.20	13.99	14.78	15.25	16.36	19.51	20.93	22.35	22.66	25.81	26.60	30.54			
24	36	1.500	13.36	14.15	15.73	16.20	17.30	18.09	18.88	19.35	20.46	23.61	25.02	26.44	26.76	29.91	30.70	34.63			
32	48	1.500	11.78	12.57	14.15	14.62	15.73	16.52	17.30	17.78	18.88	22.03	23.45	24.87	25.18	28.34	29.12	33.06			
48	72	1.500	8.58	9.37	10.96	11.43	12.54	13.33	14.12	14.60	15.70	18.86	20.28	21.70	22.01	25.17	25.96	29.90			
22	44	2.000	12.87	13.66	15.24	15.71	16.81	17.60	18.39	18.87	19.97	23.12	24.54	25.96	26.28	29.43	30.22	34.16			
24	48	2.000	12.38	13.17	14.75	15.23	16.33	17.12	17.91	18.39	19.49	22.65	24.06	25.48	25.80	28.95	29.74	33.68			
28	56	2.000	11.41	12.20	13.79	14.26	15.37	16.16	16.95	17.42	18.53	21.69	23.11	24.53	24.84	28.00	28.79	32.73			
32	64	2.000	10.43	11.22	12.81	13.29	14.40	15.19	15.98	16.46	17.56	20.73	22.15	23.57	23.88	27.04	27.83	31.77			
36	72	2.000	9.43	10.24	11.83	12.31	13.42	14.22	15.01	15.49	16.60	19.76	21.18	22.61	22.92	26.08	26.87	30.81			
40	80	2.000	8.42	9.23	10.84	11.32	12.44	13.23	14.03	14.51	15.62	18.79	20.22	21.64	21.96	25.12	25.91	29.85			
56	112	2.000	-	-	-	-	-	9.18	10.00	10.49	11.63	14.85	16.29	17.73	18.05	21.23	22.03	25.99			
72	144	2.000	-	-	-	-	-	-	-	-	-	-	12.22	13.70	14.02	17.26	18.06	22.07			
32	80	2.500	8.97	9.78	11.40	11.88	13.01	13.81	14.61	15.08	16.20	19.38	20.81	22.23	22.55	25.71	26.51	30.46			
36	90	2.500	7.71	8.55	10.19	10.68	11.82	12.62	13.43	13.91	15.03	18.22	19.66	21.09	21.40	24.58	25.37	29.32			
24	72	3.000	10.27	11.08	12.69	13.17	14.29	15.08	15.88	16.36	17.47	20.65	22.07	23.50	23.82	26.98	27.77	31.72			
30	90	3.000	8.10	8.94	10.60	11.09	12.23	13.04	13.85	14.33	15.46	18.65	20.09	21.52	21.84	25.02	25.81	29.77			
48	144	3.000	-	-	-	-	-	-	-	-	-	12.29	13.81	15.31	15.64	18.92	19.73	23.76			

TABLE 7-8 Service Factor

DriveN machine		DriveR				
The driveN machines listed below are representative samples only. Select a driveN machine whose load characteristics most closely approximate those of the machine being considered.	AC Motors: Normal Torque, Squirrel Cage, Synchronous, Split Phase, Inverter Controlled	AC Motors: High Torque, High Slip, Repulsion-Induction, Single Phase, Series Wound, Slip Ring				
	DC Motors: Shunt Wound Stepper Motors	DC Motors: Series Wound, Compound Wound Servo Motors				
	Engines: Multiple Cylinder Internal Combustion	Engines: Single Cylinder Internal Combustion, Line Shafts, Clutches				
	Intermittent Service (Up to 8 Hours Daily or Seasonal)	Normal Service (8–16 Hours Daily)	Continuous Service (16–24 Hours Daily)	Intermittent Service (Up to 8 Hours Daily or Seasonal)	Normal Service (8–16 Hours Daily)	Continuous Service (16–24 Hours Daily)
Display, Dispensing Equipment Instrumentation Measuring Equipment Medical Equipment Office, Projection Equipment	1.0	1.2	1.4	1.2	1.4	1.6
Appliances, Sweepers, Sewing Machines Screens, Oven Screens, Drum, Conical Woodworking Equipment (Light): Band Saws, Drills Lathes	1.1	1.3	1.5	1.3	1.5	1.7
Agitators for Liquids Conveyors: Belt, Light Package Drill Press, Lathes, Saws Laundry Machinery Wood Working Equipment (Heavy): Circular Saws, Jointers, Planers	1.2	1.4	1.6	1.6	1.8	2.0
Agitators for Semi-Liquids Compressor: Centrifugal Conveyor Belt: Ore, Coal, Sand Dough Mixers Line Shafts Machine Tools: Grinder, Shaper, Boring Mill, Milling Machines Paper Machinery (except Pulpers): Presses, Punches, Shears Printing Machinery Pumps: Centrifugal, Gear Screens: Revolving, Vibratory	1.3	1.5	1.7	1.6	1.8	2.0
Brick Machinery (except Pug Mills) Conveyor: Apron, Pan, Bucket, Elevator Extractors, Washers Fans, Centrifugal Blowers Generators & Exciters Hoists Rubber Calender, Mills, Extruders	1.4	1.6	1.8	1.8	2.0	2.2
Centrifuges Screw Conveyors Hammer Mills Paper Pulpers Textile Machinery	1.5	1.7	1.9	1.9	2.1	2.3
Blowers: Positive Displacement, Mine Fans Pulvertzers	1.6	1.8	2.0	2.0	2.2	2.4
Compressors: Reciprocating Crushers: Gyratory, Jaw, Rol Mills: Ball, Rod, Pebble, etc. Pumps: Reciprocating Saw Mill Equipment	1.7	1.9	2.1	2.1	2.3	2.5

These service factors are adequate for most belt drive applications. Note that service factors cannot be substituted for good engineering judgment. Service factors may be adjusted based upon an understanding of the severity of actual drive operating conditions.

- Calculate the design power by multiplying the driver rated power by the service factor.

$$\text{Design power} = P_{des} = P_{rated} \cdot SF$$

- Determine the required pitch of the belt using the belt pitch selection guide Figure (7-27). The belt pitch is based on the design power and the angular velocity of the faster (smaller) sprocket. The belt pitches available are 5 mm, 8 mm, 14 mm, and 20 mm. The design horsepower is along the x-axis and the rpm of the faster sprocket is along the y-axis. As the design power increases or the smaller sprocket angular velocity decreases, a larger belt pitch would be required. The 14-mm belt pitch is selected for the design power and angular velocity in its shaded area, but would work for any application to the left of its shaded area. This means that the 14-mm belt pitch would work for a point located in the 5-mm and 8-mm belt pitch areas, but would be considered over-designed and not an economical design choice.
- Calculate the velocity ratio VR between the driver and driven sprockets. Review Section 7-2 for this equation.

$$VR = \frac{\omega_{driving}}{\omega_{driven}} = \frac{PD_{driven}}{PD_{driving}} = \frac{N_{driven}}{N_{driving}}$$

- Select the candidate combinations using Table 7-7 of the number of teeth in the driver sprocket to that in the driven sprocket that will produce the calculated velocity ratio, VR.
- Eliminate the sprocket combinations that will not work due to space limitations and shaft diameter requirements. Some of the larger sprockets may interfere with the machine or guarding and can be eliminated due to these space limitations. The shaft diameter will dictate the minimum taper-lock bushing (Figure 7-25) that will fit on the shaft. Once the taper-lock bushing is known the minimum sprocket can be determined. This will eliminate any sprockets smaller than this minimum sprocket.
- Using the desired range of acceptable center distances, determine a standard belt length that will produce a suitable value. Table 7-7 shows that the center distance is determined by the belt length and sprocket velocity ratio. The available belt lengths are determined by the manufacturer. The belt center distance selection is influenced by the belt drive design center distance. A fixed or adjustable center distance design should be considered when selecting the proper belt length. A belt drive design that has an

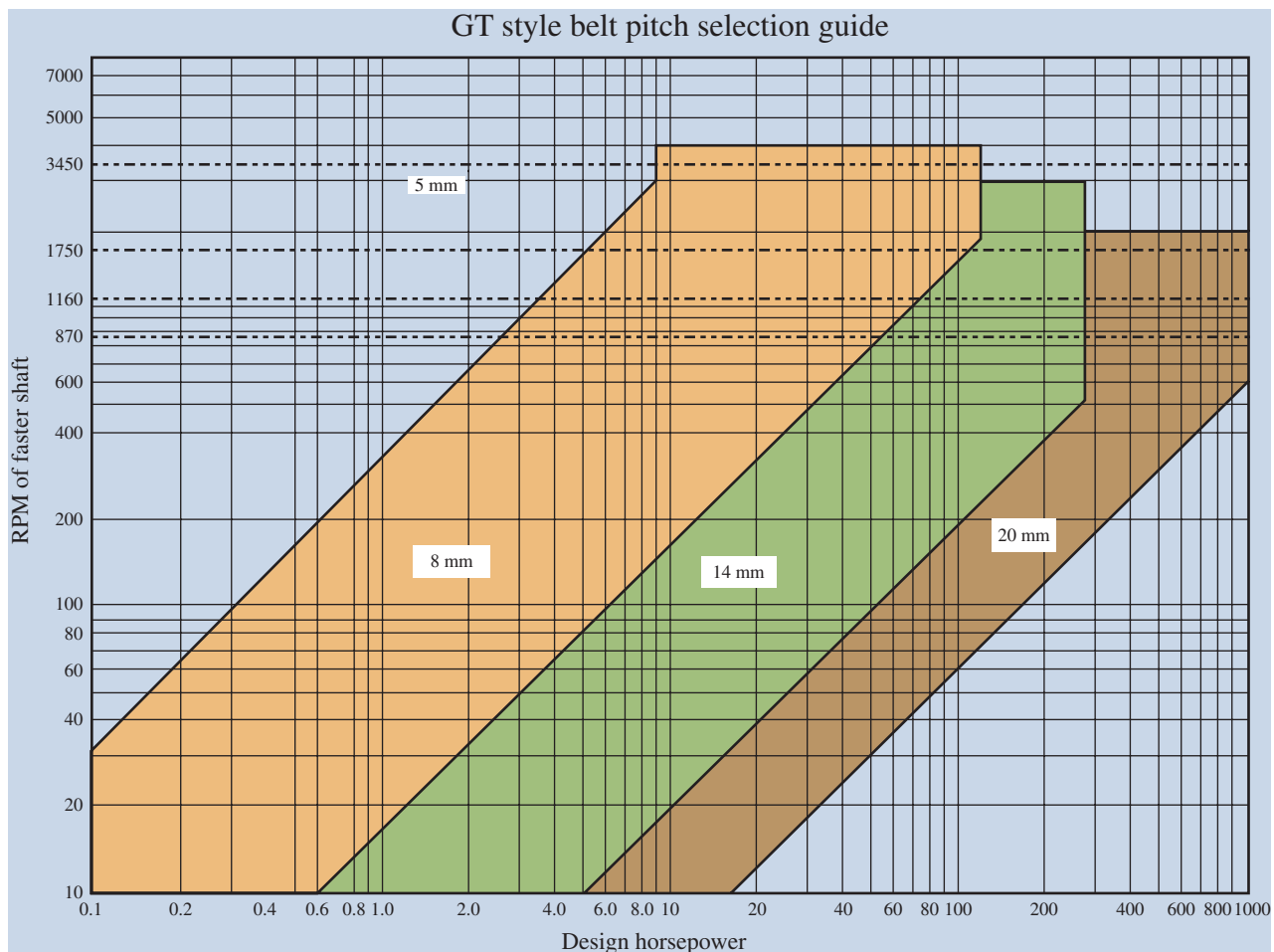


FIGURE 7-27 Belt pitch selection guide for GT style belts

adjustable center distance (Figure 7–26) will require the belt center distance to be within this range. If the belt drive design has a fixed center distance, the belt center distance must be larger than the fixed center distance. This belt drive system will require the use of a tensioner (Figure 7–30, discussed later) to take up the difference in belt lengths. An inside or outside tensioner will be selected, depending on how much the belt center distance exceeds the fixed center distance. This will require a drive belt layout to determine the best available solution.

10. Selection of the width of the belt: Although there are four belt pitches available (5 mm, 8 mm, 14 mm, and 20 mm), we will focus on the 8-mm belt

pitch. An 8-mm pitch belt is available in four different widths: 20 mm, 30 mm, 50 mm, and 85 mm. The belt width selection Tables 7–9 and 7–10 are shown for the 30-mm and 50-mm wide belts. The 20-mm and 85-mm wide belt tables can be found in the manufacturer’s website. The angular velocity of the faster (smaller) sprocket along with the number of teeth of this smaller sprocket is used to find the base rated horsepower. Let’s first look at the 30 mm belt width table. You will notice for a given sprocket size as the speed increases, the power rating of the belt increases. For a given speed of the sprocket, the belt power rating will increase as the size of the sprocket (or number of teeth) increases. The 50 mm

TABLE 7–9 8M GT Style Belt Power Rating Table—30-mm Belt Width

RPM of faster shaft	Base rated horsepower for small sprocket (Number of grooves and pitch diameter, inches)															
	22 2.206	24 2.406	26 2.607	28 2.807	30 3.008	32 3.208	34 3.409	36 3.609	38 3.810	40 4.010	44 4.411	48 4.812	56 5.614	64 6.416	72 7.218	80 8.020
10	0.10	0.12	0.13	0.15	0.16	0.17	0.19	0.20	0.22	0.23	0.26	0.29	0.34	0.40	0.45	0.51
20	0.20	0.22	0.25	0.28	0.31	0.33	0.36	0.39	0.42	0.44	0.50	0.55	0.66	0.76	0.87	0.98
40	0.37	0.43	0.48	0.53	0.59	0.64	0.69	0.75	0.80	0.85	0.96	1.06	1.27	1.47	1.68	1.88
60	0.54	0.62	0.70	0.78	0.86	0.94	1.01	1.09	1.17	1.25	1.40	1.55	1.86	2.16	2.46	2.76
100	0.87	1.00	1.12	1.25	1.38	1.51	1.63	1.76	1.89	2.01	2.26	2.51	3.00	3.49	3.98	4.47
200	1.64	1.89	2.13	2.38	2.63	2.87	3.12	3.36	3.60	3.84	4.33	4.80	5.76	6.70	7.64	8.58
300	2.37	2.74	3.10	3.46	3.82	4.18	4.54	4.90	5.25	5.61	6.32	7.02	8.42	9.80	11.2	12.5
400	3.08	3.56	4.04	4.51	4.99	5.46	5.93	6.40	6.87	7.33	8.26	9.18	11.0	12.8	14.6	16.4
500	3.77	4.36	4.95	5.54	6.13	6.71	7.29	7.87	8.45	9.02	10.2	11.3	13.6	15.8	18.0	20.2
600	4.45	5.15	5.85	6.55	7.25	7.94	8.63	9.31	10.0	10.7	12.0	13.4	16.1	18.7	21.4	24.0
700	5.11	5.93	6.74	7.54	8.35	9.15	9.95	10.7	11.5	12.3	13.9	15.5	18.6	21.6	24.7	27.7
800	5.77	6.69	7.61	8.52	9.44	10.3	11.2	12.1	13.0	13.9	15.7	17.5	21.0	24.5	27.9	31.4
870	6.22	7.22	8.22	9.20	10.2	11.2	12.2	13.1	14.1	15.1	17.0	18.9	22.7	26.5	30.2	33.9
1000	7.05	8.19	9.33	10.5	11.6	12.7	13.8	14.9	16.0	17.1	19.3	21.5	25.8	30.1	34.3	38.5
1160	8.06	9.37	10.7	12.0	13.3	14.5	15.8	17.1	18.4	19.6	22.2	24.7	29.6	34.5	39.4	44.2
1200	8.31	9.66	11.0	12.3	13.7	15.0	16.3	17.6	19.0	20.3	22.9	25.4	30.6	35.6	40.6	45.6
1400	9.54	11.1	12.7	14.2	15.7	17.3	18.8	20.3	21.8	23.3	26.3	29.3	35.2	41.0	46.8	52.4
1600	10.7	12.5	14.3	16.0	17.8	19.5	21.2	23.0	24.7	26.4	29.8	33.1	39.8	46.3	52.8	59.1
1750	11.6	13.6	15.5	17.4	19.3	21.2	23.0	24.9	26.8	28.6	32.3	36.0	43.2	50.3	57.2	64.1
2000	13.1	15.3	17.5	19.6	21.8	23.9	26.0	28.1	30.2	32.3	36.5	40.6	48.7	56.7	64.5	72.1
2400	15.4	18.0	20.5	23.1	25.6	28.1	30.7	33.1	35.6	38.1	43.0	47.8	57.3	66.6	75.6	84.4
2800	17.6	20.6	23.6	26.5	29.4	32.3	35.2	38.0	40.9	43.7	49.3	54.8	65.6	76.1	86.2	96.0
3200	19.8	23.2	26.5	29.8	33.1	36.4	39.6	42.8	46.0	49.2	55.4	61.6	73.6	85.2	96.2	
3450	21.1	24.7	28.3	31.9	35.4	38.9	42.3	45.8	49.2	52.5	59.2	65.7	78.4	90.6	102.2	
4000	24.0	28.1	32.2	36.2	40.3	44.2	48.1	52.0	55.9	59.7	67.1	74.5	88.5			
4500	26.6	31.1	35.6	40.1	44.5	48.9	53.2	57.5	61.7	65.9	74.0	82.0				
5000	29.0	34.0	39.0	43.8	48.7	53.4	58.1	62.8	67.3	71.8	80.6	89.1				
5500	31.4	36.8	42.2	47.5	52.7	57.8	62.9	67.8	72.7	77.5	86.8					

TABLE 7-10 8M GT Style Belt Power Rating Table—50-mm Belt Width

RPM of faster shaft	Base rated horsepower for small sprocket (Number of grooves and pitch diameter, inches)												
	28 2.807	30 3.008	32 3.208	34 3.409	36 3.609	38 3.810	40 4.010	44 4.411	48 4.812	56 5.614	64 6.416	72 7.218	80 8.020
10	0.25	0.28	0.30	0.33	0.35	0.38	0.40	0.45	0.50	0.59	0.69	0.78	0.88
20	0.49	0.53	0.58	0.63	0.68	0.72	0.77	0.86	0.96	1.14	1.33	1.51	1.70
40	0.93	1.02	1.11	1.21	1.30	1.39	1.48	1.66	1.84	2.20	2.56	2.92	3.27
60	1.35	1.49	1.63	1.76	1.90	2.03	2.17	2.43	2.70	3.23	3.75	4.28	4.80
100	2.18	2.40	2.62	2.84	3.06	3.28	3.50	3.93	4.36	5.22	6.08	6.92	7.77
200	4.14	4.57	4.99	5.42	5.84	6.26	6.68	7.52	8.35	10.0	11.7	13.3	14.9
300	6.02	6.65	7.27	7.90	8.52	9.14	9.75	11.0	12.2	14.6	17.0	19.4	21.8
400	7.85	8.67	9.49	10.3	11.1	11.9	12.7	14.4	16.0	19.2	22.3	25.5	28.6
500	9.63	10.7	11.7	12.7	13.7	14.7	15.7	17.7	19.7	23.6	27.5	31.4	35.2
600	11.4	12.6	13.8	15.0	16.2	17.4	18.6	20.9	23.3	28.0	32.6	37.2	41.7
700	13.1	14.5	15.9	17.3	18.7	20.1	21.4	24.2	26.9	32.3	37.6	42.9	48.2
800	14.8	16.4	18.0	19.6	21.1	22.7	24.2	27.3	30.4	36.5	42.6	48.6	54.5
870	16.0	17.7	19.4	21.1	22.8	24.5	26.2	29.5	32.9	39.5	46.0	52.5	58.9
1000	18.2	20.1	22.1	24.0	25.9	27.9	29.8	33.6	37.4	44.9	52.4	59.7	67.0
1160	20.8	23.1	25.3	27.5	29.7	32.0	34.1	38.5	42.9	51.5	60.0	68.5	76.8
1200	21.5	23.8	26.1	28.4	30.7	33.0	35.2	39.8	44.2	53.1	61.9	70.6	79.2
1400	24.7	27.4	30.0	32.7	35.3	38.0	40.6	45.8	51.0	61.2	71.3	81.3	91.2
1600	27.9	30.9	33.9	36.9	39.9	42.9	45.9	51.8	57.6	69.2	80.6	91.8	102.9
1750	30.2	33.5	36.8	40.1	43.3	46.6	49.8	56.2	62.5	75.0	87.4	99.5	111.4
2000	34.1	37.8	41.5	45.2	48.9	52.6	56.2	63.4	70.6	84.7	98.5	112.1	125.4
2400	40.2	44.6	48.9	53.3	57.6	62.0	66.2	74.7	83.1	99.7	115.8	131.5	146.8
2800	46.1	51.2	56.2	61.2	66.2	71.1	76.0	85.7	95.3	114.1	132.3	149.9	166.9
3200	51.9	57.6	63.2	68.9	74.5	80.0	85.5	96.4	107.1	128.0	148.1	167.4	
3450	55.4	61.5	67.6	73.6	79.6	85.5	91.3	102.9	114.3	136.4	157.5	177.7	
4000	63.0	70.0	76.9	83.7	90.4	97.1	103.7	116.8	129.5	154.0			
4500	69.7	77.4	85.0	92.6	100.0	107.3	114.5	128.7	142.5				
5000	76.2	84.7	92.9	101.1	109.1	117.1	124.9	140.1	154.9				
5500	82.5	91.6	100.5	109.3	117.9	126.4	134.7	150.9					

belt width will have a higher power rating than the 30 mm belt width. A larger sprocket will decrease the belt width required and yield a longer service life. The belt width should not exceed the sprocket diameter. This base rated horsepower must be adjusted by the belt length correction factor shown in Table 7-11. Catalog data will show factors less than 1.0 for shorter belt lengths and greater than 1.0 for longer belt lengths. This reflects the frequency with which a given tooth of the belt encounters a high-stress area as it enters the smaller sprocket.

Base Rated Power_{adjusted} =

Base Rated Power • *Length Correction Factor*

11. Calculate the belt linear velocity. Belt speeds above 3500 fpm increase the noise level of the synchronous belt drive. Also verify that the belt linear velocity does not exceed 6500 fpm, due to the excessive centrifugal forces that are placed on a sprocket.

$$v_{belt} = \frac{PD_1}{2} \cdot \omega_1 = \frac{PD_2}{2} \cdot \omega_2$$

12. Specify the final design details for the belt drive system. This includes all sprockets, type and bore size of taper-lock bushings, belt, and tensioner if required. Summarize the design, check compatibility with other components of the system, and prepare the purchasing documents.

TABLE 7-11 8M GT Style Belt Length Correction Factor

Pitch/Length designation	No. of teeth	Correction factor	Pitch/Length designation	No. of teeth	Correction factor	Pitch/Length designation	No. of teeth	Correction factor	Pitch/Length designation	No. of teeth	Correction factor
384-8MGT	48	0.70	920-8MGT	115	1.00	1440-8MGT	180	1.10	2600-8MGT	325	1.20
480-8MGT	60	0.80	960-8MGT	120	1.00	1512-8MGT	189	1.10	2800-8MGT	350	1.20
560-8MGT	70	0.80	1040-8MGT	130	1.00	1584-8MGT	198	1.10	3048-8MGT	381	1.20
600-8MGT	75	0.80	1064-8MGT	133	1.00	1600-8MGT	200	1.10	3280-8MGT	410	1.20
640-8MGT	80	0.90	1120-8MGT	140	1.00	1760-8MGT	220	1.10	3600-8MGT	450	1.20
720-8MGT	90	0.90	1160-8MGT	145	1.00	1800-8MGT	225	1.20	4400-8MGT	550	1.20
800-8MGT	100	0.90	1200-8MGT	150	1.00	2000-8MGT	250	1.20			
840-8MGT	105	0.90	1224-8MGT	153	1.00	2200-8MGT	275	1.20			
880-8MGT	110	0.90	1280-8MGT	160	1.10	2400-8MGT	300	1.20			

Example Problem 7-3

For the belt drive layout shown in Figure 7-28 perform the following calculations for the kinematics of the drive:

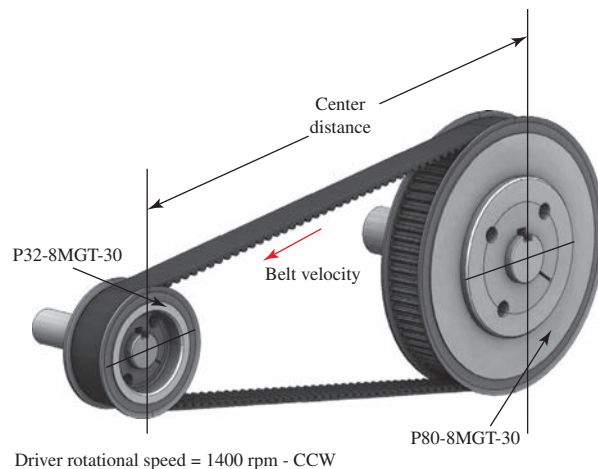


FIGURE 7-28 Synchronous belt drive for Example Problem 7-3

- Specify the number of teeth and the pitch diameter of each sheave.
- Calculate the velocity ratio of the belt drive and the angular velocity of the driven sheave.
- Calculate the belt linear velocity.
- Determine the center distance using a 1200-8MGT-30 belt.

Solution

Given Data in Figure 7-28. Input sprocket speed = 1400 rpm

- a. Pitch diameters: From Table 7-4,

P32-8MGT-30 sprocket: 32 teeth; Pitch diameter = 3.208 in
 P80-8MGT-30 sprocket: 80 teeth; Pitch diameter = 8.020 in

- b. Velocity ratio: Using Equation (7-4) in Section 7-2,

$$VR = \frac{n_{driving}}{n_{driven}} = \frac{N_{driven}}{N_{driving}} = \frac{80}{32} = 2.50$$

Driven speed of the driven sprocket:

$$n_{driven} = \frac{n_{driving}}{VR} = \frac{1400 \text{ rpm}}{2.5} = 560 \text{ rpm}$$

- c. Belt speed: Using Equation (7-3) in Section 7-2,

$$v_{belt} = \frac{PD_{driving}}{2} n_{driving} = \frac{3.208 \text{ in}}{2} \cdot 1400 \frac{\text{rev}}{\text{min}} \cdot \frac{2\pi \text{ rad}}{\text{rev}} \cdot \frac{1.0 \text{ ft}}{12 \text{ in}} = 1175.8 \text{ ft/min}$$

- d. Center distance: From Table 7-7, using 32 and 80 teeth and 1200-8MGT-30 belt,

$$CD = 14.61 \text{ in}$$

Example Problem 7-4

Figure 7-29 shows a 20-hp electric motor driving a gear pump using a synchronous belt drive. The normal torque motor speed is 1750 rpm and the gear pump speed is to be 875 rpm. The drive system is on a steel mounting plate with slots permitting an adjustment of the pump position by ± 0.375 in from the nominal center distance of 22.00 in. The motor shaft diameter is 1.625 in and the gear pump shaft diameter is 1.375 in. The sprocket on the pump shaft cannot exceed 14.00 in due to potential interference with the base plate. The pump will operate 16–24 h/day, 5 days per week. Specify the complete drive system.

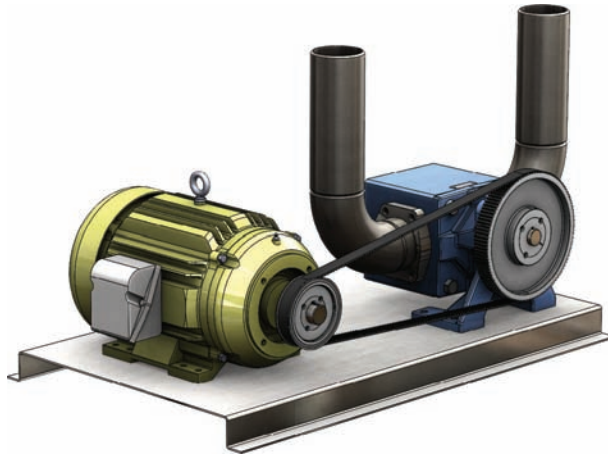


FIGURE 7-29 Gear pump drive system for Example Problem 7-4

Solution Use the General Selection Procedure for Synchronous Belt Drives

Given Motor: 20 hp; 1750 rpm; Normal torque; shaft diameter = 1.625 in
 Gear pump: 875 rpm; shaft diameter = 1.375 in; Maximum OD = 14.00 in
 Center distance between motor and pump shafts = 22.00 in ± 0.375 in; The range is 21.625 in minimum to 22.375 in maximum
 Operating conditions: 16 to 24 h/day, 5 days per week.
 Specify the synchronous belt drive system including driving and driven sprockets, bushings, and the belt.

- Results
1. Motor speed = 1750 rpm; Pump speed = 875 rpm
 2. Rated power = 20 hp
 3. Service factor: From Table 7-8, $SF = 1.7$
 4. Design power: $P_{des} = P_{rated} \cdot SF = 20 \text{ hp} \cdot 1.7 = 34 \text{ hp}$
 5. Required pitch for the belt: Using Figure 7-27, the intersection of 34 hp and 1750 rpm of the driver falls in the range of the 8 mm, 8MGT belt system.
 6. Velocity ratio:

$$VR = \frac{n_{driving}}{n_{driven}} = \frac{1750 \text{ rpm}}{875 \text{ rpm}} = 2.00$$

7. Select candidate combinations of driver and driven sprockets: Table 7-7 lists eight possible combinations having $VR = 2.00$. They are

22:44, 24:48, 28:56, 32:64, 36:72, 40:80, 56:112, 72:144

8. Eliminate the sprocket combinations that are not acceptable because of space limitations and shaft diameter requirements:

For shaft diameters, the bushing style and the maximum bore sizes must be checked using data in Tables 7-4 and 7-5.

First check the motor shaft size of 1.625 in.

Sprocket	Bushing no.	Max. Bore (in)
22	1108	1.00
24	1108	1.00
28	1108	1.00
32	1210	1.25
36	1610	1.50
40	1610	1.50
56	2012	1.875
72	2012	1.875

Not acceptable; too small

Acceptable

Now check the size of the pump sprockets; Maximum OD = 14.00 in. From Table 7-4, all pump sprockets up through 112 teeth have a flange size less than 14.00 in.

For 144 teeth, OD = 14.437 in—Too large.

Therefore, the only combination that satisfies all limitations is 56:112.

For this design: $PD_1 = 5.614$ in, $PD_2 = 11.229$ in

9. Now check the belt length and center distance for this combination:

From Table 7-7, we find that the 1800-8MGT belt will have a center distance of 22.03 in for a 56-tooth driving sprocket and a 112-tooth driven sprocket. That is within the required range. The 0.75-in long slots in the base plate will permit the installation of the belt and then setting the proper belt tension.

10. Select the belt width: Data in Tables 7-9 to 7-11 can be used. Table 7-9 is for a belt width of 30 mm and for a motor speed of 1750 rpm and 56 grooves in that sprocket, we find the rated power is 43.2 hp. To this value, a belt length correction factor must be applied, found in Table 7-11: For the 1800-8MGT belt, $C_L = 1.20$. Then the adjusted rated power is

$$P_{adjusted} = P_{rated} \cdot C_L = 43.2 \text{ hp} \cdot 1.20 = 51.8 \text{ hp}$$

This value is well above the design power of 34.0 hp found in Step 4.

11. Calculate the belt speed to ensure that it does not exceed 6500 rpm.

$$V_{belt} = \frac{PD_1}{2} \cdot \omega_1 = \frac{5.614 \text{ in}}{2} \cdot \frac{1750 \text{ rev}}{\text{min}} \cdot \frac{2\pi \text{ rad}}{\text{rev}} \cdot \frac{1 \text{ ft}}{12 \text{ in}} = 2572 \text{ ft/min} = 2572 \text{ fpm}$$

This is an acceptable belt speed.

12. Specify the final design details: [Quantity (1) each]

P56-8MGT-30 Sprocket—Motor shaft

2012-1.625 in diameter bore—Taper-lock bushing

P112-8MGT-30 Sprocket—Pump shaft

2517-1.375 in diameter bore—Taper-lock bushing

1800-8MGT-30 Belt

Alternate Configurations for Synchronous Belt Drives

Idlers and belt tensioners are used to set the correct belt length and take up belt slack if fixed centers are required between the driver and driven sprockets. Idlers do not directly drive any component and are

fixed in the belt drive system. A tensioner is an idler that is adjustable to provide the correct belt tension. The location of a tensioner should be on the slack side of the belt span. The tensioner can be located on either the inside or outside of the slack side belt span shown in Figure 7-30. Tensioners located on the inside of the

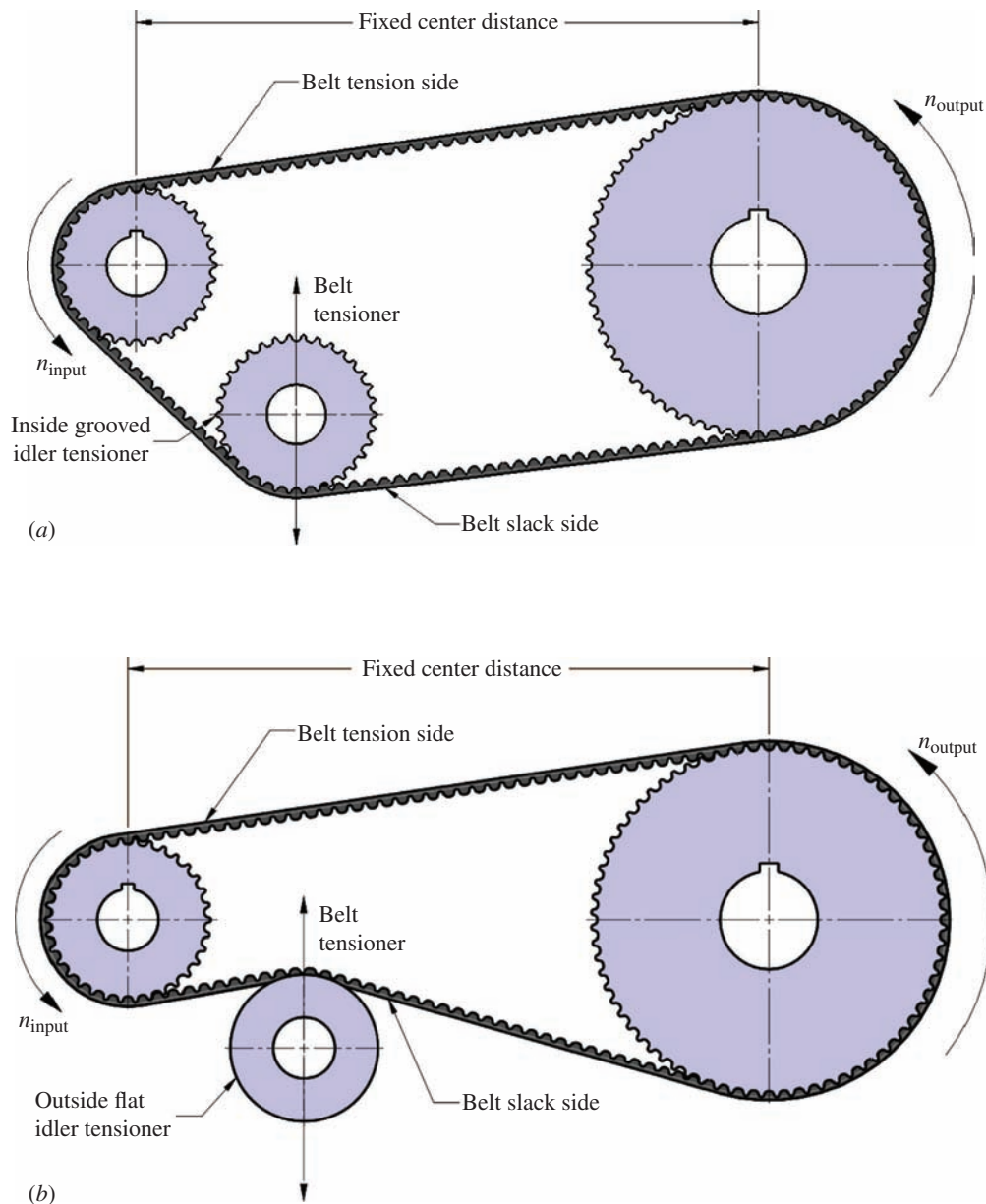


FIGURE 7-30 Belt drives with fixed center distances and added tensioners. (a) Inside grooved-idler tensioner. (b) Outside flat-idler tensioner

belt should use a grooved sprocket and a flat pulley should be used if the idler is located on the outside of the belt. The tensioner may decrease the life of the belt and the belt manufacturer should be consulted.

Belt drives can be used to transmit motion and power reliably and efficiently in a variety of configurations. Figure 7-31 shows two different multiple shaft belt drive configurations. Figure 7-31(a) shows a belt drive that has four sprockets that are the same size. The input sprocket (1) is driving two output sprockets (2 and 3) and an inside tensioner (4) is used to set the length and proper tension of the belt. The sprockets

are all rotating in the same direction at the same speed. Figure 7-31(b) shows a belt drive that has an input sprocket (1) that drives two output sprockets (2) and (3). The belt drive also has a flat pulley used as an outside belt tensioner (4). All sprockets are rotating in the same direction. The larger sprocket (2) is rotating slower than the input sprocket. Output sprocket (3) is the same size as the input sprocket and will rotate at the same speed.

Twin power belts shown in Figure 7-32 have teeth on both sides of the belt to provide a positive drive from either side of the belt. Serpentine belt drives allow

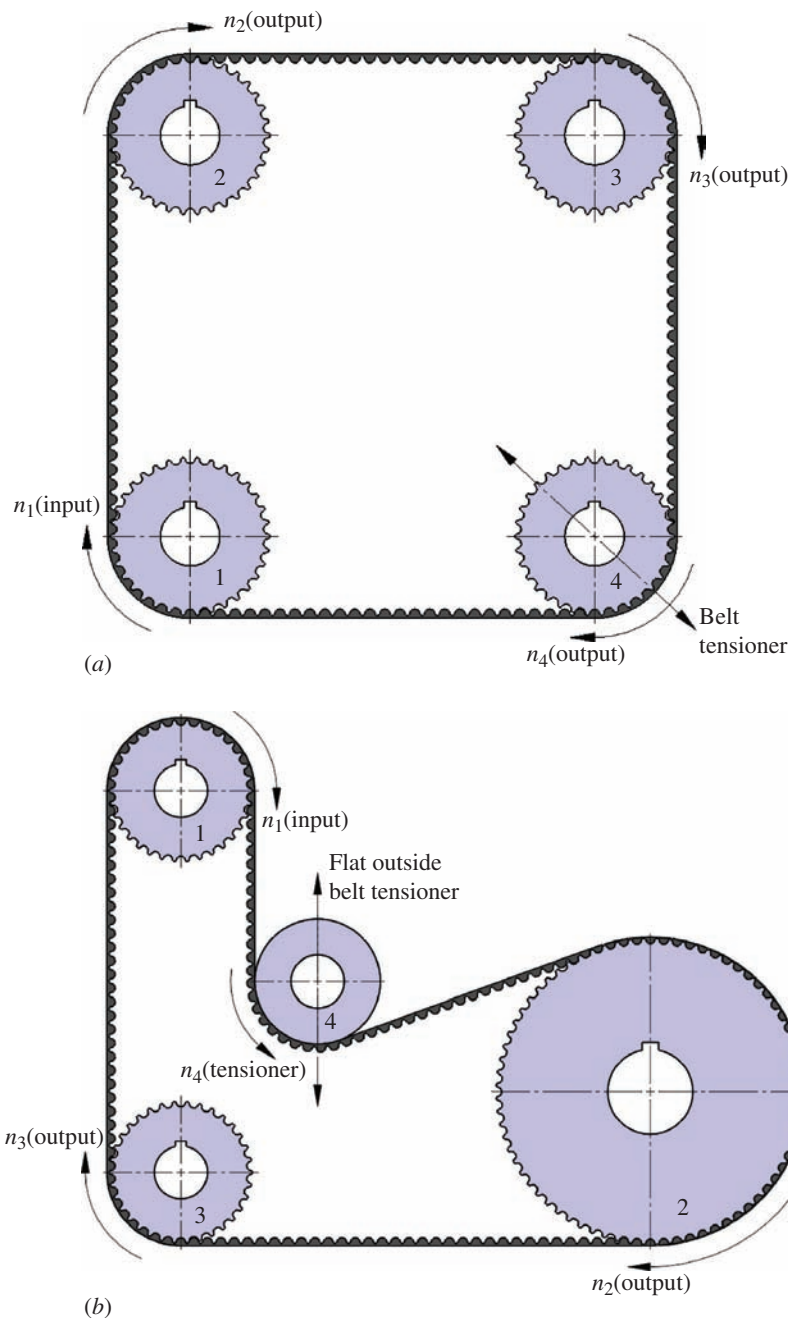


FIGURE 7-31 Multiple shaft drive configurations

designs with multiple drive points to reverse the shaft rotation. Figure 7-32(a) shows a twin tooth serpentine belt drive with the input sprocket (1), output sprocket (2), fixed idler (3), and inside belt tensioner (4). The objective of this design is to have the output sprocket (2) rotate in the opposite direction of the input sprocket (1). The fixed idler (3) does not drive anything, but it is used to wrap the belt around the output sprocket (2) to provide more teeth to carry the belt driving tension. The inside belt tensioner (4) is an idler that is movable

and is used to position the belt to wrap the output sprocket (2) and to set proper belt tension. The belt tensioner (4) also does not drive any output directly. Figure 7-31(b) shows a twin tooth serpentine belt drive with an input sprocket (1) and five driven sprockets. The input and output sprockets (2), (4), and (6) have clockwise rotation while the output sprockets (3) and (5) have opposite rotation due to the serpentine belt wrap. The speed of the five output sprockets are dependent on the input and output sprocket ratios.

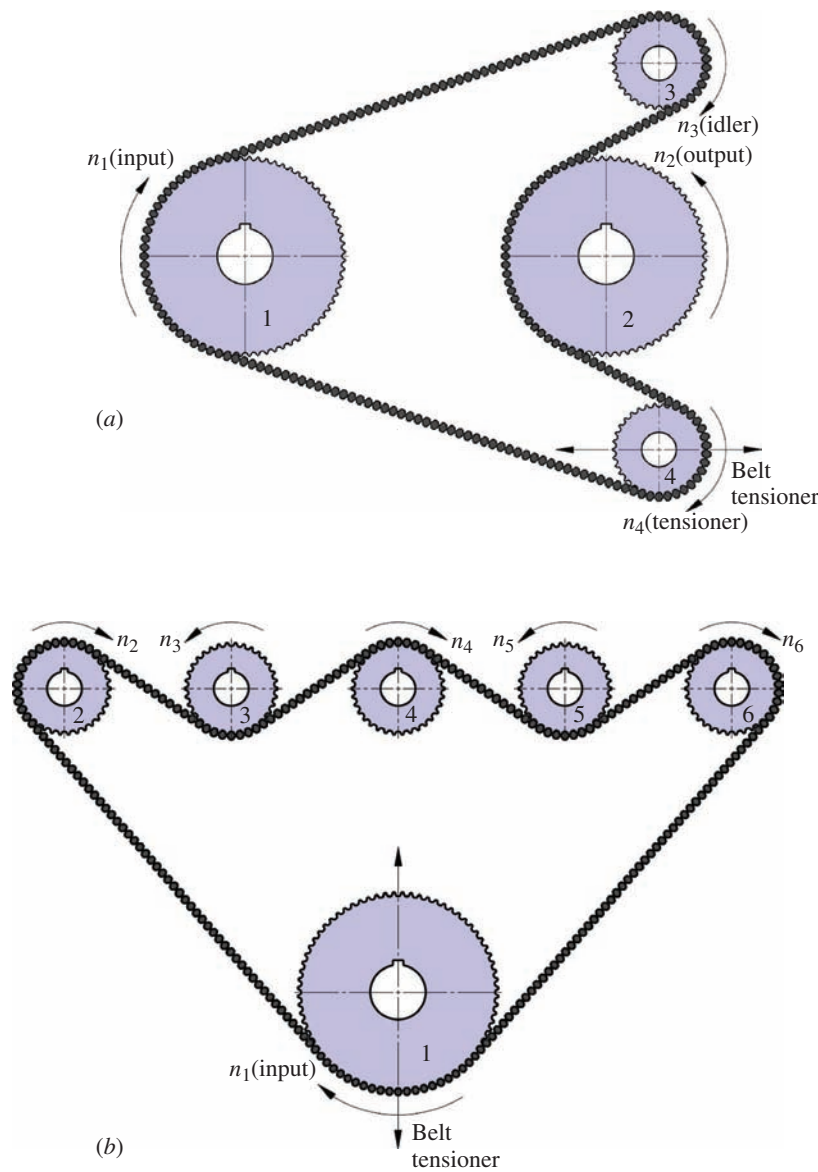


FIGURE 7-32 Serpentine belt drive configurations

7-6 CHAIN DRIVES

A chain is a power transmission element made as a series of pin-connected links. The design provides for flexibility while enabling the chain to transmit large tensile forces. See References 1–3 and Internet sites 1, 4, 6–12, 14, and 15 for more technical information and manufacturers' data.

When transmitting power between rotating shafts, the chain engages mating toothed wheels, called sprockets. Figure 7-33 shows a typical chain drive.

The most common type of chain is the *roller chain*, in which the roller on each pin provides exceptionally low friction between the chain and the sprockets.

Roller chain is classified by its *pitch*, the distance between corresponding parts of adjacent links. The pitch is usually illustrated as the distance between the centers of adjacent pins. U.S. Standard roller chain carries a size designation from 40 to 240, as listed in Table 7-11.

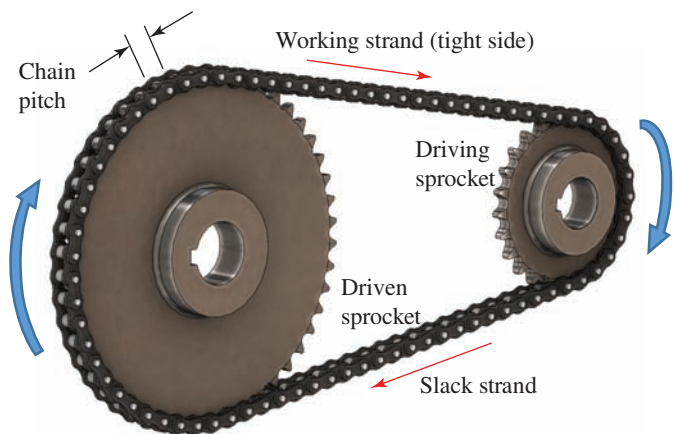


FIGURE 7-33 Basic arrangement of an industrial chain drive

See Reference 2. The digits (other than the final zero) indicate the pitch of the chain in eighths of an inch, as in the table. For example, the no. 100 chain has a pitch of 10/8 or 1 1/4 in. A series of heavy-duty sizes, with the suffix *H* on the designation (60H–240H), has the same basic dimensions as the standard chain of the same number except for thicker side plates. In addition, there are the smaller and lighter sizes: 25, 35, and 41.

The average tensile strengths of the various chain sizes are also listed in Table 7–12. These data can be used for very-low-speed drives or for applications in which the function of the chain is to apply a tensile force or to support a load. It is recommended that only 10% of the average tensile strength be used in such applications. For power transmission, the rating of a given chain size as a function of the speed of rotation must be determined, as explained later in this chapter.

ISO standards define several different chain types, data for three of which are listed in Table 7–13. One commonly used style from ISO-606 has basically the same design dimensions as for many of the standard U.S. roller chains. Then the pitch and dimensions for sprocket features and bore sizes are listed in the metric unit of mm making it more convenient to integrate familiar chain designs into an all-metric piece of equipment. ISO-3512 includes eight sizes of chain used for heavy-duty power transmission and lifting equipment. Some of the pitches for these chains are also equal to common U.S. sizes. Agricultural equipment such as tractor accessories, planters, harvesters, and mowers employ many chain drives to actuate moving systems. ISO-487 defines eight S-designations that cover a wide range of power transmission and

tension pull applications. See Reference 3 and Internet sites 7, 9, 14, and 15 for more information on metric-style chains and for manufacturers' data.

Another ISO document that is closely related to U.S. roller chain sizes is ISO 10823 and those designations are shown in Table 7–12. The designations are very similar to those listed in Table 7–13 from ISO 606, except the letter following the number is A instead of B.

Other types of chains include multiple strand designs, heavy series chains, double-pitch chains, and double-pitch conveyor chains as shown on the left side of Figure 7–34. A wide variety of attachments are available to facilitate the application of roller chain to conveying or other material handling uses. Usually in the form of extended plates or tabs with holes provided, the attachments make it easy to connect rods, buckets, parts pushers, part support devices, or conveyor slats to the chain. The right side of Figure 7–34 shows some attachment styles.

Figure 7–35 shows a variety of chain types used especially for conveying and similar applications. Such chain typically has a longer pitch than standard roller chain (usually twice the pitch), and the link plates are heavier. The larger sizes have cast link plates.

Design of Chain Drives

The rating of chain for its power transmission capacity considers three modes of failure: (1) fatigue of the link plates due to the repeated application of the tension in the tight side of the chain, (2) impact of the rollers as they engage the sprocket teeth, and (3) galling between the pins of each link and the bushings on the pins.

The ratings are based on empirical data with a smooth driver and a smooth load (service factor = 1.0) and with a rated life of approximately 15 000 h. The important variables are the pitch of the chain and the size and rotational speed of the smaller sprocket. Lubrication is critical to the satisfactory operation of a chain drive. Manufacturers recommend the type of lubrication method for given combinations of chain size, sprocket size, and speed. Details are discussed later.

Tables 7–14 to 7–16 list the rated power for three sizes of standard chain: no. 40 (1/2 in), no. 60 (3/4 in), and no. 80 (1.00 in). These are typical of the types of data available for all chain sizes in manufacturers' catalogs and can be used for problems in this book. When making final designs and specification, you should consult the catalog data for the particular manufacturer you are using. Notice these features of the data:

1. The ratings are based on the speed of the smaller sprocket and an expected life of approximately 15 000 h.
2. For a given speed, the power capacity increases with the number of teeth on the sprocket. Of course, the larger the number of teeth, the larger the diameter of the sprocket. Note that the use of a chain with a small pitch on a large sprocket produces the quieter drive.

TABLE 7–12 U.S. Roller Chain Sizes

Chain number	ISO 10823	Pitch (in)	Average tensile strength (lb)
25	4A	1/4	925
35	6A	3/8	2100
41		1/2	2000
40	8A	1/2	3700
50	10A	5/8	6100
60	12A	3/4	8500
80	18A	1	14 500
100	20A	1 1/4	24 000
120	24A	1 1/2	34 000
140	28A	1 3/4	46 000
160	32A	2	58 000
180	36A	2 1/4	80 000
200	40A	2 1/2	95 000
240	48A	3	130 000

Reference: ANSI Standard B29.1.

TABLE 7-13 Metric Roller Chain Sizes and Strength Ratings**General-Purpose power transmission****ISO-606**

Chain number	Pitch		Breaking strength		Alternate designation
	in	mm	lb	kN	
04B	0.250	6.350	—	—	—
05B	0.315	8.000	989	4.4	—
06B	0.375	9.525	2 001	8.9	Metric 35
08B	0.500	12.700	4 002	17.8	Metric 40
10B	0.625	15.88	4 991	22.2	Metric 50
12B	0.75	19.05	6 497	28.9	Metric 60
16B	1.00	25.40	9 510	42.3	Metric 80
20B	1.25	31.75	14 501	64.5	Metric 100
24B	1.50	38.10	22 010	97.9	Metric 120
28B	1.75	44.45	29 002	129.0	Metric 140
32B	2.00	50.8	37 995	169.0	Metric 160
40B	2.50	63.5	58 993	262.4	Metric 200
48B	3.00	76.2	89 996	400.3	Metric 240
56B	3.50	88.9	122 010	542.7	—
64B	4.00	101.6	160 004	711.7	—
72B	4.50	114.3	202 001	898.5	—

Heavy-Duty power transmission**ISO-3512**

Chain number	Pitch		Breaking strength	
	in	mm	lb	kN
2010	2.500	63.5	58 903	262.0
2512	3.067	77.9	84 982	378
2814	3.500	88.9	116 007	516
3315	4.073	103.5	133 993	596
3618	4.500	114.3	183 004	814
4020	5.000	127.0	236 960	1054
4824	6.000	152.4	341 951	1521
5628	7.000	177.8	464 928	2068

Power transmission for agricultural uses**ISO-487**

Chain number	Pitch		Breaking strength	
	in	mm	lb	kN
S32	1.150	29.2	1 799	8.0
S42	1.375	34.9	6 003	26.7
S45	0.843	21.4	4 002	17.8
S52	1.500	38.1	4 002	17.8
S55	1.630	41.4	4 002	17.8
S62	1.650	41.9	6 003	26.7
S77	2.297	58.3	10 004	44.5
S88	2.609	66.3	10 004	44.5

Notes: Not all suppliers offer all sizes.

Breaking strength data must be verified with specific supplier.

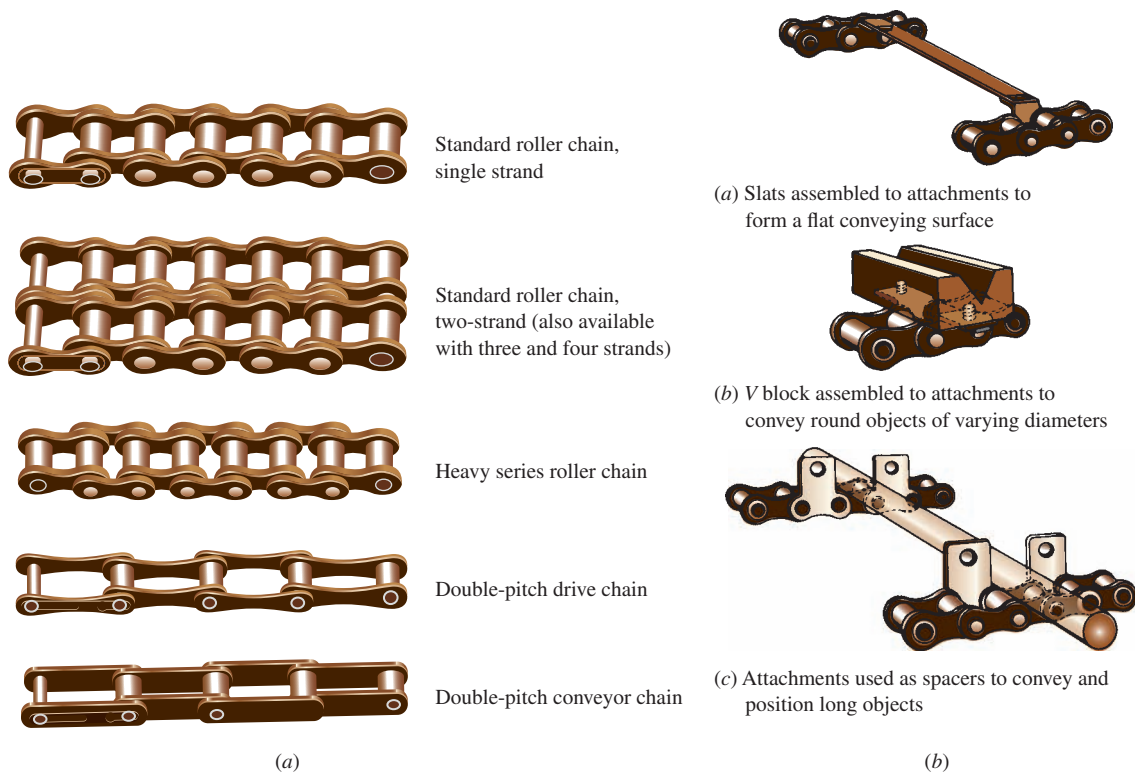


FIGURE 7-34 Other roller chain and examples of attachments

- For a given sprocket size (a given number of teeth), the power capacity increases with increasing speed up to a point; then it decreases. Fatigue due to the tension in the chain governs at the low to moderate speeds; impact on the sprockets governs at the higher speeds. Each sprocket size has an absolute upper-limit speed due to the onset of galling between the pins and the bushings of the chain. This explains the abrupt drop in power capacity to zero at the limiting speed.
- The ratings are for a single strand of chain. Although multiple strands do increase the power capacity, they do not provide a direct multiple of the single-strand capacity. Multiply the capacity in the tables by the following factors.

Two strands: Factor = 1.7

Three strands: Factor = 2.5

Four strands: Factor = 3.3

- The chain manufacturer's ratings are for a service factor of 1.0. Specify a service factor for a given application as shown in Table 7-16. The combination of the nature of the driving member and the driven machine permit the selection of the service factor. Then compute the design power, P_{des} , from,

$$P_{des} = SF(P)$$

where P is the power delivered to the chain drive.

DESIGN GUIDELINES FOR CHAIN DRIVES ▼

The following are general recommendations for designing chain drives:

- The minimum number of teeth in a sprocket should be 17 unless the drive is operating at a very low speed, under 100 rpm.
- The maximum speed ratio should be 7.0, although higher ratios are feasible. Two or more stages of reduction can be used to achieve higher ratios.
- The center distance between the sprocket axes should be approximately 30 to 50 pitches (30–50 times the pitch of the chain).
- The larger sprocket should normally have no more than 120 teeth.
- The preferred arrangement for a chain drive is with the centerline of the sprockets horizontal and with the tight side on top.
- The chain length must be an integral multiple of the pitch, and *an even number of pitches is recommended*. The center distance should be made adjustable to accommodate the chain length and to take up for tolerances and wear. Excessive sag on the slack side should be avoided, especially on drives that are not horizontal. A convenient relation between center distance (CD), chain length (L_c), number of teeth in the small sprocket (N_1), and number of teeth in the large sprocket (N_2), expressed in pitches, is

Chain Length in Pitches

$$L_c = 2CD + \frac{N_2 + N_1}{2} + \frac{(N_2 - N_1)^2}{4\pi^2 CD} \quad (7-18)$$

The center distance for a given chain length, again in pitches, is

Center Distance in Pitches

$$CD = \frac{1}{4} \left[L_c - \frac{N_2 + N_1}{2} + \sqrt{\left[L_c - \frac{N_2 + N_1}{2} \right]^2 - \frac{8(N_2 - N_1)^2}{4\pi^2}} \right] \quad (7-19)$$

The computed center distance assumes no sag in either the tight or the slack side of the chain, and thus it is a *maximum*. Negative tolerances or adjustment must be provided. Adjustment for wear must also be provided.

- The pitch diameter of a sprocket with N teeth for a chain with a pitch of p is

Pitch Diameter of Sprocket

$$PD = \frac{p}{\sin(180^\circ/N)} \text{ inches or mm} \quad (7-20)$$

- The minimum sprocket diameter and therefore the minimum number of teeth in a sprocket are often limited by the size of the shaft on which it is mounted. Check the sprocket catalog.
- The arc of contact, θ_1 , often called the *angle of wrap*, of the chain on the smaller sprocket should be greater than 120° .

Angle of Wrap Smaller Sprocket

$$\theta_1 = 180^\circ - 2\sin^{-1}[(PD_2 - PD_1)/2CD] \quad (7-21)$$

- For reference, the arc of contact, θ_2 , on the larger sprocket is

Angle of Wrap Large Sprocket

$$\theta_2 = 180^\circ + 2\sin^{-1}[(PD_2 - PD_1)/2CD] \quad (7-22)$$



Mill, narrow series
(drive and conveyor sizes)
Offset cast-link chain used primarily in the lumber industry for conveyor applications.



Combination mill
(wide conveyor sizes)
Cast block links and steel sidebar construction for drag conveyor applications.



Heavy-duty drag chain
Cast steel offset block links. Used in ash and clinker conveyors.



Pintle chain
Chain constructed of a series of cast offset links coupled by steel pins or rivets. Suitable for slow-to-moderate-speed drive, conveyor and elevator service.



Roller-top transfer
Cast links with top rollers used in several strands to convey material transversely.



Roof-top
Cast roof-shaped links used in several strands on transfer conveyors.



Detachable
Consists of unit links, each with an open-type hook that flexes on the end bar of the adjacent link. Used for slow- to moderate-speed drive and conveyor application.



Drop-forged
Drop-forged inner and outer links coupled by headed pins. Used for trolley, scraper, flight, and similar conveyors.

FIGURE 7-35 Conveyor chains (Rexnord Industries, LLC, Milwaukee, WI)

TABLE 7-14 Horsepower Ratings—Single-Strand Roller Chain No. 40

No. of teeth	0.500-in pitch										Rotational speed of small sprocket, rev/min														
	10	25	50	100	180	200	300	500	700	900	1000	1200	1400	1600	1800	2100	2500	3000	3500	4000	5000	6000	7000	8000	9000
11	0.06	0.14	0.27	0.52	0.91	1.00	1.48	2.42	3.34	4.25	4.70	5.60	6.49	5.57	4.66	3.70	2.85	2.17	1.72	1.41	1.01	0.77	0.61	0.50	0.00
12	0.06	0.15	0.29	0.56	0.99	1.09	1.61	2.64	3.64	4.64	5.13	6.11	7.09	6.34	5.31	4.22	3.25	2.47	1.96	1.60	1.15	0.87	0.69	0.57	0.00
13	0.07	0.16	0.31	0.61	1.07	1.19	1.75	2.86	3.95	5.02	5.56	6.62	7.68	7.15	5.99	4.76	3.66	2.79	2.21	1.81	1.29	0.98	0.78	0.00	0.00
14	0.07	0.17	0.34	0.66	1.15	1.28	1.88	3.08	4.25	5.41	5.98	7.13	8.27	7.99	6.70	5.31	4.09	3.11	2.47	2.02	1.45	1.10	0.87	0.00	0.00
15	0.08	0.19	0.36	0.70	1.24	1.37	2.02	3.30	4.55	5.80	6.41	7.64	8.86	8.86	7.43	5.89	4.54	3.45	2.74	2.24	1.60	1.22	0.97	0.00	0.00
16	0.08	0.20	0.39	0.75	1.32	1.46	2.15	3.52	4.86	6.18	6.84	8.15	9.45	9.76	8.18	6.49	5.00	3.80	3.02	2.47	1.77	1.34	0.00	0.00	0.00
17	0.09	0.21	0.41	0.80	1.40	1.55	2.29	3.74	5.16	6.57	7.27	8.66	10.04	10.69	8.96	7.11	5.48	4.17	3.31	2.71	1.94	1.47	0.00	0.00	0.00
18	0.09	0.22	0.43	0.84	1.48	1.64	2.42	3.96	5.46	6.95	7.69	9.17	10.63	11.65	9.76	7.75	5.97	4.54	3.60	2.95	2.11	1.60	0.00	0.00	0.00
19	0.10	0.24	0.46	0.89	1.57	1.73	2.56	4.18	5.77	7.34	8.12	9.66	11.22	12.64	10.59	8.40	6.47	4.92	3.91	3.20	2.29	0.09	0.00	0.00	0.00
20	0.10	0.25	0.48	0.94	1.65	1.82	2.69	4.39	6.07	7.73	8.55	10.18	11.81	13.42	11.44	9.07	6.99	5.31	4.22	3.45	2.47	0.00	0.00	0.00	0.00
21	0.11	0.26	0.51	0.98	1.73	1.91	2.83	4.61	6.37	8.11	8.98	10.69	12.40	14.10	12.30	9.76	7.52	5.72	4.54	3.71	2.65	0.00	0.00	0.00	0.00
22	0.11	0.27	0.53	1.03	1.81	2.01	2.96	4.83	6.68	8.50	9.40	11.20	12.99	14.77	13.19	10.47	8.06	6.13	4.87	3.98	2.85	0.00	0.00	0.00	0.00
23	0.12	0.28	0.56	1.08	1.90	2.10	3.10	5.05	6.98	8.89	9.83	11.71	13.58	15.44	14.10	11.19	8.62	6.55	5.20	4.26	3.05	0.00	0.00	0.00	0.00
24	0.12	0.30	0.58	1.12	1.98	2.19	3.23	5.27	7.28	9.27	10.26	12.22	14.17	16.11	15.03	11.93	9.18	6.99	5.54	4.54	0.87	0.00	0.00	0.00	0.00
25	0.13	0.31	0.60	1.17	2.06	2.28	3.36	5.49	7.59	9.66	10.69	12.73	14.76	16.78	15.98	12.68	9.76	7.43	5.89	4.82	0.00	0.00	0.00	0.00	0.00
26	0.13	0.32	0.63	1.22	2.14	2.37	3.50	5.71	7.89	10.04	11.11	13.24	15.35	17.45	16.95	13.45	10.36	7.88	6.25	5.12	0.00	0.00	0.00	0.00	0.00
28	0.14	0.35	0.67	1.31	2.31	2.55	3.77	6.15	8.50	10.82	11.97	14.26	16.53	18.79	18.94	15.03	11.57	8.80	6.99	5.72	0.00	0.00	0.00	0.00	0.00
30	0.15	0.37	0.72	1.41	2.47	2.74	4.04	6.59	9.11	11.59	12.82	15.28	17.71	20.14	21.01	16.67	12.84	9.76	7.75	6.34	0.00	0.00	0.00	0.00	0.00
32	0.16	0.40	0.77	1.50	2.64	2.92	4.31	7.03	9.71	12.38	13.66	16.30	18.89	21.48	23.14	18.37	14.14	10.76	8.54	1.41	0.00	0.00	0.00	0.00	0.00
35	0.18	0.43	0.84	1.64	2.88	3.19	4.71	7.69	10.62	13.52	14.96	17.82	20.67	23.49	26.30	21.01	16.17	12.30	9.76	0.00	0.00	0.00	0.00	0.00	0.00
40	0.21	0.50	0.96	1.87	3.30	3.65	5.38	8.79	12.14	15.45	17.10	20.37	23.62	26.85	30.06	25.67	19.76	15.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00
45	0.23	0.56	1.08	2.11	3.71	4.10	6.06	9.89	13.66	17.39	19.24	22.92	26.57	30.20	33.82	30.63	23.58	5.53	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Type A: Manual or drip lubrication

Type B: Bath or disc lubrication

Type C: Oil stream lubrication